

Geometric Computation: Twin Helices and the Algebra of Flow

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Abstract

This work constructs a manifold consisting of twin helices to investigate the existence of computational geometric bordisms, as part of the broader program of the Cobordism Hypothesis. We use these structures to define a class of deterministic computational substrates, termed *Geometrically Computable Manifolds*, and study their geometric, algebraic, and computational properties. We show that, in a discrete regime, these manifolds encode arbitrary cyclic groups and support universal computation. This leads to a theorem describing the structural trade-offs in physical computation across distinct computational phases. We then analyze a regime in which abelian gauge computation is modeled with ordinary differential equations, linking the construction to continuous geometric dynamics. This signals a new research direction in physical mathematics at the intersection of geometry, gauge theory, and computability theory, termed *Geometric Computability Theory*, in which computation is treated as a property of geometric compositions. This perspective opens avenues for future investigation, motivating the development of a constructive computational gauge theory with applications to Yang–Mills theory. As of this writing, no specific algebraic calculus yet exists to formalize the behavior of these manifolds; future work will develop this formalism.

Keywords: geometric computation, higher gauge theory, prequantum geometry, Yang–Mills theory, cobordism hypothesis, helices, topological quantum field theory.

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Introduction

“In the long run, we hope that some theory of language expressiveness develops into a formal theory of the programming language design space.”

— Mattias Felleisen, *On the Expressive Power of Programming Languages* (1991)

“A twisted version of four-dimensional supersymmetric gauge theory is formulated...”

— Edward Witten, *Topological Quantum Field Theory* (1988)

The following work constructs a geometric model of computation.

The gap between computer science and physics has been a long-standing one. While various attempts to bridge the divide have been made over the years, a full resolution remains out of reach. On the one hand, familiar models of computation are fundamentally discrete, while much of the symmetries observed in fundamental physics are continuous. So, are the laws of physics truly computable? Penrose and others have argued that certain key physical (and cognitive) phenomena may elude algorithmic description; by contrast, mainstream practice treats numerical approximation on a continuum as acceptable. Here we bridge this divide.

Section 1 constructs a deterministic geometric substrate from manifolds built out of joined helices. This leads naturally to coherence conditions imposed by chirality, torsion, and gluing, and from these conditions emerges a class of symbolic flow manifolds with non-cancellative monoidal structure. We then analyze their basic group-theoretic, geometric, and algebraic behavior.

Section 2 introduces the main computational objects of the paper: the *Geometrically Computable Manifold* and its associated state object, the *geobit*. The goal is to reconcile semantic expressiveness in computation with the structural stability and reversibility of geometric processes. Instead of treating semantics syntactically, we recast it geometrically in terms of moduli spaces of helices and their categorical behavior under symbolic flow. This leads to a framework in which symbolic computation is expressed through geometry itself, and from this analysis, we arrive at the central result of the paper: a conservation law governing computational behavior.

Section 3 studies the algebraic behavior of these manifolds as computation evolves toward increasingly gauge-like regimes. We examine foundational issues in continuous computation, establish formal bounds, and give explicit models for computational behavior in this setting. An extended homotopy-theoretic framework is then outlined, together with an abelian gauge-theoretic phase formalism and a discussion of the relevant moduli spaces.

Section 4 situates the framework within a broader categorical and physical setting. There, we examine how computational randomness may arise as a quotient of geometric complexity, through coarse-graining of torsion and chirality across moduli. Holonomy-based semantics are introduced as the theory passes into non-abelian gauge-theoretic behavior, and we analyze the behavior of the resulting moduli space under unrestricted analog flow. We conclude by outlining possible connections to LQG, string-theoretic, and twistor-based programs, together with a broader interpretation of quantum theory in light of our results.

In the end, we remain hopeful that a new form of constructive computational gauge theory may provide a way forward in the study of non-perturbative field theory.

Our task here, by its nature, is a work of synthesis as much as rigor. If its scope sometimes appears unruly, or its exposition occasionally ventures beyond traditional boundaries, it is only because the questions at hand resist treatment by standard means. I have aimed for clarity without sacrificing depth. So, if at times, arguments feel *unconventional*?

I hope that patient reading and engagement will reward the effort.

Ultimately, my intent is not to assert authority, but to invite the reader to explore together an unexpected landscape that may, in time, help reshape our understanding of computation itself.

1 Twin Helical Manifolds

1.1 Volumetric Helices

For a concrete realization, fix parameters

$$R > 0, \quad h \neq 0, \quad \theta_{\min} < \theta_{\max},$$

and define the helical spine

$$\gamma(\theta) = (R \cos \theta, R \sin \theta, h\theta), \quad \theta \in [\theta_{\min}, \theta_{\max}].$$

Let $(T(\theta), N(\theta), B(\theta))$ denote the Frenet–Serret frame of γ [Bis75].

Definition 1.1 (Volumetric Helix). Let ρ_{base} and ρ_{apex} satisfy

$$0 < \rho_{\text{apex}} < \rho_{\text{base}},$$

and define the normalized parameter

$$u = \frac{\theta - \theta_{\min}}{\theta_{\max} - \theta_{\min}}.$$

The cross-sectional radius profile is

$$\rho(\theta) = \rho_{\text{apex}} + (\rho_{\text{base}} - \rho_{\text{apex}}) \cos^2\left(\frac{\pi}{2}u\right).$$

The corresponding tubular body is the image of

$$\Phi(\theta, r', \phi) = \gamma(\theta) + r'(\cos \phi N(\theta) + \sin \phi B(\theta)),$$

for

$$\theta \in [\theta_{\min}, \theta_{\max}], \quad 0 \leq r' \leq \rho(\theta), \quad 0 \leq \phi < 2\pi.$$

The *base* is the distinguished disk at $\theta = \theta_{\min}$, and the *apex* is the terminal end at $\theta = \theta_{\max}$.

The profile $\rho(\theta)$ is monotone decreasing, attains its maximal value ρ_{base} at the base, attains its minimal value ρ_{apex} at the apex, and satisfies

$$\rho'(\theta_{\min}) = \rho'(\theta_{\max}) = 0.$$

Thus, the tube tapers smoothly from base to apex.

To remove the terminal boundary disk at the apex and obtain a smooth capped body, we attach a dome-shaped cap.

Definition 1.2 (Apical Cap). Let $(T_{\max}, N_{\max}, B_{\max})$ be the Frenet–Serret frame of γ evaluated at $\theta = \theta_{\max}$. The *apical cap* is the surface

$$S_{\text{cap}}(u, v) = \gamma(\theta_{\max}) + \rho_{\text{apex}} \sin(u) T_{\max} + \rho_{\text{apex}} \cos(u) (\cos v N_{\max} + \sin v B_{\max}),$$

for

$$u \in [0, \pi/2], \quad v \in [0, 2\pi).$$

At $u = 0$, this cap matches the terminal circle of the tube, and because $\rho'(\theta_{\max}) = 0$, the tangent plane of the cap agrees with the tangent plane of the tubular body along the junction. Hence, the capped volumetric helix has C^1 boundary.

Remark 1.1. We impose

$$\rho_{\text{base}} < R \quad \text{and} \quad \rho_{\text{base}} < \pi|h|$$

to ensure that the thickened tube [Gra04] is embedded and does not intersect either the helical axis or adjacent turns. We separate the *abstract topology* of the object from its *embedded geometric realization*. Abstractly, a capped finite helix is a compact smooth 3-manifold diffeomorphic to the 3-ball D^3 , equipped with a distinguished embedded base disk and a distinguished core arc. The helical geometry enters through the embedding in $\mathbb{R}^3$.

1.2 Chirality and Alignment

Definition 1.3 (Chirality). A subset $O \subset \mathbb{R}^n$ is said to have *chirality* if it is not properly congruent to its mirror image. Equivalently, O is chiral if for every reflection r across a plane in $\mathbb{R}^n$, there exists no proper isometry, i.e., orientation-preserving Euclidean motion (translations or rotations), g such that $g(O) = r(O)$. We say that helices may possess either left-handed or right-handed chirality. A helix is left-handed if $h < 0$ and right-handed if $h > 0$.¹

Lemma 1.1 (Spiral Matching Lemma). *Two Archimedean spirals, S_1 and S_2 , with distinct centers and the same pitch, can intersect to form a C^0 curve with a matching phase if and only if they have opposite chirality.*

Proof. Let the two spirals be defined in $\mathbb{R}^2$ by the vector equations:

$$S_i(\theta_i) = C_i + a_i \theta_i \begin{pmatrix} \cos \theta_i \\ \sin \theta_i \end{pmatrix}, \quad i = 1, 2$$

We are given that the centers are distinct ($C_1 \neq C_2$) and the pitches are equal ($|a_1| = |a_2| = a > 0$). The lemma requires two conditions to be met simultaneously:

- (a) **C^0 Intersection.** The spirals meet at a common point. $S_1(\theta_1) = S_2(\theta_2)$
- (b) **Phase Matching.** The local polar angles are the same. $\theta_1 = \theta_2 =: \theta$

Substitute the phase-matching condition into the intersection equation:

$$C_1 + a_1 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} = C_2 + a_2 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

Rearranging the terms to isolate the vector between the centers gives our primary relation:

$$C_2 - C_1 = (a_1 - a_2) \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

We analyze this relation for both chirality cases.

- (a) **Same Chirality.** ($a_1 = a_2$) If the spirals have the same chirality, the term $(a_1 - a_2)$ is zero. The relation becomes

$$C_2 - C_1 = 0 \implies C_1 = C_2$$

This contradicts the given condition that the centers are distinct. Therefore, an intersection with matching phase is impossible for spirals of the same chirality.

- (b) **Opposite Chirality.** ($a_2 = -a_1$) If the spirals have opposite chirality, the term $(a_1 - a_2)$ becomes $(a_1 - (-a_1)) = 2a_1$. The relation becomes

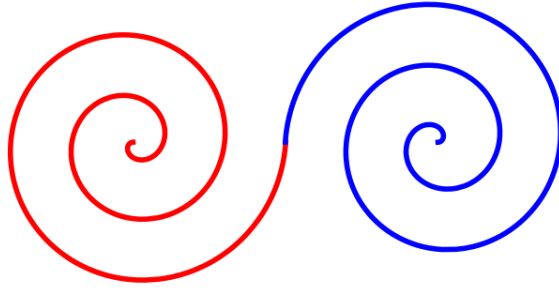
$$C_2 - C_1 = 2a_1 \theta \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

Thus, an intersection with matching phase is possible if and only if the spirals have opposite chirality. $\square$

Remark 1.2. We impose the phase-matching condition $\theta_1 = \theta_2$ so that the radial directions—and hence the unit tangent directions—agree at the intersection, removing any kink, the necessary first step toward any C^1 or C^2 splice.

Corollary 1.2 (Coaxial Spiral Chirality Constraint). *Let S_1 and S_2 be Archimedean spirals centered at the same point. Then, a C^0 bounded path connecting them with matching phase at a common point can exist only if their chiralities are identical.*

¹Here, *chirality* means literal geometric handedness of a helix: noncongruence with its mirror image under orientation-preserving Euclidean motions. It should not be confused with fermionic chirality in relativistic quantum field theory, which refers to left- and right-handed spinor representations or γ^5 -eigenspaces.



Proof. This is a direct consequence of the Spiral Matching Lemma. As proven above, with $C_1 = C_2$, equation:

$$(C_2 - C_1) = (a_1 - a_2)\theta(\cos \theta, \sin \theta)$$

which gives

$$(a_1 - a_2)\theta(\cos \theta, \sin \theta) = 0$$

for $\theta \neq 0$, this forces $a_1 = a_2$, which in turn requires the spirals to have the same chirality. $\square$

Remark 1.3. This illustrates a fundamental principle in geometric design: adding a simple local constraint, such as matching phase, can enforce a relationship between the curves' global properties (chirality). In this case, the seemingly mild phase-matching condition is sufficient to rule out a connection entirely for same-chirality spirals with offset centers.

1.3 Twin Helical Manifolds

We now form pairs of volumetric helices by gluing them along their base disks.

Definition 1.4 (Binary Orientation of a Helix). Let H be a helix in $\mathbb{R}^3$ with axis of symmetry parallel to the z -axis. The orientation of H is defined by the monotonicity of its z -coordinate as a function of its intrinsic parameter u , increasing from base to apex:

- (a) **Upwards-oriented.** $z(u)$ is strictly increasing from base to apex.
- (b) **Downwards-oriented.** $z(u)$ is strictly decreasing from base to apex.

Definition 1.5 (Type 1 Twin Helix). Fix $R > 0$ and $h > 0$. Let the first spine be

$$\gamma_A(\theta) = (R \cos \theta, R \sin \theta, h\theta), \quad \theta \in \mathbb{R},$$

and define the second spine by rotation through π about the x -axis:

$$\gamma_B(\theta) = (R \cos \theta, -R \sin \theta, -h\theta).$$

This produces the basic *Type 1* twin configuration. The remaining types are obtained from this model by applying the Euclidean isometries recorded in the final column of table 1.

Definition 1.6 (Twin Helical Manifold). Let H_A and H_B be volumetric helices, and let

$$B_A \subset \partial H_A, \quad B_B \subset \partial H_B$$

be their distinguished base disks. Given a diffeomorphism

$$f : B_A \rightarrow B_B,$$

the *Twin Helical Manifold* associated to (H_A, H_B, f) is the quotient

$$H_{AB} = (H_A \sqcup H_B) / \sim, \quad p \sim f(p) \quad (p \in B_A).$$

This quotient defines the abstract manifold obtained by gluing the two helices along their bases. In the present paper, we typically take f to be the identity in the natural cylindrical coordinates on the base disks. To realize this quotient as an embedded C^1 manifold in $\mathbb{R}^3$, we may interpolate between the two bases by an explicit bridge.

Definition 1.7 (Embedded Bridge Realization). Let P_A and P_B be the centers of the two base disks, and let $\mathbf{t}_A, \mathbf{t}_B$ be the corresponding unit tangent vectors of the spines at the base. Fix a tension parameter $\ell > 0$ and define

$$\mathbf{m}_A = \ell \mathbf{t}_A, \quad \mathbf{m}_B = \ell \mathbf{t}_B.$$

The bridge spine is the cubic Hermite curve [dB78]

$$\Gamma(s) = h_{00}(s)P_A + h_{01}(s)P_B + h_{10}(s)\mathbf{m}_A + h_{11}(s)\mathbf{m}_B, \quad s \in [0, 1],$$

where

$$\begin{aligned} h_{00}(s) &= 2s^3 - 3s^2 + 1, \\ h_{01}(s) &= -2s^3 + 3s^2, \\ h_{10}(s) &= s^3 - 2s^2 + s, \\ h_{11}(s) &= s^3 - s^2. \end{aligned}$$

By construction,

$$\Gamma(0) = P_A, \quad \Gamma(1) = P_B, \quad \Gamma'(0) = \mathbf{m}_A, \quad \Gamma'(1) = \mathbf{m}_B,$$

so the bridge matches the endpoint positions and tangent data.

For Types 2 and 4, where the naive Hermite curve may develop a small hair-pin, we replace Γ by the bumped curve

$$\tilde{\Gamma}(s) = \Gamma(s) + \alpha s^2(1-s)^2 \mathbf{n},$$

where $\mathbf{n}$ is a unit vector orthogonal to the relevant tangent directions and

$$\alpha > 2\rho_{\text{base}}.$$

This perturbation preserves the endpoint position and tangent data while moving the bridge clear of the parent tubes. A tubular neighborhood of Γ (or $\tilde{\Gamma}$) of radius at most ρ_{base} then gives an explicit C^1 bridge between the two volumetric helices. This produces an embedded realization of the abstract quotient H_{AB} inside $\mathbb{R}^3$.

Remark 1.4. The spline bridge is *not* part of the abstract definition of a Twin Helical Manifold. It is only one convenient geometric realization used to exhibit a C^1 embedding in $\mathbb{R}^3$. The underlying manifold is determined by the gluing of the base disks, while the particular bridge affects only the chosen embedded representative.

We call any such glued object a *Twin Helical Manifold* (THM), or more informally, a *Twin Helix*.

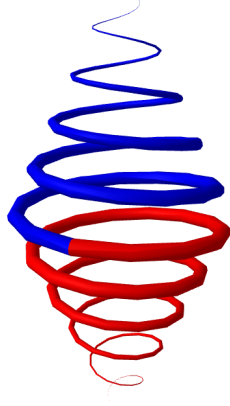
Definition 1.8 (Twin Helical Manifold and Unit Helices). A *THM* consists of two embedded helical surfaces $\Sigma_{\pm} \subset M$ whose boundary circles are identified along a smooth annulus $A \simeq S^1 \times I$, with orientations matched so that the resulting subset $\Sigma_+ \cup_A \Sigma_-$ is a smooth submanifold (no cone angle or cusp). A connection $A \in \Omega^1(M; \mathfrak{g})$ restricts smoothly across A ; its curvature F is bounded (no measure concentration). The two helices composing a single *THM* are termed *unit helices*, denoted H_A and H_B .

By combining different chiralities and orientations, we define a geometric family of four canonical manifolds, denoted THM_4 . This yields a total of 4 distinct invariant configuration states. However, our Type 2 twin helix has a small problem. A simple drawing shows that constructing a twin helix with the specified attributes becomes problematic in the previously defined $\mathbb{R}^3$ space. A closer investigation leads to the following conclusion.

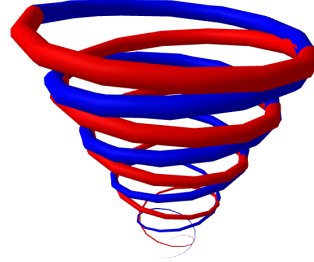
Lemma 1.3 (Twin Helix Intersection Lemma). *No twin helical manifold whose unit helices with turns $N \geq 1/2$, sharing both axial alignment, orientation, and opposing chirality, can exist without intersection of its respective unit helices in $\mathbb{R}^3$.*

Table 1: Essential type chart for twin helical manifolds

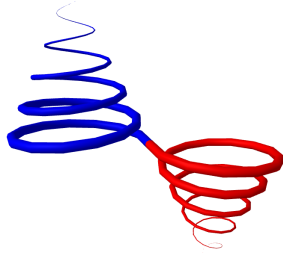
Type	Chirality	Orientation	Axiality	Mnemonic	Map $\Rightarrow H_B$
1	Opposite	Opposite	Coaxial	Helical Orb	$R_x(\pi)$
2	Opposite	Same	Coaxial	Twisted Pair	Σ_y
3	Same	Opposite	Offset	Symmetrical Flow	$R_y(\pi), T$
4	Same	Same	Offset	Co-propagating Twins	$R_z(\pi), T$



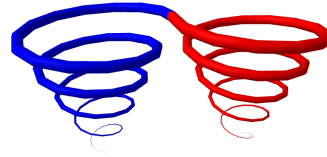
(a) Type 1



(b) Type 2



(c) Type 3



(d) Type 4

Figure 1: The THM_4 geometric family.

Proof. Let H_A and H_B be two coaxial helices parameterized by $t \in [0, 1]$. Both helices share the same radial function $r(t)$ and the same axial function $z(t)$. Their distinction lies in chirality and a possible phase offset.

The angular position of a point on helix H_x at parameter t is given by

$$\alpha_x(t) = 2\pi N \chi_x t + \phi_x,$$

where:

- (a) N is the number of turns in one unit helix,
- (b) χ_x is the chirality of H_x , with $\chi_A = -1$ and $\chi_B = 1$ for Type 2,
- (c) ϕ_x is the phase offset. Without loss of generality, we may set $\phi_A = 0$.

The center lines of the two helices intersect if, for some $t \in [0, 1]$, their angular positions coincide modulo 2π . That is

$$\alpha_B(t) - \alpha_A(t) = 2k\pi$$

for some integer k . Substituting the expressions for $\alpha_A(t)$ and $\alpha_B(t)$, we obtain:

$$\begin{aligned}(2\pi N\chi_B t + \phi_B) - (2\pi N\chi_A t) &= 2k\pi \\ 2\pi Nt(\chi_B - \chi_A) + \phi_B &= 2k\pi \\ 4\pi Nt + \phi_B &= 2k\pi.\end{aligned}$$

Define the function $f(t) = 4\pi Nt + \phi_B$. An intersection occurs if there exists a $t \in [0, 1]$ such that $f(t) = 2k\pi$ for some integer k .

The function $f(t)$ is continuous and monotonic on $[0, 1]$ (assuming $N > 0$). The values it takes range from

$$f(0) = \phi_B, \quad f(1) = 4\pi N + \phi_B.$$

This interval has length $4\pi N$. For $f(t)$ to contain at least one multiple of 2π , it suffices that

$$4\pi N \geq 2\pi \quad \Rightarrow \quad N \geq \frac{1}{2}.$$

By the Intermediate Value Theorem, since $f(t)$ is continuous on $[0, 1]$, it must attain all values in the interval $[\phi_B, \phi_B + 4\pi N]$. If this interval has length at least 2π , it must contain at least one value $2k\pi$. Therefore, for $N \geq \frac{1}{2}$, there always exists a $t \in [0, 1]$ for which $\alpha_B(t) \equiv \alpha_A(t) \pmod{2\pi}$, and thus the helices intersect.

Hence, for any $N \geq \frac{1}{2}$, it is mathematically impossible to choose a constant angular phase offset ϕ_B that avoids the intersection of the center lines of the Type 2 twin helices on $[0, 1]$. $\square$

Twin helices and the Whitney paradigm. In $\mathbb{R}^3$, a closed orientable surface has a line normal bundle; demanding a global 2-frame (to preserve both chirality and phase across the join) overconstrains codimension 1. Equivalently, any such immersion develops transverse self-intersections at the Type-2 junction, so the framed join cannot be realized without crossings. Passing to $\mathbb{R}^4$ removes this extrinsic obstruction: the normal bundle is rank 2 and can be trivialized, so one obtains a V_4 -equivariant, self-intersection-free embedding that preserves the twist data. This matches the general Whitney [Whi44] paradigm: every smooth 2-manifold embeds in $\mathbb{R}^4$, and the extra normal direction carries the required framing.

Chirality rule-reversal between 2- and 3-dimensional gluing. A curious but systematic reversal occurs when one passes from planar curves to spatial curves concerning the handedness required for a C^2 splice. In the planar setting (spirals), the data to be matched at an intersection consists of position, unit tangent, and signed curvature. Because curvature is the only first-order invariant beyond the tangent, one free phase parameter suffices to adjust its magnitude but not its sign. Consequently, two offset spirals can be C^0 -glued iff their curvature signs are opposite.

In the spatial setting (helices), a space curve adds a second scalar invariant (torsion), whose sign records right-handed versus left-handed screw motion. The gluing constraints now depend on the relative placement of the helical axes:

Proposition 1.4 (Rules for Smooth Gluing of Helices). *Let*

$$\gamma_i(t) = c_i + R_i(\cos t \mathbf{u}_i + \sin t \mathbf{v}_i) + p_i t \mathbf{e} \quad (i = 1, 2)$$

be two circular helices in $\mathbb{R}^3$ whose axes are parallel to the same unit vector $\mathbf{e}$.

Define curvature and torsion in the usual way:

$$\kappa_i = \frac{R_i}{R_i^2 + p_i^2}, \quad \tau_i = \frac{p_i}{R_i^2 + p_i^2},$$

so $\text{sign}(\tau_i)$ records right- or left-handedness.

- (a) **Coaxial case.** *If the centres c_1, c_2 differ only along the common axis $(c_2 - c_1) \parallel \mathbf{e}$, then one can splice γ_1 to γ_2 by a C^2 curve through an intersection point P iff*

$$\tau_1 + \tau_2 = 0, \quad \text{equivalently} \quad \text{sign}(\tau_1) = -\text{sign}(\tau_2).$$

- (b) **Offset-axis case.** If the axis directions coincide but the centres are displaced sideways, $(c_2 - c_1) \perp \mathbf{e}$, then a C^2 splice exists iff

$$\tau_1 = \tau_2, \quad \text{equivalently} \quad \text{sign}(\tau_1) = \text{sign}(\tau_2).$$

Proof. Identify $P = \gamma_1(t_1) = \gamma_2(t_2)$. Matching a 2-jet at P requires:

$$\mathbf{J}^0(\gamma_1) = \mathbf{J}^0(\gamma_2), \quad \mathbf{J}^1(\gamma_1) = \mathbf{J}^1(\gamma_2), \quad \mathbf{J}^2(\gamma_1) = \mathbf{J}^2(\gamma_2).$$

Explicitly this means (position) P coincides, (tangent) $\dot{\gamma}_1(t_1)$ is a positive multiple of $\dot{\gamma}_2(t_2)$, and (curvature + torsion) the Frenet frames agree.

Coaxial case. The equality of normals forces $t_2 = t_1 + \pi$ up to 2π , leaving a single phase parameter. The curvature equality $\kappa_1 = \kappa_2$ fixes R_2 once R_1 is given. The remaining torsion constraint is $\tau_1 + \tau_2 = 0$, hence opposite signs.

Offset case. Because $d \neq 0$ there are now *two* independent phase angles (t_1, t_2) available. These can be chosen so that the binormals (and hence normals) align, provided $\tau_1 = \tau_2$. If the torsion signs differ, the binormals rotate in opposite senses and cannot be made parallel.

Details are an elementary exercise in solving the linear constraints on $\dot{\gamma}_i$ and $\ddot{\gamma}_i$. $\square$

Remark 1.5. Passing from $\mathbb{R}^2$ to $\mathbb{R}^3$ promotes the matching problem from the 1-jet to the 2-jet bundle: curvature sign is now unsigned (since $\kappa > 0$), while torsion sign becomes the relevant orientation datum. Whether this new constraint binds or is absorbed by an extra phase degree of freedom hinges on the axial configuration. In effect, the added dimension introduces a framing (normal-binormal plane) whose sign can obstruct or permit the splice depending on how many phase parameters survive the geometric symmetry.

Proposition 1.5 ($THM_4 \cong V_4$). *The group of transformations generated by the Chirality Swap (X) and Orientation Swap (Y) operators, which acts on the set of four canonical THM types, is isomorphic to the Klein four-group, V_4 .*

Proof. Let $H_{\text{types}} = \{\text{Type 1, Type 2, Type 3, Type 4}\}$, representing the four configurations of unit helices determined by combinations of chirality (Same vs Opposite) and orientation (Same vs Opposite).

Define the following transformations acting on H_{types} :

- Identity I : Trivial transformation. $I(T_i) = T_i$ for all $T_i \in H_{\text{types}}$.
- Chirality Swap X : Swap the chirality relationship (Same $\leftrightarrow$ Opposite), while preserving orientation.
- Orientation Swap Y : Swap the orientation relationship (Same $\leftrightarrow$ Opposite), while preserving chirality.
- Composite $Z = X \circ Y$: Acts by applying both swaps.

The action on the four manifold types proceeds as follows:

Type	(Chirality, Orientation)	X	Y
H_1	(O, O)	H_3	H_2
H_2	(O, S)	H_4	H_1
H_3	(S, O)	H_1	H_4
H_4	(S, S)	H_2	H_3

Composition rules:

$$\begin{aligned}
Z &= X \circ Y = Y \circ X \\
Z(H_1) &= X(Y(H_1)) = X(H_2) = H_4 \\
Z(H_2) &= X(Y(H_2)) = X(H_1) = H_3 \\
Z(H_3) &= X(Y(H_3)) = X(H_4) = H_2 \\
Z(H_4) &= X(Y(H_4)) = X(H_3) = H_1
\end{aligned}$$

All non-identity elements have order 2:

$$X^2 = I, \quad Y^2 = I, \quad Z^2 = (X \circ Y)^2 = X^2 \circ Y^2 = I$$

Pairwise compositions:

$$X \circ Y = Z, \quad Y \circ X = Z \quad \Rightarrow \quad X \circ Y = Y \circ X$$

$$X \circ Z = X \circ (X \circ Y) = (X \circ X) \circ Y = I \circ Y = Y$$

$$Y \circ Z = Y \circ (X \circ Y) = (Y \circ Y) \circ X = I \circ X = X$$

The set of transformations $G = \{I, X, Y, Z\}$ forms a group under composition. All elements have order 1 or 2, and the group is abelian. Thus, $G \cong V_4$. $\square$

Table 2 displays the calculated finite-symmetry-group progressions computed with GAP [GAP].

Table 2: Group progression of the THM_4

Group Structure	Order	Defined by...	Type
$V_4 \cong C_2 \times C_2$	4	Chirality and orientation combinations of the 4 canonical THM_4 types	abelian
$D_4 \cong (C_4) \rtimes C_2$	8	Chirality flips + swapping identity of the two helices	non-abelian
C_2^4	16	Independent flips of (c_A, c_B, d_A, d_B) for each unit helix	abelian
$C_2^4 \rtimes C_2$	32	Above + helix identity swapping symmetry	non-abelian
$(C_2^4 \rtimes C_2) \times C_4$	128	Adds 90° global rotation symmetry around z -axis	non-abelian
$(C_2^4 \rtimes C_2) \times C_{4h} \cong (C_2^4 \rtimes C_2) \times (C_4 \times C_2)$	256	Full group including xy -plane reflection symmetry (σ_h)	non-abelian

Table 3: Generators of the THM_{256} symmetry group

Generator	Action
c_1	Flip chirality of Helix A
c_2	Flip chirality of Helix B
d_1	Flip orientation of Helix A (up/down)
d_2	Flip orientation of Helix B (up/down)
S	Swap identity of Helix A and Helix B
R_{90}	Global 90° rotation of the twin helix around the z -axis
σ_h	Reflect the entire twin helix across the xy -plane

In addition, the generators of THM_{256} and lower orders are shown in table 3. These symmetries exhibit a progression toward non-abelian behavior as more generators and state permutations are introduced. We note that this group structure may not be unique to the present construction. Similar generator sets arise in various contexts of symmetry and braid groups. For our purposes here, we do not require the full symmetry group, so the explicit relations and whole structure of THM_4 , including its precise order, are left open for further investigation.

Definition 1.9 (Regularity Principle for Geometric Computation). A geometric computational state consists of an admissible manifold configuration together with a specified local evolution law (flow, pulse update, or hybrid dynamics). A necessary condition for well-posed computation is that the geometric data and the evolution law possess sufficient regularity to prevent singular breakdown and to make local updates unambiguous.

At minimum, the state manifold must admit the geometric structures used by the model (e.g., tangent directions, phase variables, curvature), and the associated vector field or update operator must be at least locally Lipschitz on the relevant state space. This ensures local existence and uniqueness of the induced ordinary differential evolution whenever the model is expressed in ODE form.

Definition 1.10 (Minimal Regularity by Computational Regime). The required differentiability depends on which geometric invariants actively participate in the computation:

- **Abelian regimes** (symbolic and quasi-fluidic). It is generally sufficient that the geometric state be at least C^1 and that the local evolution law be locally Lipschitz (or C^1) in the active variables. This guarantees well-defined tangent/phase transport and deterministic local evolution. Higher regularity is often convenient but not essential.
- **Non-abelian regimes** (fully fluidic and gauge). When the computation depends explicitly on curvature, torsion, holonomy, or other higher-order geometric data, one typically requires at least C^2 regularity of the underlying geometric configuration, together with compatible regularity of the evolution law. In practice, C^k regularity for $k \geq 2$ (and often smoother) is preferable for stable geometric evolution.

Remark 1.6. Geometric computation must be regular enough for its update rules to be well-defined. In the symbolic and quasi-fluidic bands, this usually means controlling tangent and phase data; in the higher regimes, curvature and higher-order structure enter directly, forcing stronger regularity assumptions. Thus, singularities are not merely geometric defects: they mark a breakdown of the computational model itself.

2 Geometrically Computable Manifolds

We now begin the construction of a Turing Machine (“TM”) from a lattice of THM . The feasibility of this approach rests on the fact that THM , unlike most manifolds, provides a rich but controlled symmetry space. It is precisely because most well-known shapes in this metric (e.g., circles, tori, spheres) possess too much symmetry—not too little—that they fail to offer the structural asymmetry needed for computational behavior. The THM construction breaks this impasse.

2.1 Geometrically Computable Manifolds

Definition 2.1 (Geometrically Computable Manifold). A **GCM** on a space is:

$$\mathcal{G} = (\mathcal{M}, \{R_i\}_{i \in I}, \pi : E \rightarrow \mathcal{M}, \mathcal{I}, \text{Ev})$$

consisting of:

- (a) **Smooth Manifold $\mathcal{M}$.** A smooth manifold (often a finite working region, possibly with boundary). A finite family of compact submanifolds $\{R_i\}_{i \in I}$ (cells) covers the working region and provides locality (finite overlap pattern).
- (b) **State bundle $\pi : E \rightarrow \mathcal{M}$.** A smooth fiber bundle with finite-dimensional fibers. A *global state* is a section $\sigma \in \Gamma(E)$. Discrete/symbolic, toroidal/phase, and integer-valued components may all live in the fiber.
- (c) **Interfaces $\mathcal{I}$.** For each nonempty overlap $R_i \cap R_j$, an interface map:

$$I_{ij} : (\sigma|_{R_i}, \sigma|_{R_j}, \text{local data}) \mapsto (\sigma'|_{R_i}, \sigma'|_{R_j}),$$

attached to a portion of $\partial R_i \cap \partial R_j$. Interfaces couple neighboring cells and enforce locality: updates at R_i depend only on σ over R_i and finitely many overlaps $R_i \cap R_j$.

- (d) **Evolution Ev .** Either:

- discrete time: an update operator $U : \Gamma(E) \rightarrow \Gamma(E)$ that is local with respect to $\{R_i\}$, or
- continuous time: a flow of sections $(\sigma_t)_{t \in \mathbb{R}}$ solving $\dot{\sigma}_t = F(\sigma_t)$ for a smooth, fiber-wise vector field F over $\mathcal{M}$;

possibly accompanied by a base diffeomorphism $\Phi_t : \mathcal{M} \rightarrow \mathcal{M}$. In either case, evolution is non-singular; it exists for all steps/times on admissible initial states and remains in a fixed finite-energy subset of E .

What a GCM is. A GCM is a manifold that can encode its computable properties. A single THM can qualify as a GCM. A lattice of THM is also a GCM—just with more computing capacity. We call $\mathcal{G}$ a GCM when its evolution is local, interface-driven, and closed under composition: finite products of such systems and gluing along matched interfaces yield new systems of the same kind.

2.2 Geobits—Bits, Geometric Bits, and State Space

Definition 2.2 (Geometric Bits — Symbolic and Enriched). Fix an ambient manifold region $\mathcal{M} \subset \mathbb{R}^n$ and a twin helix template H (a pair of embedded tubular helices) together with a framing: a preferred axial direction $\hat{z}$ and a reference scale r_0 . Let $\text{Emb}^{\text{fr}}(H, \mathcal{M})$ denote the set of framed embeddings $e : H \hookrightarrow \mathcal{M}$ modulo framed ambient isotopy that preserves the axial framing and the scale r_0 so that radius and orientation remain part of the state.

Let $H_{\text{ext}} \leq O(n)$ be a chosen ambient symmetry group (allowing reflections so that chirality changes are representable) and let H_{int} be a finite internal symmetry group acting on the labels of the twin (e.g. the swap of the two helices). Set $G := H_{\text{ext}} \times H_{\text{int}}$, acting on $\text{Emb}^{\text{fr}}(H, \mathcal{M})$ by pre/post-composition.

(a) **Symbolic geobit.** Define the symbolic geobit state set:

$$\mathbf{S}_{\text{sym}} := \left\{ [e; \chi, \omega] \mid e \in \text{Emb}^{\text{fr}}(H, \mathcal{M}), \chi, \omega \right\} / G,$$

where $[e; \chi, \omega]$ denotes a framed embedding decorated by chirality (χ) and axial orientation (ω) relative to $\hat{z}$, and the quotient is by the G -action, so states are defined up to global symmetry. A *symbolic geobit* is an element $s \in \mathbf{S}_{\text{sym}}$. The finite subgroup generated by “flip chirality” C , “flip orientation” D , and (optionally) “swap” $S \in H_{\text{int}}$ acts on $\mathbf{S}_{\text{sym}}$; in the canonical twin helix unit this induces a $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ action on the labels (χ, ω) .

(b) **Enriched (quasi-fluidic) geobit.** To allow compact phase and quantized charges, extend the label by a finite-type fiber:

$$P \cong (S^1)^m \times \mathbb{Z}^p \times F,$$

where the S^1 -coordinates encode bounded phases (e.g. axial angle), the $\mathbb{Z}$ -coordinates encode twist/winding/linking numbers, and F is a finite set of discrete tags (e.g. mode, parity). Set

$$\mathbf{S}_{\text{enr}} := \left\{ [e; \chi, \omega, p] \mid e \in \text{Emb}^{\text{fr}}(H, \mathcal{M}), \chi, \omega, p \in P \right\} / G,$$

with G acting equivariantly on the S^1 -factors (by rotation) and preserving the $\mathbb{Z}$ -charges. An *enriched geobit* is an element of $\mathbf{S}_{\text{enr}}$. The G -action need not be transitive; its orbits in $\mathbf{S}_{\text{enr}}$ are the *geobit types*. The symbolic case is recovered by taking $m = p = 0$ and F trivial.

In both cases, the geobit is a G -set (not necessarily a group): the finite operations generated by C, D (and S when present) act on states, yielding the canonical V_4 action in the symbolic unit. The enrichment by P provides the quasi-fluidic degrees of freedom (compact phases and integer counters) used elsewhere in the text.

For the time, we work only with $\mathbf{S}_{\text{sym}}$. Now, let:

$$(H_1, H_2, H_3, H_4) \mapsto (00, 01, 10, 11)$$

where each H_n is a configuration from the THM_4 symmetry schema defined in table 1, preserving the order of the original type chart. We assign bit values based on symmetry: if the chirality and orientation of the unit helices are the same, assign bit value 1; if they differ, assign 0. This yields a total of 2 bits per THM_4 configuration.

Definition 2.3 (Geobit Capacity). Let L be the finite set of *internal labels* for a GCM cell (e.g. chirality/orientation choices before identifying symmetries), and let the *encoding symmetry* O act on L by intrinsic transformations (excluding extrinsic rigid motions). The set of *observable* states is the orbit set L/O , and the *geobit capacity* is

$$\text{Cap}(L, O) := \log_2 |L/O|.$$

When $L = O$ with the left-regular action, this reduces to $\text{Cap} = \log_2 |O|$.

Example 2.1 (Embedding Relationships for Helices). Let G act on a configuration space and let $O \subseteq G$ be the intrinsic encoding subgroup; put L for the raw label set and compute observable states via L/O .

- (a) THM_{16} . $O \cong C_2^4$, $L = O$ with the left-regular action. Then $|L/O| = |O| = 16$, so

$$\text{Cap} = \log_2 16 = 4 \text{ bits} \quad (\text{“2 bits from } \chi, 2 \text{ from } \omega\text{”}).$$

- (b) THM_4 . Impose identifications (e.g., identity-swap/pairing symmetries) modeled by a normal subgroup $H \trianglelefteq O$ of order 4; the effective encoding collapses to $O_{\text{eff}} \cong O/H \cong C_2^2$. With $L = O_{\text{eff}}$ (regular action),

$$\text{Cap} = \log_2 |O_{\text{eff}}| = \log_2 4 = 2 \text{ bits}.$$

- (c) THM_2 . If only orientation is swappable, the effective group is $O_{\text{eff}} \cong C_2$, hence

$$\text{Cap} = \log_2 2 = 1 \text{ bit}.$$

- (d) THM_8 . Let the raw labels be $L = C_2 \times C_4$ with elements (χ, ω) (chirality $\chi \in \{\pm 1\}$, orientation $\omega \in \mathbb{Z}/4$). Let $O = D_4 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle$ act by

$$r \cdot (\chi, \omega) = (\chi, \omega + 1), \quad s \cdot (\chi, \omega) = (-\chi, -\omega).$$

Then $|L| = 8$, but the action has nontrivial stabilizers (reflections identify (χ, ω) with $(-\chi, -\omega)$), yielding $|L/O| = 4$ orbits. Hence

$$\text{Cap} = \log_2 4 = 2 \text{ bits} \quad \text{while} \quad \log_2 |O| = 3,$$

illustrating the reduction caused by non-commuting symmetries (orbit collisions).

This leads to a key structural pattern. Let O act on the finite label set L :

- **Abelian Case.** If $O \cong (C_2)^n$ acts by independent flips on L with a free action (e.g. the regular action on $L = O$), then $|L/O| = |O| = 2^n$ and the capacity is n bits.
- **Non-abelian Case.** When O is non-abelian, typical elements introduce larger stabilizers and merge labels across conjugate directions; consequently, $|L/O| < |L|$ and often $\text{Cap} < \log_2 |O|$ (as in the D_4 example).

Table 4: Relationship between geobits and their groups

Group	Order	Dimension	# Bits	Comments
C_2	2	2	1	Classical bit
$(C_2)^2 = V_4$	4	4	2	Base level 2-geobit
$(C_2)^4$	16	4	4	Canonical abelian 4-geobit
D_4	8	3	≤ 3	Non-abelian, simple global swap
$D_4 \ltimes (C_2)^4$	128	≥ 5	< 8	Non-abelian, global rotations

Why the focus on abelian vs non-abelian groups? In an abelian group, every state can be constructed as the direct sum of independent generators. That is, every geobit state is uniquely addressable by an n -tuple over $\mathbb{C}_2^n$. There are no collisions. The total number of distinguishable geobit states is exactly 2^n , and the encoding is non-redundant.

In contrast, a non-abelian group acts less cleanly. Its action “mixes” states: some group elements act non-trivially, creating identifications among configurations. Orbits form under conjugation or subgroup stabilizers, and the full state space becomes quotient-like. Redundancy becomes intrinsic—some geobit states are indistinguishable under the group action. As a result, the number of distinct encodable states in the orbit space may be strictly less than the order of the group itself.

This distinction becomes crucial when interpreting a GCM as a computational substrate: *abelian structure maximizes symbolic resolution, while non-abelian structure imposes constraints that must be managed or exploited.*

Remark 2.1. The embedding dimension for a *THM* symmetry group is the minimal integer n such that all distinct geometric configurations corresponding to the elements of the group can be realized as non-self-intersecting embedded submanifolds in Euclidean space. It is essential to note that global reflections or entire rotational symmetry groups do not contribute to the intrinsic geobit capacity in $\mathbb{R}^4$. These operations require action on the global geometry and do not alter the intrinsic state of the unit helices. Thus, group orders beyond 16 that arise from external symmetry do not increase state space in a computable sense. Table 5 compares geobits with their close cousin [NC10].

Table 5: Qubit vs. Geobit

Property		Quantum qubit	Geometric geobit
Pure state space		$\mathbb{C}P^1 \cong S^2$ (Bloch sphere)	Finite labels $\times$ continuous geometric parameters (e.g. chirality/orientation, phase)
Dynamics		Unitaries (e.g. $SU(2)$ rotations)	Geometric transformations (flips, rotations, deformations)
Measurement/Readout		Born rule (probabilistic)	Geometric probe/partition (deterministic given microstate)
Accessible info	classical	≤ 1 bit per use	Depends on chosen encoding and symmetries
Elementary operations	opera-	Single-/two-qubit gates; entanglement via tensor product	Local geometric updates; composition by connecting/gluing components
Key feature		Coherence and entanglement central	Robustness tied to analog stability

Definition 2.4 (Geometric Kets). We refer to an individual geobit ket array as “ n -geobit,” directly paralleling the usage of qubit arrays with ket-style notation. Following the same notation, we employ a modified bra-ket notation.

Let the microstate of a twin helix cell be

$$\alpha = (C_A, O_A, C_B, O_B) \in C_2^4.$$

Where C is chirality and O is orientation. We write the *state ket* $|\alpha\rangle := |C_A O_A C_B O_B\rangle$. Intrinsic flips act by the Klein-four group

$$V_4 = \{e, s_A, s_B, s_A s_B\},$$

where s_A flips (C_A, O_A) and s_B flips (C_B, O_B) coordinatewise. A 1-bit readout is any fixed function $r : C_2^4 \rightarrow C_2$ (e.g. parity $r = o_A \oplus o_B$ or equality $r = [o_A = o_B]$).

Example 2.2 (Update Step with V_4). Take parity readout $r = o_A \oplus o_B$ and the microstate

$$|\alpha\rangle = |1101\rangle.$$

Then $b = r(x) = 1 \oplus 1 = 0$. Suppose the next bit should be $b' = 1$. Apply any toggler in V_4 , e.g. s_A :

$$s_A \cdot |\alpha\rangle = |1001\rangle, \quad r(s_A \cdot \alpha) = 0 \oplus 1 = 1 = b'.$$

(If instead $b' = b$, use a parity-preserver e or $s_A s_B$.) Optionally, a coarse projection $\pi : C_2^4 \rightarrow C_2^2$ such as

$$\pi(C_A, O_A, C_B, O_B) = (C_A \oplus C_B, O_A \oplus O_B)$$

can be displayed as $|\pi(\alpha)\rangle$; note that s_A or s_B need not descend to operators on the quotient, so actions are applied on C_2^4 before projecting. The resulting ket would then become $|11\rangle$.

2.3 The GCM Lattice

Definition 2.5 (GCM Lattice). A *GCM Lattice* or *GL* is a finite or locally finite assembly of GCMs arranged over a graph. Formally, it is the data:

$$GL = (\mathcal{L}, G = (I, E), \{\mathcal{G}_i\}_{i \in I}, \{\iota_i\}, \mathcal{I}^{\text{lat}}, \text{Ev}, \mu)$$

with:

- (a) **Ambient region** $\mathcal{L}$. Smooth manifold (often a bounded working region) into which cells will be embedded.
- (b) **Lattice graph** $G = (I, E)$. A finite or locally-finite graph (e.g. a box of $\mathbb{Z}^d$) with uniformly bounded degree; vertices index the cells, edges encode adjacency.
- (c) **Cells** $\{\mathcal{G}_i\}$. For each $i \in I$, a GCM $\mathcal{G}_i = (\mathcal{L}_i, \{R_\alpha^i\}, \pi_i : E_i \rightarrow \mathcal{L}_i, \mathcal{I}_i, \text{Ev}_i)$ as in definition 2.1.
- (d) **Placements** $\{\iota_i\}$. Smooth embeddings $\iota_i : \mathcal{L}_i \hookrightarrow \mathcal{L}$ with pairwise disjoint interiors. The support of cell i is $\text{supp}(i) := \iota_i(\mathcal{L}_i)$.
- (e) **Lattice interfaces** $\mathcal{I}^{\text{lat}}$. For each edge $(i, j) \in E$, a family of interface maps attached to portions of $\partial \text{supp}(i) \cap \partial \text{supp}(j)$,

$$I_{ij} : (\sigma|_{\text{supp}(i)}, \sigma'|_{\text{supp}(j)}, \text{local flow data}) \mapsto (\sigma'|_{\text{supp}(i)}, \sigma'|_{\text{supp}(j)}),$$

which are local (depend only on a finite stencil around i, j) and equivariant with respect to the chosen cell symmetries.

- (f) **Evolution** Ev . Either (a) a discrete, synchronous update $U : \Gamma(E) \rightarrow \Gamma(E)$ on the global state bundle $E := \bigsqcup_{i \in I} \iota_i E_i$, or (b) a continuous flow of sections $(\sigma_t)_{t \in \mathbb{R}}$ solving $\dot{\sigma}_t = F(\sigma_t)$ and compatible with all interfaces. Evolution is local and non-singular on admissible initial states.
- (g) **Symbolic observation** μ . A readout map $\mu : \Gamma(E) \rightarrow \Sigma^N$ that, at each cell i , selects a coding $q_i : E_i \rightarrow \Sigma^{n_i}$ of its geobits; the total capacity is $N = \sum_{i \in I} n_i$.

We say a *GL* is *computationally coherent* if the symbolic dynamics induced by the geometric evolution commutes with readout, i.e., there exists $\Phi_s : \Sigma^N \rightarrow \Sigma^N$ with:

$$\mu(U(\sigma)) = \Phi_s(\mu(\sigma)) \quad (\text{discrete time}), \quad \text{or} \quad \frac{d}{dt} \mu(\sigma_t) = F_s(\mu(\sigma_t)) \quad (\text{continuous time}).$$

Remark 2.2. The architecture of geometric computability is decidedly a non-Von Neumann one. A GL is necessary to compute programs that are not trivial, since one can't discretely read out results on a single GCM with symbolic or quasi-fluidic computation, not without a projection layer or a symbolic "slice." In a GCM there is no separation of program and data, no instruction pointer, or fetch-decode-execute cycle, the geometry *is* the computation. Computation is intrinsic, continuous, and spatially distributed. While one can clearly see the similarity to cellular automata in the discrete case, parallels can be drawn to neural fields or reaction-diffusion systems if an outside example is required.

2.4 Achieving Turing Completeness in R4

Finite state machines (FSMs) are the building blocks of classical computation [Sip13]. A FSM consists of a quintuple $(S, \Sigma, \delta, s_0, F)$: a finite set of states S , a finite input alphabet Σ , a transition function $\delta : S \times \Sigma \rightarrow S$, an initial state $s_0 \in S$, and a designated accepting set $F \subseteq S$ (or an output map). Given an input word over Σ , the machine updates its state by iterating δ ; acceptance (or output) is determined by the terminal state.

Proposition 2.1 (FSM Embedding in Symbolic Mode). *In the symbolic regime, any finite-state machine $(S, \Sigma, \delta, s_0, F)$ embeds into a GCM tile with finite internal labels and symbolic interfaces.*

Table 6: Classical vs. Geometric Computation

Feature	Classical Computation	Geometric Computation
Basic substrate	Strings, tapes, registers, graphs of discrete symbols	Embedded geometric cells/manifolds arranged in a lattice or interface network
Primitive state	Symbol/bit / finite register value	Local geometric state (e.g. chirality, orientation/phase, and optional topological counters)
State space	Finite alphabet or discrete configuration space	Structured local fiber, typically discrete or hybrid: symbolic labels and, in richer regimes, continuous phase data and quantized invariants
Elementary operations	Rewrite rules, gate updates, transition functions	Admissible geometric transformations, local interface updates, and symmetry-respecting deformations
Dynamics	Discrete step semantics (FSM/TM/CA update)	Local evolution on geometric states: symbolic updates, driven pulses, or continuous flows with induced return maps
Composition	Product of machines, circuit wiring, cellular adjacency	Gluing/coupling of cells through interfaces, adjacency graphs, and local geometric compatibility conditions
Input	External symbol stream or initial tape/register contents	Boundary conditions, perturbations, forcing pulses, or initialized geometric configurations
Output	Tape contents, accepting state, register readout	Readout of geometric observables via a chosen projection/partition of the state space
Symmetry	Usually implicit or combinatorial	Explicit: finite groups in the symbolic regime, abelian phase symmetries and richer geometric symmetries beyond
Constraint structure	Logical consistency and finite update rules	Smoothness constraints, and admissibility of local geometric moves
Computational viewpoint	Computation is manipulation of discrete symbols	Computation is evolution of structured geometry, with symbolic data recovered as a readout of the substrate

Sketch. Let a cell carry a finite internal label set $L \cong S$ (e.g. $L = C_2^k$ with a fixed encoding of S). Inputs are a finite set of discrete interface signals Σ (boundary toggles). Specify a local update table $\Delta : L \times \Sigma \rightarrow L$ implementing δ . Fix an initial label $\ell_0 \in L$ encoding s_0 . Choose a readout predicate $R : L \rightarrow \{0, 1\}$ and define the accepting set $F = \{\ell \in L : R(\ell) = 1\}$.

A single tile thus realizes the FSM; a single-dimensional lattice with nearest-neighbor symbolic communication realizes synchronous products. $\square$

Proposition 2.2. *A one-dimensional GL whose cells are THM_4 tiles with symbolic nearest-neighbor communication can simulate Rule 110.*

Proof. Each cell contains a twin helix with two binary orientation labels

$$(o_A, o_B) \in C_2 \times C_2,$$

and intrinsic orientation flips act by the Klein four group

$$V_4 \cong C_2 \times C_2 = \{e, s_A, s_B, s_A s_B\},$$

where s_A flips o_A , s_B flips o_B , and $s_A s_B$ flips both.

Readout/encoding. Fix the bit readout as the parity (XOR)

$$r(o_A, o_B) := o_A \oplus o_B \in \{0, 1\}.$$

Any fixed local encoding is acceptable; parity is chosen because s_A or s_B toggles r , while $s_A s_B$ preserves r .

Cellular update. Let $b_i^t := r(o_{A,i}^t, o_{B,i}^t)$ be the bit at site i and time t . Compute the Rule 110 local rule

$$b_i^{t+1} = \delta(b_{i-1}^t, b_i^t, b_{i+1}^t), \quad \delta = 110 \rightarrow 1, 101 \rightarrow 1, 011 \rightarrow 1, 010 \rightarrow 1, 001 \rightarrow 1, \text{ else } 0.$$

Now choose a group element $g_i^t \in V_4$ to enforce the new readout:

$$g_i^t = \begin{cases} e \text{ or } s_A s_B, & \text{if } b_i^{t+1} = b_i^t \quad (\text{preserve parity}); \\ s_A \text{ or } s_B, & \text{if } b_i^{t+1} \neq b_i^t \quad (\text{toggle parity}). \end{cases}$$

Apply the intrinsic update

$$(o_{A,i}^{t+1}, o_{B,i}^{t+1}) = g_i^t \cdot (o_{A,i}^t, o_{B,i}^t).$$

This uses only nearest-neighbor bits to compute b_i^{t+1} and a local V_4 action to realize it; hence, the induced dynamics on the readout $\{b_i^t\}_{i \in \mathbb{Z}}$ is exactly Rule 110.

By Cook's theorem [Coo04], such a lattice simulates a cyclic tag system and therefore a Turing machine. Consequently, the GL with THM_4 cells and the above encoding is Turing complete. $\square$

2.5 The Shape of Computation

A classical bit is typically modeled as a two-point discrete state space, while a qubit has a canonical continuous pure-state space, namely the Bloch sphere. A geobit, by contrast, does not come equipped with a unique ambient state manifold of this kind. Its state space is determined only up to a chosen *encoding*: one may represent the same symbolic transition structure by different Cayley graphs, polytopes, or orthogonal symmetry realizations.

In the symbolic regime, the relevant object is therefore not a canonical “geobit sphere,” but a finite transition system equipped with a symmetry action. For THM_4 , this structure is governed by the Klein four group and may be visualized by a 3-dimensional Cayley graph. For the full symbolic model, one may ask for the smallest orthogonal representation in which the entire symmetry group acts faithfully.

Proposition 2.3 (Minimal Orthogonal Embedding Dimension for $(C_2)^4$). *A faithful embedding of the symbolic symmetry group $(C_2)^4$ into an orthogonal rotation group requires at least $SO(5)$.*

Proof. Let $E \subset SO(n)$ be an elementary abelian 2-subgroup. Since every element of E has order 2, each element is an orthogonal involution and hence diagonalizable over $\mathbb{R}$ with eigenvalues in $\{\pm 1\}$. Because the elements of E commute, they are simultaneously diagonalizable. Therefore, after conjugation, E is contained in the subgroup of diagonal sign matrices

$$\left\{ \text{diag}(\varepsilon_1, \dots, \varepsilon_n) : \varepsilon_i \in \{\pm 1\}, \varepsilon_1 \cdots \varepsilon_n = 1 \right\} \subset SO(n),$$

which is isomorphic to $(C_2)^{n-1}$.

It follows that every elementary abelian 2-subgroup of $SO(n)$ has rank at most $n-1$. In particular, $(C_2)^4$ cannot embed in $SO(4)$, since the largest such subgroup there has rank 3. On the other hand, in $SO(5)$ the determinant-1 diagonal sign subgroup is exactly

$$\left\{ \text{diag}(\varepsilon_1, \dots, \varepsilon_5) : \varepsilon_i \in \{\pm 1\}, \varepsilon_1 \cdots \varepsilon_5 = 1 \right\} \cong (C_2)^4,$$

so the bound is sharp. $\square$

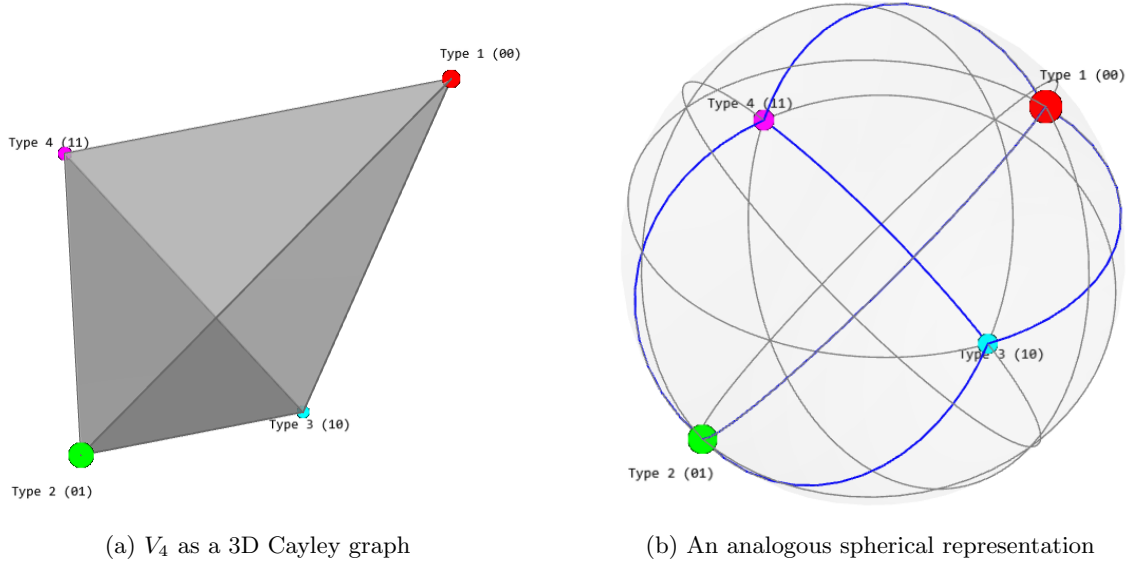


Figure 2: Encoding-dependent realizations of the V_4 symbolic state space.

Thus, although the underlying twin-helix geometry is naturally constructed in low dimension, a faithful orthogonal realization of its full symbolic symmetry requires a higher-dimensional ambient representation. This should not be interpreted as an intrinsic physical dimension of the manifold itself, but rather as a statement about the geometry of *state-space realization*. In particular, the “shape” of a geobit is not canonical: it depends on which observables, generators, and equivalence relations one chooses to encode. For visualization, we therefore restrict attention to the smallest symbolic cases, such as V_4 , whose Cayley graph admits a simple 3-dimensional rendering.

Remark 2.3. The tetrahedral appearance of the V_4 state space is notable in light of Baez’s discussion of the quantum tetrahedron [BB99] where similar state collapses were noted. In the present setting, however, the tetrahedral structure arises from a purely combinatorial symbolic transition system rather than from spin-network quantization. We return to issues of state space comparison in section 4.3.

2.6 The Logic of Geometry

In a GCM, logic may be encoded geometrically rather than symbolically. The most basic mechanism is *orientation*: once a preferred axis has been fixed, the phase of a unit helix around that axis determines a logical value. This enables a passage from binary logic to finite n -ary logic [Pos21, Zad65].

Orientation as a geometric alphabet. Fix a helical axis and let $\theta \in S^1 \cong \mathbb{R}/2\pi\mathbb{Z}$ denote the corresponding orientation phase. For each integer $n \geq 2$, partition the circle into n equal sectors and define the discrete readout

$$b(\theta) := \left\lfloor \frac{n\theta_{\text{norm}}}{2\pi} \right\rfloor \in \{0, 1, \dots, n-1\}, \quad \theta_{\text{norm}} \in [0, 2\pi).$$

This induces an n -valued logic on the orientation channel, which we call ω -*logic*. Equivalently, one may regard the underlying continuous phase as

$$s(\theta) := \frac{\theta_{\text{norm}}}{2\pi} \in [0, 1),$$

with the discrete alphabet obtained by thresholding s into n bins. In this way, binary, ternary, quaternary, and higher n -ary logics all arise from the same geometric mechanism: a cyclic phase together with a chosen readout.

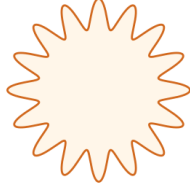
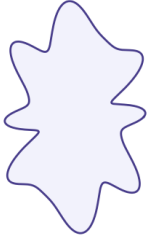
Definition 2.6 (ω -logic). Let $n \geq 2$. The n -ary orientation logic of a unit helix is the map

$$\omega_n : S^1 \longrightarrow \{0, 1, \dots, n-1\}, \quad \omega_n(\theta) = \left\lfloor \frac{n\theta_{\text{norm}}}{2\pi} \right\rfloor.$$

When $n = 2$ this recovers binary logic; when $n > 2$ it gives finite n -ary logic; and if one retains θ itself rather than thresholding, one obtains the continuous $U(1)$ phase corridor. Table 7 organizes a table of models.

Table 7: The chart of n -logics

Geometric Expressiveness	Join Symmetry	Algebraic Closure Group	Result
Extreme	—	$\text{Diff}(M)$ (Maximal)	Computational gauge field
High	1-fold (360°)	C_1	Geometric chaotic analog flow
Medium	2-fold (180°)	C_2	Classical Boolean logic
Low	n -fold ($2\pi/n$), finite	C_n	Finite n -ary logic
Minimal	Full rotational symmetry	$U(1)$ (Lie)	Algebraic fuzzy analog flow



(a) Expressiveness in binary vs. hexadecimal

(b) n -logical gluing maps for 3, 4, and 5-ary logic

Figure 3: Cross-sectional shape families and the n -logic tradeoff.

A spectrum of physical logic. From the computational point of view, low-order cyclic logics such as C_2 or C_4 occupy a particularly stable region: they retain enough discrete algebraic structure to support verification and symbolic composition, while still allowing genuinely geometric realizations. At one extreme, the continuous $U(1)$ limit becomes an abelian phase medium; at the other, the loss of usable discrete closure drives the system toward geometric or chaotic analog behavior. In this sense, Boolean and low-order finite logic form a kind of *logical habitable zone* for physically stable symbolic computation.

Hence, even in an idealized, noiseless world, n -ary logical computers are “sandwiched” between two analog extremes. For n -ary logic, the cross-section of the *THM* can be any smooth closed manifold with at least n distinguished points on its boundary that realize the requisite “logical angularities.” The primary family consists of sinusoidal perturbations (“clovers”) that can be globally smooth and are easily visualized in polar coordinates—see fig. 3b. However, one can readily imagine richer forms that realize the required angular structure. Figure 3a illustrates such alternatives and compares expressiveness across different logical regimes.

A geometric tradeoff. The passage from C_1 to larger cyclic groups and finally to $U(1)$ exhibits a *systematic tradeoff*. Increasing rotational symmetry enlarges the continuity of the orientation channel, but at the same time restricts the available shape space and weakens finite algebraic closure. Conversely, low-order discrete symmetry yields fewer phase values but stronger symbolic verifiability. This motivates the following lemma.

Lemma 2.4 (Geometric Logic Conservation Lemma). *Fix a THM G together with a chosen readout. Let $\mathcal{C}(G)$ denote the complexity of the induced algebra of local symbolic operations (for example, the size of the effective gate clone or finite symmetry algebra visible under the readout), and let $\mathcal{E}(G)$*

denote the corresponding geometric expressiveness (for example, the number of orbit-distinguishable cross-sectional shapes or phase patterns available at fixed regularity). Then one expects a bounded tradeoff of the schematic form

$$\mathcal{C}(G) + \mathcal{E}(G) \lesssim \Lambda(G),$$

where $\Lambda(G)$ depends on the ambient geometric constraints and symmetry class of the model.

Remark 2.4. The essential point is that the symmetry of the system's configuration space directly induces an invariant quantity in the system's dynamics. Every symmetry break in topology introduces a logical ambiguity, and vice versa. Although we are only in the initial regime of computation, we will eventually encounter higher-order manifolds that devolve into pure fluidic computability, leaving discrete logic altogether and entering increasingly abstract and chaotic forms of analog computation.

Definition 2.7 (Geometric Expressiveness at Finite Cutoff). Let $\mathcal{F}_K$ denote the space of real trigonometric polynomials of degree at most K ,

$$r(\theta) = a_0 + \sum_{k=1}^K (a_k \cos(k\theta) + b_k \sin(k\theta)),$$

interpreted as smooth 2π -periodic radial functions. Let $\mathcal{F}_{n,K} \subseteq \mathcal{F}_K$ be the subspace of functions invariant under rotation by $2\pi/n$:

$$r(\theta + 2\pi/n) = r(\theta) \quad \forall \theta.$$

The *relative geometric expressiveness* at symmetry level n and cutoff K is

$$E_K(n) := \frac{\dim(\mathcal{F}_{n,K})}{\dim(\mathcal{F}_K)}.$$

Proposition 2.5 (Symmetry Collapses Cross-sectional Expressiveness). *For a fixed cutoff K ,*

$$\dim(\mathcal{F}_{n,K}) = 2 \left\lfloor \frac{K}{n} \right\rfloor + 1, \quad E_K(n) = \frac{2 \lfloor K/n \rfloor + 1}{2K + 1}.$$

In particular, for fixed K , $E_K(n)$ decreases monotonically with n , and

$$E_K(n) \rightarrow \frac{1}{2K + 1} \quad \text{as } n \rightarrow \infty.$$

At the level of smooth functions (without cutoff), full rotational symmetry forces the radial function to be constant, hence the cross-section is a circle.

Proof. If

$$r(\theta) = a_0 + \sum_{k=1}^K (a_k \cos(k\theta) + b_k \sin(k\theta))$$

satisfies $r(\theta + 2\pi/n) = r(\theta)$ for all θ , then only those Fourier modes with frequency divisible by n may remain. Hence the allowed modes are $k = 0, n, 2n, \dots, \lfloor K/n \rfloor n$, giving

$$\dim(\mathcal{F}_{n,K}) = 1 + 2 \left\lfloor \frac{K}{n} \right\rfloor.$$

Since $\dim(\mathcal{F}_K) = 2K + 1$, the formula for $E_K(n)$ follows immediately. In the full smooth category, invariance under all rotations implies that only the constant Fourier mode survives, so the radial profile is constant, and the cross-section is a circle. $\square$

Progression to algebraic analog flow. This is the essential phenomenon as the size of the rotational symmetry group grows. Even as the symmetry group becomes continuous, the system's logic remains identical to its geometry. This is the core idea behind geometric computation, and why such systems are not limited to discrete operations.

Definition 2.8 (χ -logic). Let $\mathcal{C}$ be a class of framed geometric configurations equipped with a continuous *chiral observable*

$$\kappa : \mathcal{C} \longrightarrow \mathbb{R},$$

where κ may be realized, for example, by signed torsion, signed twist density, or another orientation-sensitive local invariant. The associated *chirality logic* is the map

$$\chi : \mathcal{C} \longrightarrow \{-1, 0, +1\}, \quad \chi(x) := \text{sgn}(\kappa(x)).$$

Thus:

- $\chi(x) = +1$ denotes a right-handed configuration,
- $\chi(x) = -1$ denotes a left-handed configuration,
- $\chi(x) = 0$ denotes the non-chiral or degenerate locus where the chiral invariant vanishes.

When the value 0 is admitted as a stable readable state, χ -logic is ternary. When the degenerate locus is excluded from the admissible state space, χ -logic reduces to a binary left/right channel.

The role of chirality. Unlike orientation, chirality is not naturally a cyclic n -ary phase coordinate. Geometrically, changing handedness is not a small rotation in a compact group; it amounts to changing the sign of a chiral invariant, such as torsion for a framed centerline or the sign of a local twist density. Consequently, any continuous interpolation between left- and right-handed configurations must pass through the non-chiral locus on which the relevant invariant vanishes.

At the level of coarse readout, this produces a natural three-way distinction: left-handed, right-handed, and degenerate/non-chiral. In the present paper, however, the twin helix register itself is taken to operate only on the nondegenerate binary sector

$$\{-1, +1\} \subset \{-1, 0, +1\},$$

so that the value 0 functions only as a boundary or transition state rather than as a stable logic value. This does not preclude other classes of GCMs from using the full ternary chirality space as a genuine computational channel; indeed, later gauge-theoretic examples (see appendix A.1) may best be understood in precisely that broader sense. This is the sense in which chirality forms a qualitatively different channel from orientation: ω -logic supports a rich cyclic phase structure, whereas χ -logic is organized by sign and degeneracy.

2.7 Syntax, Semantics, and Computation

We now synthesize three convergent lines of thought, alongside our geometric model.

First, following Felleisen’s account of expressiveness [Fel91], we distinguish *micro-* from *macro-expressiveness*: adding new local states or notations versus adding genuinely new global, compositional power.

Second, in the compositional semantics of processes of Categorical Quantum Mechanics: Abramsky–Coecke, with graphical calculi surveyed by Selinger, the meaning of a composite is determined by the wiring of its parts; contextuality and entanglement arise as *compositional* phenomena rather than add-ons [AC04, Sel11]. This gives us a precise, syntax-agnostic way to talk about how meanings compose.

Third, our geometric/groupoid models instantiate these ideas concretely: GCMs and decorated paths make composition, context, and macrostate emergence *geometric*. Macrostates are equivalence classes of geometric/computational microstates; macro-expression corresponds to traversing or constructing new geometric cobordisms.

Finally, in the spirit of Baez–Stay’s “Rosetta Stone” perspective on physics, topology, logic, and computation [BS10], we regard computation itself as a process theory: objects are interfaces, morphisms are computations, and composition is gluing. In this language, the familiar tradeoffs become geometric/categorical phenomena rather than informal slogans. We do not merely interpret computation using geometry; we *construct the geometry of computation itself*. The following definitions guide our landscape.

Definition 2.9 (Micro-Expression). A language extension is *micro-expressive* w.r.t. a base language L if every program P in the extended language L' can be mechanically translated (compiled/macro-expanded) to a program P_0 in L without changing its semantic structure. Formally: for all $P \in L'$, there exists a derivable $P_0 \in L$ with equivalent semantics; the extension is syntactic sugar, not added power.

Definition 2.10 (Macro-Expression). A language extension is *macro-expressive* if it enables programs or control/data abstractions that cannot be obtained from the base language by any mechanical or structure-preserving translation. Formally: there exists $P \in L'$ such that no equivalent $P_0 \in L$ can be constructed by a uniform, semantics-preserving translation.

Felleisen’s framework makes this distinction precise and operational [Fel91], and we adopt it as the semantic backbone for what follows. They define precise criteria for when a language extension is merely syntactic (micro-expressive, or “eliminable by mechanical translation”) versus when it adds genuine new expressive power (macro-expressive, or “semantically irreducible”). Just as in programming, where some features create behaviors not reducible to simpler forms—gluing submanifolds can create new topological invariants only if the gluing is smooth (singularity-free).² As systems grow more complex, obstructions to smooth composition proliferate, paralleling the limits of macro-expression in computation.

Definition 2.11 (Verifiability). Let $\mathcal{S}$ be a computational/semantic system with predicates $\mathcal{P}$ over its states/behaviors. The *verifiability* $\mathcal{V}(\mathcal{S})$ is a normalized measure of how many *nontrivial* properties in $\mathcal{P}$ admit effective, pre-execution decision procedures. Equivalently: $\mathcal{V}(\mathcal{S})$ measures the size/structure of the subset of $\mathcal{P}$ consisting of properties decidable by algorithmic, structural, or analytic means (e.g., checkable invariants or geometric signatures), without brute-force evaluation.

Verifiability captures the capacity to certify properties *by construction*. In algebraic regimes, it correlates with rigid “ledgers” (e.g., line bundles/cohomology classes that distinguish states). Still, it is not identical to micro-expression: the former is semantic/algorithmic, whereas the latter is syntactic/generative. High verifiability corresponds to systems where many global or local properties can be decided by construction; low verifiability signals the presence of undecidable, chaotic, or non-algorithmic behavior.

In the process semantics of Abramsky–Coecke, contextual/entangled behavior is a *compositional* effect of how processes wire together. Our geometric program inverts the usual narrative: start from deterministic geometric computation, then see probabilistic/quantum-like structure *emerge* as the statistical/compositional shadow of complex gluing and flow.

Definition 2.12 (Tractability). Let $\mathcal{S}$ be a formal system (e.g., a programming language, a computational model, or a geometric computation system). Tractability is the property that there exists a procedure (algorithmic or semantic) to:

- For machines: Execute or evaluate a statement/program/configuration in $\mathcal{S}$ with feasible computational resources (e.g., polynomial time or some agreed practical bound).
- For humans: Understand, reason about, or verify the intent and behavior of statements in $\mathcal{S}$ without overwhelming cognitive complexity.

More precisely: A property or decision problem in $\mathcal{S}$ is tractable if it can be decided or realized with computational resources (time, space, etc.) bounded by a polynomial (or similarly ‘practical’) function of the input size. A system is tractable if most of its intended computations and the process of programming/understanding them fall within humanly or machine-manageable complexity.

Tractability, in our framework, refers to the ease with which both a machine can execute and a human can understand or reason about a computational statement in a formal system. For a programming language or system of symbols, tractability is both a computational and a cognitive phenomenon. The degree of tractability depends not only on algorithmic complexity but also on the alignment between the system’s formal structure and the way meaning is constructed and communicated among its users.

With these parts in place, we can now *macro-express* a new theorem.

²The geometric mismatches noted earlier explain why, in the language of Feynman diagrams, certain gluings can fail: the logic of the bordisms must agree.

Conjecture 2.6 (CORE THESIS: Expressiveness/Verifiability/Tractability (“EVT” Hypothesis)). *The conjecture formally comprises two parts:*

- (A) **Unverifiability of Macro-expressive Systems.** *Any sufficiently macro-expressive system of physical computation generically loses global algorithmic verifiability unless additional structural restrictions are imposed.*
- (B) **Expressiveness/Verifiability/Tractability Trilemma.** *Within such a system, every deterministic computational model is fundamentally constrained by a three-way tradeoff among expressiveness (geometric or semantic generality), verifiability (algebraic checkability or formal runtime assurance), and tractability (algorithmic or computational efficiency). These attributes satisfy the following trilemma:*

No system can simultaneously maximize all three properties. For any given model, at most two of the three can be optimized beyond a critical threshold, while the third must necessarily be sacrificed.

Formally, the achievable combinations of these properties are bounded by a 2-dimensional simplex in the space of attributes. The precise tradeoff is determined by the system’s local symmetries and structural constraints.

This is the central motivating result of this work. We take the phenomenon isolated in lemma 2.4 and generalize it from our geometric setting to physical deterministic computation more broadly. The claim is not that *all* computation is uniformly irreducible, nor that undecidability appears by mere complexity or by the use of a particular alphabet. Rather, we claim that there is a *structural threshold* in the design space of deterministic computational substrates.

The first half of the EVT Hypothesis should be read as a selection principle on substrate classes. It identifies which structural features allow local verification to compose into global verification, and which features obstruct that passage. The central issue is not self-reference by itself, nor the size of the local alphabet, but the failure of local checks to commute once a substrate becomes sufficiently macro-expressive. Later, in definition 4.2, we give a more precise geometric formulation of *macro-expressiveness*; for now, the term should be understood in the sense suggested by Felleisen: a substrate is macro-expressive when it supports genuinely new global compositional behavior rather than merely new local notation.

In abelian regimes, verification is comparatively stable because local operations commute and can often be checked independently before being assembled. In non-abelian regimes, this ceases to hold. Joins become order-sensitive, transport becomes path-dependent, and gauge-invariant observables are no longer values of individual local states but class functions of holonomy or other global data. Thus, the obstruction to verification is not merely that there are many states, but that the meaning of a state depends on the history and context by which it is composed.

To state the thesis informally:

Whenever a deterministic computational substrate acquires nontrivial global compositional structure—especially non-abelian transport, path-dependent joins, or context-sensitive readouts—local verification no longer automatically glues to global verification. Recovering verifiability then requires additional algebraic, type-theoretic, topological, or resource-sensitive constraints.

Rice’s theorem [Ric53] is the sharp symbolic mirror of this phenomenon. In ordinary computability theory, Rice tells us that every non-trivial extensional semantic property of partial computable functions is undecidable. In the present framework, this is not treated as the root mechanism of the EVT, but as its classical discrete manifestation: once symbolic computation admits unrestricted composition, extensional semantics, and self-interpretation, the failure of global verification appears as diagonalization. In geometric computation, the same pressure appears first as noncommuting transport, holonomy dependence, moduli obstruction, and the failure of local certificates to compose.

Why macro-expression? Felleisen’s distinction between *micro-* and *macro-expressiveness* is essential for our purposes because the present framework is *not* restricted from the outset to digital syntax, natural number objects, or Turing-style encodings. If one begins directly with Kleene-style recursion-theoretic machinery, one has already assumed a symbolic regime rich enough to support self-interpretation. Our problem is prior to that: we need a language for describing when a *physical* or *geometric* substrate acquires genuinely new global computational power, independently of whether that power is presented discretely, continuously, or in mixed form.

Felleisen provides exactly this level of abstraction. His notion of *macro-expressiveness* isolates those extensions that do not merely add more local states or notation, but enlarge the global compositional possibilities of the system itself. This is the right conceptual tool for our purposes, because the loss of verifiability in our setting is not driven by the size of the local alphabet, nor by the mere richness of local geometry, but by the appearance of new *global* forms of composition. In our framework, this occurs precisely at the transition into non-abelian regimes, where order-sensitive transport, path dependence, and conjugation-sensitive observables become available.

Microstates, macrostates, micro-expression, and macro-expression. It is crucial to distinguish the state-based and expressive notions. A *microstate* is a local configuration of the substrate: a specific geometric, algebraic, or dynamical arrangement. A *macrostate*, by contrast, is an equivalence class of such configurations under the observational or symmetry relations relevant to the model. Thus, the micro/macrostate distinction is *ontic* or *configurational*: it concerns what states the system can occupy and how those states are grouped.

The distinction between *micro-expression* and *macro-expression* is different. It is *operational* and *compositional*. Micro-expression concerns the creation of new local states, labels, or local update rules without changing the global expressive capacity of the substrate. Macro-expression concerns the introduction of genuinely new global compositional freedoms: new ways of combining processes, transporting information, or building observables that cannot be reduced to the previous regime by a uniform local translation. In this sense, microstates/macrostates describe *what exists*, whereas micro-/macro-expression describes *what can be done*.

This distinction is exactly what the first half of the EVT must track. The relevant threshold is not the proliferation of microstates; a system may admit a very large or even continuous local state space while remaining compositionally tame. What matters is the onset of macro-expression: the point at which the substrate acquires new global compositional freedoms such as order-sensitive control, path-dependent transport, conjugation-sensitive readout, or, in the symbolic limit, self-reference.

Accordingly, the role of Felleisen in this paper is not to supply a recursion-theoretic proof of Rice’s theorem. Rather, it is to identify the correct *threshold variable*. Rice-style undecidability does not appear simply because a substrate is rich, continuous, or high-dimensional. It appears in the symbolic case when a macro-expression reaches the point of unrestricted semantic self-application. In our geometric setting, the analogous jump occurs when macro-expression coincides with the onset of non-abelian compositional structure, and it is there that verifiability begins to fail in the precise sense articulated by the EVT.

Why alphabet size is not the issue. Recall the spectrum of n -logics generated from join symmetry in table 7. It is tempting to attribute the ubiquity of undecidability to binary logic alone: a two-symbol alphabet offers high compositional freedom while imposing very little local algebraic discipline. But the arity of the local alphabet is not the source of the Rice-style barrier; this is driven by *macro-expressiveness*, not by the number of local symbols.

Any finite n -ary digital substrate recovers the same Rice-style barriers as soon as it supports conditionals, memory, self-description, and unbounded composition. Ternary or quaternary hardware may enlarge the local state space, but they do not, by themselves, neutralize diagonalization. Put differently: richer alphabets change *micro-expressiveness*, while Rice is controlled by *macro-expressiveness*.

In principle, one could seek to purchase more verifiability in hardware by enforcing a richer local symbolic structure. In practice, device physics, noise margins, and error-correction heavily favor binary encodings. More importantly, even when local alphabets are enriched, universality and semantic opacity reappear as soon as the substrate admits unrestricted global composition. To mitigate Rice in a principled way, one must instead withdraw some macro-expressive feature: forbid general recursion, stratify effects, impose linear/affine discipline, or adopt total calculi. This is exactly how systems

inspired by Coquand’s Calculus of Constructions [CH88] recover semantic control: they do not eliminate local richness, but they constrain the global forms of self-application and composition that would otherwise make undecidability ambient.³

Mathematics and computation are not symmetric. There is a genuine parallel here with Gödel’s incompleteness theorem, though the two results live on different sides of the logic/computation divide. Gödel says that sufficiently strong recursively axiomatized theories cannot prove all true arithmetical statements; Rice says that sufficiently expressive symbolic computational substrates cannot decide all non-trivial semantic properties of their own behaviors. Both involve failures of global certification under sufficient expressive power. The point of the EVT is not to collapse these theorems into one another, but to place Rice in the broader context of substrate design: undecidability becomes generic not simply because “Turing machines are strange,” but because certain structural freedoms prevent local checks from assembling into a global certificate.

Rice’s theorem feels harsher than Gödel’s in practice. In mathematics, undecidability is often buffered by deliberate axiomatic discipline: many working theories are complete or decidable, and even incomplete ones frequently permit large islands of stable reasoning. In computation, by contrast, universal systems are often engineered to collapse program and data into a common semantic domain and to permit unbounded composition. That design choice places computation much closer to the edge, where global verification fails.

This is the real “impedance mismatch” between the two domains. It is not that computation is somehow metaphysically more dangerous than mathematics, but that mainstream computational models were deliberately engineered to maximize expressive freedom. Once one notices this, Rice no longer looks like an arbitrary theorem about machine languages; it looks like a warning label attached to a certain region of the design space.

Symbolic self-reference as a limiting case. Self-reference is not the general mechanism of the EVT; it is the form the obstruction takes in symbolic universal computation. In classical recursion theory, the failure of global verification appears through evaluator-like mechanisms, self-application, and diagonalization. Kleene’s Recursion Theorem [Kle52] and the s - m - n theorem make this explicit, while Turing’s halting argument provides the canonical example. But in geometric computation, the corresponding obstruction need not appear first as an evaluator fed its own code. It may appear as noncommuting joins, path-ordered transport, conjugation-sensitive observables, or moduli data that cannot be certified by independent local checks.

Accordingly, the central question is not whether a system can literally quote its own syntax, but whether its local semantic data compose in a way that preserves global checkability. Abelian substrates tend to admit such a composition because the relevant operations commute. Non-abelian substrates generally do not: the result of a computation can depend on the order, path, and gluing history by which it is assembled.

Rice’s theorem is often stated more broadly than its hypotheses warrant. Its full force depends on extensionality together with a sufficiently rich self-referential structure. When those assumptions are withdrawn—for example, in total, strongly normalizing, staged, or heavily stratified systems—large classes of semantic questions become decidable, even though some equivalence problems may remain hard. This is not an exception to the EVT, but an instance of its calibration: verifiability is recovered by imposing a structure that prevents uncontrolled global composition.

Why gauge theory appears. This is also the point at which gauge theory becomes structurally unavoidable. As long as local observables commute, one can often approximate a global state by assigning compatible values to local pieces. Once the relevant observables cease to commute, this picture fails: local data depend on the frame, the order of comparison, and the path along which they are transported. A gauge field is precisely the mathematical device that records this dependence. It replaces naive global state assignment by local trivializations, transition functions, connections, curvature, and holonomy. In this sense, gauge theory is not an optional physical embellishment added after the fact; it is the natural language forced on us when local computational observables no longer glue as ordinary commuting data.

³One may heuristically regard an n -ary geometric computer as a kind of “hardware proof assistant”: the local state space may be rich, but semantic control comes only from global discipline, not from alphabet size alone.

For this reason, the first half of the EVT should be read as a calibration principle. Increasing macro-expressiveness enlarges the space of admissible compositions, but beyond the abelian or locally checkable corridor, those compositions need not preserve global verifiability. Recovering verifiability requires imposing structure: commutativity, totality, stratification, type discipline, gauge-invariant readouts, topological invariants, or resource bounds. In this sense, the claim is not that all interesting systems are irreducible. Rather, it is that certain architectural choices place a system in an irreducibility-like regime, while others—typing disciplines, totality restrictions, staged semantics, strong normalization, abelianization, or invariant-preserving quotienting—restore islands of semantic control by limiting the forms of composition available to the substrate.

The second half of the EVT Hypothesis sheds light on the odd “dual nature” observed in functional programming languages. On the one hand, languages with strong static typing offer stronger compile-time reasoning principles (e.g. type safety and, in suitable fragments, parametricity), at the cost of meta-linguistic restrictions during construction (Haskell). On the other hand, dynamically typed languages maximize meta-linguistic expressiveness at construction time (macros, reflection), while giving up static guarantees at runtime and shifting assurance to testing and contracts (Scheme).

Bridging attempts proceed along multiple axes: *gradual typing* [ST06] (Typed Racket) inserts types with higher-order contracts at typed/untyped boundaries, and *staging* [TS97] (MetaOCaml) separates compile-time code generation from runtime execution. From the proof-theoretic direction (Coquand), total dependent type theories recover strong guarantees by forbidding general recursion. From the pragmatic language research side, (Felleisen) shows how large untyped codebases can be migrated toward sound types via occurrence typing and higher-order contracts—at the cost of explicit blame and potential runtime overhead—an archetypal EVT tradeoff between meta-linguistic freedom and verifiability [THF06]. Here, the goal is not to re-formalize the trilemma, but to show how the second half of the EVT manifests concretely in real language design. When expressive features are pushed far enough, the same structural tension repeatedly reappears in practice through a small number of recognizable engineering symptoms:

The limits of expressive semantics. These bridges are inherently delicate. When many powerful features coexist, three recurring pressure points tend to appear:

- (a) **Static complexity.** Extremely rich type features (GADTs, type families, dependent types) that can render inference undecidable, forcing annotations; combining dependency with effects or staging threatens phase separation unless carefully constrained.
- (b) **Compile-time liveness.** Macros or multi-stage generation to diverge at compile time; expressive compile-time evaluators reintroduce the very fixed-point phenomena the EVT warns about.
- (c) **Runtime boundaries.** Gradual typing preserves type safety via contracts, but introduces blame and potentially unbounded overhead; interoperation across typed/untyped modules trades static guarantees for dynamic checks.

These are not accidents of particular implementations; they are manifestations of the EVT tradeoff: increased *macro-expressiveness* (reflection, self-interpretation, unconstrained composition) strains *global verifiability*, unless restrictions restore structure. Hence, an explicitly stated *conservation law for physical computational behavior* explains these difficulties.⁴

The reason this theorem has long gone unformulated is that its essential tradeoff has long been hidden in plain sight—embedded in the very architecture of classical computation we all use in our day-to-day lives. The Von Neumann architecture, by treating program and data uniformly and allowing maximal flexibility (any memory can store code or data, and programs can modify themselves), enables *extraordinary* expressiveness. However, this very flexibility comes at the cost of global verifiability: reasoning about whole-system properties becomes fragile, partial, or expensive unless additional structure is imposed. When the hardware is instead fixed algebraically—restricting the set of possible

⁴A fuller account of how the EVT Trilemma manifests in macroscopic programming practice is given in [Kam26]. This outlines the stated connections to resource theory and can be read as a companion piece.

states, operations, or transitions—the system gains predictability and becomes globally checkable, but loses expressive power.

The tradeoff does not assert a single conservation equality. There exist regimes in both classical (language-based) and physical (geometric) computation with both low expressiveness *and* low verifiability—the infamous “Turing tarpit,” (e.g., bash, Malbolge), where everything is theoretically possible but practically inexpressible. The second half of the EVT is best read as a *no-free-lunch inequality* with phases: beyond a threshold expressiveness, Rice-type barriers are pervasive; and to regain verifiability, one must withdraw some of that structure. Between these extremes lie transitional corridors in which small feature additions can cause qualitative shifts in what can be verified.

Because these tradeoffs depend on ambient structure, the EVT treats *local* symmetries of a model or architecture rather than postulating a single global invariant across all substrates. In a Von Neumann setting, some of the most important local regularities are cognitive and conventional: programming idioms, abstractions, and composition patterns induce de facto invariants that temporarily “abelianize” parts of the system, while reflective or unrestricted features reintroduce non-commuting structure. Pushing too far past the barriers above yields systems that remain formally expressive but become progressively less usable in physical and human practice.

Removing barriers to viewing computation. Regarding tractability, the present paper intentionally abstracts away many ordinary engineering bottlenecks. In an idealized geometric regime, one may treat the substrate as reconfigurable enough that the sharp distinction between hardware and program becomes less natural than it is in fixed Von Neumann architectures. This does *not* remove physical resource limits; rather, it allows the expressiveness/verifiability relation to be studied more directly, without immediately collapsing into familiar questions about instruction sets, compilers, and implementation overhead. For this reason, although the full trilemma must be stated as such, we do not pursue tractability in equal depth here: in the present paper, it functions chiefly as a reminder that human-built machines impose additional constraints beyond those intrinsic to the underlying geometric substrate.

So in our GCM-based setting, the expressiveness/verifiability relationship can be studied more transparently because there is *no fixed separation between logic and memory* of the usual Von Neumann sort. A GL is an idealized *non*-Von Neumann substrate defined directly in geometric and algebraic terms. By fusing logic and language into a single analog-geometric object, one can *see* how the operational semantics deforms the computational substrate, and model how the algebraic closure group constrains the geometry with every increasing bit!

This tension between expressive freedom and structural discipline is not unique to programming languages; it is a recurring theme whenever algebraic power begins to dominate geometric intuition. As Baez [Bae22] writes in *Getting to the Bottom of Noether’s Theorem*:

“Atiyah [Ati02] warned us of the dangers of algebra: “...*algebra is to the geometer what you might call the Faustian offer. As you know, Faust in Goethe’s story was offered whatever he wanted (in his case the love of a beautiful woman), by the devil, in return for selling his soul. Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul: give up geometry and you will have this marvellous machine.*”

Baez notes that “While this is sometimes true, algebra is more than a computational tool: it lets us *express* concepts in a very clear and distilled way.” We agree that the finest mathematics arises when algebra and geometry are brought into balance. But our contention here is that computation is one of the settings in which this balance becomes unusually unforgiving. Within an ordinary formal background such as ZFC, geometry and algebra can coexist without tension; but once a geometric system is required to *compute*, the act of doing so (in our case—the “gluing”) *enforces constraints* on *how* the flow can be structured. The bargain, in this case, is not rhetorical but structural, and the devil shall not be cheated.

But why physics? It is reasonable for readers to ask why a theorem about computation is applied in a mathematical physics paper. Why do this? Two pragmatic reasons guide this choice. First, differential geometry and geometric analysis provide a language in which *flows, invariants, and gluing*

are native: variational principles, conservation laws, and well-posed evolution already live there. Second, many of the sharpest constraints on computation are *physical*—finite precision, energy/entropy budgets, and measurement back-action—so a formulation that is testable against laboratory and numerical practice is preferable to one that remains purely syntactic (e.g. in terms of abstract syntax trees and Galois connections).

This does not diminish the interest of the EVT Hypothesis for computer science, education, or the philosophy of natural laws. Versions of the underlying tension have appeared in many domains, though rarely isolated as a single, explicit selection principle. A few recurring patterns (illustrative, not exhaustive) are:

- **Logic and programming.** As previously stated, total/strongly normalizing calculi gain verification power at the price of general recursion; reflective/metaprogramming gains flexibility at the cost of global guarantees.
- **Numerics and PDEs.** Stability constraints (e.g. CFL-type conditions) and structure-preserving schemes (e.g. symplectic integrators) enforce invariants but restrict admissible updates; aggressive damping improves robustness while destroying time-reversibility and fine invariants [CFL28].
- **Many-body/field theory.** Integrable corners admit exact charges and factorization but are measure-zero in parameter space; generic regimes are expressive yet analytically resistant. Undecidability phenomena (e.g. spectral-gap questions) underscore that some semantic properties of dynamics cannot be decided uniformly [CPGW15].

But the most pressing applications of a *computational* selection principle are presently in *quantum gravity* and *lattice gauge theory*, where one must extract dynamics from large, ongoing physical computations (experiments or simulations) under resource and precision constraints. The EVT Trilemma is intended to bound *how* such processes can behave and what can be recovered from them, independently of any particular quantization scheme.

Mathematics has Gödel–Tarski style incompleteness, and computer science has the halting problem and Rice-type no-go results. Physics, despite its rich family of conservation laws, lacks a comparably explicit bound on the *simultaneous* attainability of global verifiability and unrestricted expressive power in *physical* information-processing. Various desiderata for such a bound have been voiced (e.g. calls for a “Gödelization” of fundamental theory in public remarks), and adjacent programs (such as Constructor Theory [MVK⁺22]) have sought axioms phrased in terms of counterfactual capabilities; our starting point diverges in ontology but shares the motivation.

The EVT Hypothesis is proposed as a *selection principle* for the behavior and modeling of computational processes: physical computation—discrete or analog—is not substrate-independent magic; it is a *gauge-constrained flow* carried by a geometric substrate, with invariant quantities and resource monotones that cannot be simultaneously optimized without limit. From this viewpoint, the structures that make computation possible (*how* we compute) are as consequential as the extensional outcomes (*what* we compute). Any physical theory of computation should, at a minimum, respect this geometric constraint.

3 Abelian Gauge Dynamics and Phase Geometry

Recent work on the computability and complexity of fluid flows [CMPSP21], together with Tao’s conjectural program on universality phenomena for the Euler equations [Tao18], highlights a recurring theme: *continuous media can support computation*, but often only through dynamics whose analytic behavior is difficult to control.

In this section, we introduce the *quasi-fluidic band* as an *abelian gauge* mode of computation. Informally: a GL is a lattice of geometric cells whose internal degrees of freedom carry abelian phase data (typically $U(1) \simeq S^1$, or a finite cyclic approximation C_n), together with discrete labels and quantized counters.

As established earlier, a geometric computer is neither strictly a Turing machine nor a quantum computer. Quantum computation is organized around a global state vector and coherence constraints; locality and scaling are mediated by entanglement. By contrast, a GL scales by composition: cells are wired together by interfaces and boundary conditions, and the operative degrees of freedom are abelian phases and discrete readouts that compose functorially across the lattice. There are no complex

amplitudes and no Born-rule collapse; instead, one reads gauge-invariant features of geometry (e.g. phase differences and winding).

As a motivating example, consider the classic experiment in which a hose filled with running water is placed across a vibrating speaker. The resulting jet forms a visible oscillating pattern: the driving frequency controls coil tightness, amplitude controls the spiral radius, and phase controls the initial offset. This is an instructive caricature of the quasi-fluidic regime: periodic forcing induces an effective $U(1)$ -phase along a geometric substrate, and the observable shape provides a coarse readout of that phase. In the language adopted later (3.1), the phase offset belongs to the T -layer (S^1 data), while winding/twist counts correspond to quantized Z -invariants; discrete readouts arise from coarse partitions of these quantities. Helical and twisted structures appear here not as an aesthetic choice, but as energetically natural carriers of phase and twist under periodic driving. A visual demonstration by [bru13] makes this phenomenon concrete.⁵

Computational geometric bordisms. The vibrating-hose experiment may be viewed as an analog of a *computational geometric bordism*: a smooth evolution mediating between boundary conditions (fixed endpoints, imposed phase/frequency, initial and final configurations). Here, the bordism is “internal” to the geometry of the evolving flow, and the computable content is not the PDE solution itself but the stable, gauge-invariant features one can read from it. See appendix A.5 for cobordism formalisms.

Proposition 3.1. *The geometric topology of a GCM constrains its information-bearing degrees of freedom, but the effective information content is relative to a chosen readout (coarse-graining). In particular, geometric computation is deterministic at the microstate level, while information is defined relative to an encoding.*

Unlike in classical computation—where the bit is treated as an absolute, context-free primitive—in geometric computation, the notion of a “geobit” is always specified with respect to a readout map (or partition) on the underlying state space. Concretely, one fixes an observable or coarse-graining

$$\rho : \Sigma \longrightarrow B,$$

where Σ is the (possibly structured) microstate space of the GCM and B is a finite label set used for symbolic computation. Once ρ is fixed, information becomes an objective property of the pair (Σ, ρ) : the dynamics may be fully deterministic on Σ , yet the induced evolution on B depends on which features are chosen as readable and which are treated as hidden or irrelevant.

This is directly analogous to physics, where measurements and reference choices enter through specified observables rather than through subjectivity. The boundary between symbolic and quasi-fluidic computation is therefore not a matter of taste. Still, of modeling choice: symbolic regimes arise when the operative degrees of freedom are purely discrete (a finite B with static invariants), while the quasi-fluidic regime appears when one enlarges the local fiber to include compact abelian phases and quantized topological counters (the $B/T/Z$ structure of definition 3.1). Geometric deformation then permits interpolation between discrete and continuous logic *without* introducing probabilistic amplitudes.

This does not mean the quasi-fluidic regime is vague. In many ways, it has already appeared implicitly: even in earlier symbolic constructions, evolution was driven by an explicit action stream (global pulses and local update schedules) acting on finite geometric labels. The quasi-fluidic mode makes this action stream geometric by allowing abelian phase transport and thresholded topological events, while retaining the core principle of GCT: computability is realized by local, deterministic transition structure rather than by solving global differential equations.

3.1 A New Kind of Computation

A central motivation of GCT is to avoid the classical reliance on global PDE evolution as the *definition* of computation over continuous media (fluids, fields, elastic bodies). As Tao has repeatedly emphasized, even well-posed PDE models may develop loss of regularity, extreme sensitivity to initial conditions, or finite-time singular behavior; these analytic pathologies are not merely technicalities, but can become *computationally relevant* in unfavorable ways.

⁵Regarding an unconventional reference; the point is purely pedagogical: it is a compact way to visualize phase, winding, and coarse readout in a physical system.

The paradox of analog logic flow. Historically, analog and fluidic devices exploited continuous physics to compute. Mechanical differential analyzers and wartime fire-control machines, as well as later fluidic and hydraulic integrators, demonstrate that a physical medium can implement nontrivial transforms. Yet this tradition also exposes a basic paradox of analog logic: the computational substrate is not insulated from the mathematics it is tasked with realizing. If the governing model admits blow-up, stiffness, or chaotic amplification, then the physical device inherits the same fragility. In contrast to digital machines—which often fail noisily via discrete faults— analog computation can fail by gradual loss of identifiability: the state remains *well-defined physically* while becoming *unreadable* with respect to the intended encoding.

GCT responds by shifting the locus of computability from *solving equations* to *enforcing structure*. A GCM is designed so that the fundamental computational step is a local, deterministic update at an interface: an equivariant transformation of a structured state fiber (in the quasi-fluidic band, the $B/T/Z$ decomposition of definition 3.1), driven by controlled inputs (pulses, boundary perturbations, or local couplings). The ambient dynamics may be smooth, but the computation is carried by the admissible morphisms of the system—what changes, what cannot change, and what invariants must persist. In this sense, GCMs are naturally piecewise smooth: not because we approximate a PDE, but because computation is organized around local transitions and thresholded events rather than global analytic integration. In a GL, the computer is the problem, and the problem is the solution.

Accordingly, we do not attempt to compute Euler solutions by “bit-flips” internal to the manifold. Instead, we encode logic into the topology, phase structure, and allowable deformations of embedded geometry, and we read outputs through fixed observables (coarse-grainings) of that geometry. In the coming sections, we will model how complex behavior can emerge from simple shapes.

This structural viewpoint is not without precedent. Goriely and collaborators, whose earlier work on nonlinear geometry and helices we will cite later, have recently advocated a related shift in applied modeling: replacing monolithic PDE descriptions by networked, local interaction rules in neuroscience and biophysics. Their work illustrates how complex global behavior can arise from local coupling laws on a discrete architecture, providing a concrete example of how one may trade PDE-centric descriptions for robust, compositional dynamics [TCK⁺20].

3.2 Boundaries in Quasi-Fluidic GCMs

At scale, computation in a GL emerges from the coordinated action of many such units coupled by local interfaces. The issue in this regime is not partial recursion in the classical sense, but the passage from discrete symbolic updates to continuous, phase-carrying geometric evolution. The quasi-fluidic band marks exactly this transition: local rules remain computable and structured, while the underlying state space begins to acquire continuous abelian degrees of freedom. We now make this phase boundary precise.

Definition 3.1 (Quasi-Fluidic Computation and Regime Phases). Given a GCM, we say it operates in the *quasi-fluidic regime* if each region R_i carries a finite-dimensional state fiber:

$$\mathcal{M} \cong B_i \times T_i \times Z_i,$$

as in definition 2.1 with the following structure:

- (QF1) **Discrete core (symbolic).** B_i is finite, typically $B_i \cong (C_2)^{k_i}$; in the twin helix unit one has $k_i = 4$ for chirality/orientation bits.
- (QF2) **Phase layer.** T_i is a continuous but compact abelian Lie group (a torus), $T_i \cong (S^1)^{m_i} \simeq U(1)^{m_i}$, encoding axial phase(s), orientation angles, or other bounded “analog” degrees of freedom.
- (QF3) **Quantized invariants.** Z_i is a finitely generated abelian group (e.g. $\mathbb{Z}^{p_i}$), recording twist/winding/linking numbers or other topological charges.
- (QF4) **Abelian symmetry and equivariance.** The local symmetry acting on Σ_i is

$$K_i \cong (C_2)^{k_i} \times (S^1)^{m_i},$$

and every interface map $I_{ij} \in \mathcal{I}$ and the induced flow on Σ_i are K -equivariant, that is, they depend on phase differences and commute with flips/rotations.

- (QF5) **Flow coupling.** The ambient flow $\{E v_t\}$ transports the T -coordinates continuously (phase evolution) and can modify Z only by isolated, energy-threshold events (phase slips) prescribed by the interface rules; B updates by the same local rules. No non-abelian monodromy is present.
- (QF6) **Non-singularity and bounded energy.** Trajectories have no finite-time blow-ups; curvature/torsion remain bounded; state updates at interfaces are well defined for all t .

A system is symbolic when $m_i = 0$ and Z_i is static; it is purely fluidic when B_i is trivial and non-abelian holonomy appears. The quasi-fluidic band sits between these: it combines a finite discrete core with $U(1)$ -type phases and integer topological counters, all governed by abelian symmetry.

Composition of GCMs. In the symbolic model, a GL is composed of *discrete interconnect*: neighboring GCMs exchange a finite readout and update by a local rule, precisely as in a cellular automaton with richer internal state. In the quasi-fluidic regime, this picture remains valid. Still, each cell now carries an additional compact phase and a quantized invariant structure (the $B/T/Z$ fiber of definition 3.1), so the internal dynamics of a single unit are no longer adequately described by a purely discrete transition table.

There is also a second, more geometric notion of composition which becomes natural in this band: *flow-circuit composition*. Rather than identifying adjacency inside a single ambient lattice, one regards a GCM as a module with an *input interface* and an *output interface*, together with a local update law. The module, therefore, induces a (generally state-dependent) transfer map on the relevant readouts and phase variables, and connecting two modules corresponds to wiring output interfaces into input interfaces. At the level of these transfer maps, composition is ordinary function composition.

Operationally, “wiring” is encoded by the choice of interfaces $\mathcal{I}$ and their coupling strength: one may couple two neighboring regions by a nontrivial interface map I_{ij} , or effectively *disconnect* them by selecting a trivial (e.g. identity/no-op) interface, or by imposing a boundary condition that forces the coupling term to vanish. In this way, reconfiguration is expressed within the same functorial framework: one modifies the *interface structure* (the allowed local morphisms). This viewpoint aligns with the guiding principle of the quasi-fluidic regime: computation is carried by the admissible local moves and their compositions, while the ambient drive (pulses, boundary perturbations) supplies controlled input to those moves.

3.3 Abelian Lattice Gauge Dynamics in the Quasi-Fluidic Mode

In the QF band, the appropriate “computational” physics we should work with is not a discrete logic model as the symbolic band but an *abelian lattice gauge dynamics*: compact phases transported and coupled locally, with evolution governed by a bounded local energy and (optionally) a driven-damped relaxation law. This section is *interpretive*; it provides a stable dynamical model for analyzing QF behavior, without elevating differential equations to the definition of geometric computability.

Lattice variables (QF state). Let $\Gamma = (V, E)$ be the interaction graph of a GL (cells/modules are vertices, interfaces are edges). For each cell $i \in V$, we take an internal phase (orientation layer)

$$\theta_i \in U(1) \cong \mathbb{R}/2\pi\mathbb{Z},$$

and for each interface $(i, j) \in E$ we take a *link phase* (a discrete connection 1-form)

$$A_{ij} \in \mathbb{R}/2\pi\mathbb{Z}, \quad A_{ji} = -A_{ij},$$

equivalently $U_{ij} := \exp(iA_{ij}) \in U(1)$. (If one wishes to remain in the finite cyclic subband, replace $U(1)$ by C_N and interpret all equations below as a continuous approximation to a stroboscopically driven C_N update.)

Gauge redundancy. A $U(1)$ gauge transformation is a choice of angles $\alpha_i \in \mathbb{R}/2\pi\mathbb{Z}$ at each cell,

$$\theta_i \mapsto \theta_i + \alpha_i, \quad A_{ij} \mapsto A_{ij} + \alpha_i - \alpha_j,$$

under which the *gauge-invariant phase difference*

$$\Delta_{ij} := \theta_i - \theta_j - A_{ij} \pmod{2\pi}$$

is unchanged. In this sense, “independence of local modules” is a choice of gauge/frame, while couplings are encoded invariantly by Δ_{ij} .

Local energy (interface functional). A minimal gauge-invariant energy on Γ is

$$\mathcal{E}(\theta, A) = \sum_{(i,j) \in E} J_{ij} (1 - \cos(\Delta_{ij})) + \sum_{p \in \mathbf{P}} \kappa_p (1 - \cos((dA)_p)), \quad (1)$$

where $J_{ij} \geq 0$ are interface couplings and $\mathbf{P}$ is a chosen set of plaquettes (2-cells) with discrete curvature

$$(dA)_p := \sum_{(i,j) \in \partial p} s_{p,ij} A_{ij} \pmod{2\pi}, \quad s_{p,ij} \in \{\pm 1\}$$

the oriented sum around ∂p . The first term penalizes a mismatch between neighboring phases modulo the link variable; the second penalizes nonzero curvature (“flux”) [Kog79].

Gradient-flow dynamics. A simplest well-behaved evolution law is driven by gradient flow on the compact state space:

$$\dot{\theta}_i = -\gamma_\theta \frac{\partial \mathcal{E}}{\partial \theta_i} + \eta_i(t), \quad \dot{A}_{ij} = -\gamma_A \frac{\partial \mathcal{E}}{\partial A_{ij}} + \eta_{ij}(t), \quad (2)$$

with mobilities $\gamma_\theta, \gamma_A > 0$ and external drives $\eta_i(t), \eta_{ij}(t)$ (global pulses, boundary forcing, or localized injections). Concretely, the Euler–Lagrange components are

$$\frac{\partial \mathcal{E}}{\partial \theta_i} = \sum_{j \sim i} J_{ij} \sin(\Delta_{ij}), \quad (3)$$

$$\frac{\partial \mathcal{E}}{\partial A_{ij}} = -J_{ij} \sin(\Delta_{ij}) + \sum_{p \ni (i,j)} \kappa_p s_{p,ij} \sin((dA)_p), \quad (4)$$

so (2) relaxes toward gauge-invariant consistency ($\Delta_{ij} \rightarrow 0$) and low curvature ($(dA)_p \rightarrow 0$) unless forced otherwise. Because the variables are compact and $\mathcal{E}$ is bounded below, this regime is dynamically stable and aligns with the QF “no blow-ups” philosophy.

Remark 3.1 (Scope and Interpretation). Equations (1)–(2) are not proposed as fundamental laws of GCT; they are a *normal form* for the abelian corridor: local couplings, compact phases, gauge redundancy, and dissipative relaxation. In particular, continuous flows of this type do *not* implement jump-style updates by themselves; any discrete “computational” layer is induced later by readout/-partitioning (return maps).

Auxiliary geometric ODE variables. To retain an explicit connection to helix geometry, one may attach slowly varying, bounded geometric parameters to each cell,

$$r_i(t) \in \mathbb{R}_{>0} \quad (\text{effective radius/scale}), \quad (5)$$

$$\tau_i(t) \in \mathbb{R} \quad (\text{effective twist rate}), \quad (6)$$

$$\mathcal{L}_i(t) \in \mathbb{R}_{>0} \quad (\text{effective length}) \quad (7)$$

and couple them to *local energy density* and/or *local phase current*. A simple choice is:

$$\dot{r}_i = -\lambda_r (r_i - r_0) + \alpha_r \sum_{j \sim i} (1 - \cos(\Delta_{ij})), \quad (8)$$

$$\dot{\tau}_i = -\lambda_\tau (\tau_i - \tau_0) + \alpha_\tau \sum_{j \sim i} \sin(\Delta_{ij}), \quad (9)$$

$$\dot{\mathcal{L}}_i = -\lambda_L (\mathcal{L}_i - \mathcal{L}_0) + \alpha_L r_i |\tau_i|, \quad (10)$$

with $\lambda_\bullet > 0$ ensuring relaxation to baseline values $(r_0, \tau_0, \mathcal{L}_0)$ and $\alpha_\bullet$ setting coupling strength. These equations encode the informal idea that mismatch/coupling stress can thicken the effective geometry (8), phase transport can bias twist (9), and geometry co-scales (10). One may also let the mobilities

in (2) depend on τ_i (twist as a local “clock” factor), e.g. $\gamma_\theta = \gamma_\theta(\tau_i)$, without changing the gauge structure.

Synchronization limit: Kuramoto reduction. A useful special case of the QF phase dynamics is obtained by restricting to nearest-neighbor coupling on the $U(1)$ phase layer and passing to a co-rotating frame. In that case, the local phase evolution reduces to a Kuramoto-type system [ABV⁺05, Kur84]

$$\frac{d\theta_i}{dt} = \Omega_i + K \sum_{j \sim i} \sin(\theta_j - \theta_i),$$

where $\theta_i \in S^1$ is the phase of cell i , Ω_i is its intrinsic frequency, and K is the coupling strength.

This should be understood as a *synchronization limit* of the abelian gauge dynamics, not as the full semantics of quasi-fluidic computation. Physically, it describes the tendency of neighboring helices to phase-lock under local coupling; computationally, it captures the emergence of a shared clock or coherent phase carrier across a region of the lattice. In particular, when the $U(1)$ corridor is used only as a transport layer, Kuramoto synchronization provides a natural mechanism by which continuous phase variables may stabilize before a stroboscopic readout is applied.

Equivalently, if the local energy is taken to be of XY type,

$$E(\theta) = \sum_{(i,j) \in E(\Lambda)} \kappa_{ij} (1 - \cos(\theta_i - \theta_j)),$$

then the dissipative phase flow

$$\dot{\theta}_i = -\frac{\partial E}{\partial \theta_i}$$

recovers the Kuramoto form above up to the choice of sign convention and the addition of the drift terms Ω_i . Thus, the Kuramoto system is a familiar, physically interpretable subcase of the general QF lattice dynamics.

Example 3.1 (Plaquette Phase Slip and Integer Counter Update). Let p be an oriented plaquette in the lattice, and let

$$\Delta_e = \theta_{s(e)} - \theta_{t(e)} - A_e \pmod{2\pi}$$

denote the gauge-invariant phase difference on each oriented boundary link $e \in \partial p$. Along a continuous trajectory, choose continuous real lifts $\tilde{\Delta}_e(t) \in \mathbb{R}$ and define the lifted circulation

$$\tilde{\Gamma}_p(t) := \sum_{e \in \partial p} s_{p,e} \tilde{\Delta}_e(t), \quad s_{p,e} \in \{\pm 1\}.$$

Because each Δ_e is gauge invariant, so is $\tilde{\Gamma}_p$ once its initial lift has been fixed.

At every time, decompose the lifted circulation uniquely as

$$\tilde{\Gamma}_p(t) = \Gamma_p(t) + 2\pi z_p(t),$$

where

$$\Gamma_p(t) \in (-\pi, \pi], \quad z_p(t) \in \mathbb{Z}.$$

The variable Γ_p is the principal compact $U(1)$ flux observed on the plaquette, while z_p records the number of complete 2π windings that have passed through the chosen branch.

As long as $\tilde{\Gamma}_p(t)$ remains within a single branch interval, z_p is constant and the evolution lies entirely in the smooth phase layer. Suppose, however, that an applied drive causes $\tilde{\Gamma}_p$ to cross a branch boundary at a time t_* . Then the principal phase wraps by 2π , while the integer counter changes by one:

$$z_p(t_*^+) - z_p(t_*^-) = \begin{cases} +1, & \dot{\tilde{\Gamma}}_p(t_*) > 0, \\ -1, & \dot{\tilde{\Gamma}}_p(t_*) < 0. \end{cases}$$

The compact quantity

$$\exp(i\tilde{\Gamma}_p(t)) = \exp(i\Gamma_p(t))$$

remains continuous throughout the event; the discrete change occurs in the lifted winding ledger z_p . This is a *phase slip*.

In the phase-only reduction $A_e = 0$, the same construction measures the winding of neighboring cell phases around the plaquette. In a continuum physical realization, the event corresponds to the passage or nucleation of a vortex through the plaquette; the phase-only description may then fail briefly at the defect core while the surrounding $U(1)$ phase remains well-defined. Thus, the Z -coordinate is not introduced independently of the continuous dynamics: it records isolated threshold events generated by the evolution of the T -layer.

Remark 3.2 (Smooth Core and Thresholded Boundary Events). The strict smooth QF core is obtained by restricting to trajectories for which every z_p remains fixed. Allowing the isolated events of example 3.1 enlarges this core by adjoining transitions between integer-labeled strata. Such phase slips do not by themselves introduce non-abelian holonomy: the structure group remains $U(1)$, but the evolution is no longer confined to a single smooth lift of the compact phase space. This thresholded extension supplies the natural bridge from abelian phase transport to vortex and defect dynamics.

3.4 Vector Cellular Automata and Stroboscopic Readout

The QF dynamics of section 3.3 are continuous (compact phases with local energy and gradient flow). As such, they do *not* natively implement jump-style updates. The bridge back to classical symbolic computation is therefore *stroboscopic*: we sample the system once per pulse and apply a *discretized readout* (a quantizer) to obtain an induced update on a finite alphabet. In this sense, discrete computation in the QF band is a *return-map phenomenon*, not an ODE phenomenon.

Definition 3.2 (Pulse Times and Stroboscopic Map). Fix a pulse period $\Delta t > 0$ and define $t_n := n\Delta t$. Let $P : L \rightarrow L$ denote the time- Δt map on the QF state space L induced by one pulse:

$$P := \text{Ev}_{\Delta t}.$$

We write $x^n \in L$ for the post-pulse state at time t_n , so $x^{n+1} = P(x^n)$.

Because the abelian corridor carries gauge redundancy, any robust readout should be phrased in terms of *gauge-invariant local observables*. In the lattice-gauge normal form, the basic invariants are:

$$\Delta_{ij} = \theta_i - \theta_j - A_{ij} \pmod{2\pi} \quad \text{and} \quad (dA)_p = \sum_{(i,j) \in \partial p} s_{p,ij} A_{ij} \pmod{2\pi}.$$

We therefore define a cellwise observable $y_i = y_i(x)$ as a finite list of such invariants in a bounded neighborhood (e.g. the Δ_{ij} on links incident to i , and optionally nearby plaquette fluxes $(dA)_p$).

Definition 3.3 (Discretized Readout/Quantizer). Fix a finite alphabet B and a (piecewise constant) quantizer

$$Q : \mathcal{Y} \longrightarrow B,$$

where $\mathcal{Y}$ is the range of local invariants y_i . The induced discrete configuration for a state $x \in L$ is

$$b = \text{Read}(x) \in B^V, \quad b_i := Q(y_i(x)).$$

For example, one may partition S^1 into arcs and threshold a chosen invariant angle, or partition the flux $(dA)_p$ into bins. The specific partition is a modeling choice.

Definition 3.4 (Vector Cellular Automata on an Induced Symbolic Return Map). Write x_i^n for the local continuous state in cell i at pulse n (phases θ , links A , and any auxiliary variables such as $r, \tau, \mathcal{L}$). Because the QF energy and interfaces are finite-range, there exists a finite neighborhood $\mathcal{N}(i)$ such that the stroboscopic update is local:

$$x_i^{n+1} = \mathbf{U}\left(\{x_j^n\}_{j \in \mathcal{N}(i)}\right), \quad x_i^n \in \mathbf{X},$$

for some compact state manifold $\mathbf{X}$ (a product of circles and bounded geometric variables). This is precisely a *vector cellular automaton* (continuous alphabet, local rule, synchronous clock) [TM87].

Composing with the readout gives an induced discrete return map on B^V :

$$\mathcal{F}_B := \text{Read} \circ P \circ \text{Lift} : B^V \longrightarrow B^V,$$

where Lift selects a representative microstate $x \in L$ consistent with a discrete labeling (e.g. by choosing canonical phases inside each bin).

Remark 3.3 (What is “Computed”?). In the QF corridor, the *continuous* dynamics are the physical substrate, and the *symbolic* computation appears only after (a) stroboscopic sampling and (b) a discretized readout. This is the same conceptual move familiar from dynamical systems and engineered digital devices: continuous dynamics are real, while symbolic logic is the stable quotient obtained by thresholding and clocking [Str94].

The $U(1)$ Corridor and an E/V Paradox. It is at this point necessary to issue some disclaimers regarding the use of the full S^1 state spaces, that is, the $U(1)$ corridor. As established, within the quasi-fluidic band, the ω -logic layer may be taken either as a finite cyclic alphabet C_N or as a continuous phase $U(1)$. Passing to the full S^1 limit is mathematically tidy (abelian, compact), but it also oddly backfires, creating a regime with atypical E/V trade-off behavior. The following remarks record the current hypothesis of this paper:

- **Low geometric expressiveness.** Pure $U(1)$ phase composition is commutative. As a result, sequencing power collapses: one cannot exploit non-commuting compositions to build richer path-dependent behavior, and the phase layer alone tends to support only limited families of smooth, energetically bounded dynamics. In this corridor, most “computational complexity” must therefore be carried either by the discrete core B (via readout and interface thresholds) or by leaving the abelian setting (e.g. via threshold events/topological defects or non-abelian holonomy).
- **Low algebraic verifiability.** Any *finite* readout must discretize phase (explicitly or implicitly) by thresholds. In the $U(1)$ corridor, near-resonant phase evolution makes this discretization brittle: small phase perturbations can move trajectories across readout boundaries, producing significant changes in the inferred discrete state. Moreover, phase-locking reduces redundancy (many degrees of freedom share a common “clock”), weakening independent cross-checks.

Transition, not destination. For these reasons, the $U(1)$ phase limit is best regarded as a *critical corridor*: visually orderly and analytically tractable, yet brittle under discretized readout and shallow in purely geometric expressiveness unless supplemented by discrete structure. It serves as a staging layer for robust computation on B and as a boundary marker before the onset of genuinely non-abelian (fluidic/gauge) behavior.

3.5 Application of Lie Groupoids in Geometric Computation

As the finite cyclic phase spaces used in large abelian geobit groups are refined, they naturally approach a continuous $U(1)$ phase description. At the same time, the local nature of geometric computation forces admissibility and composability to be built into the formalism from the outset: not every transformation is available at every state, and meaningful updates must respect the underlying interface structure. These requirements are brought together naturally by the notion of a *Lie groupoid*—introduced originally by [Ehr59] and with further developments by [Pra66]—which combines smooth structure with local, state-dependent symmetry [Mac05].

Informally, a Lie groupoid is a groupoid in which the space of objects and the space of morphisms are smooth manifolds, and all structural maps (source, target, identity, inversion, and composition) are smooth. Unlike a single global group action, a Lie groupoid permits invertible morphisms only where they are geometrically meaningful. This makes it well-suited to GCMs, whose computation is local, interface-driven, and governed by admissible reversible moves rather than by a single global symmetry.

We begin with a general decorated formalism, and then specialize it to the quasi-fluidic abelian regime.

Definition 3.5 (*S*–decorated Groupoid). Let $(S, *)$ be a monoid with unit e , and let $\mathbf{B}S$ denote the one-object category whose endomorphism monoid is S . An *S*–decorated groupoid is a pair $(\mathcal{G}, d)$ consisting of a small groupoid $\mathcal{G}$ together with a functor

$$d : \mathcal{G} \longrightarrow \mathbf{B}S.$$

Equivalently, each morphism $f \in \text{Mor}(\mathcal{G})$ is assigned a label $d(f) \in S$ such that

$$d(g \circ f) = d(g) * d(f), \quad d(\text{id}_x) = e$$

for all composable f, g and all objects x . Since functors preserve isomorphisms, every decoration of an arrow lies in the unit group $S^\times \subseteq S$.

Definition 3.6 (Morphisms of *S*–decorated Groupoids). A morphism

$$(\mathcal{G}, d) \longrightarrow (\mathcal{H}, \delta)$$

of *S*–decorated groupoids is a functor $F : \mathcal{G} \rightarrow \mathcal{H}$ such that

$$\delta \circ F = d.$$

This defines the category DecGrpd_S of *S*–decorated groupoids.

Remark 3.4. The use of the term “decorated” follows modern geometric and categorical usage [Pen87, Fon15]. The point is to enrich a base compositional structure with auxiliary data while preserving functorial composition. In the present context, the decorations record geometric or computational quantities associated with paths, interfaces, or local transformations.

General decorated path data. Let M be a smooth manifold (or a stratified manifold when needed), and let $\mathcal{S}$ denote a specified collection of geometric or computational structures on M —for example, twist counts, phase accumulation, chirality labels, or local field values. The classical fundamental groupoid remembers only path homotopy classes. In geometric computation, however, one often needs a refinement that also records such auxiliary data.

Definition 3.7 (Decorated Fundamental Groupoid). Let M be a smooth manifold and let S be a monoid. An *S*–decorated fundamental groupoid of M is an *S*–decorated groupoid

$$(\Pi_1^*(M), d)$$

whose underlying groupoid $\Pi_1^*(M)$ is chosen from a path groupoid naturally associated to M (for example, the ordinary fundamental groupoid, or the thin fundamental groupoid when differential invariants are to be preserved), and whose decoration functor

$$d : \Pi_1^*(M) \rightarrow \mathbf{B}S$$

records the specified geometric data along paths. Composition is induced by path concatenation together with the monoid law in S .

Remark 3.5. When the decoration depends only on ordinary homotopy classes, one may work with the usual fundamental groupoid $\Pi_1(M)$. When the decoration depends on differential data such as connection holonomy or thin geometric transport, one should instead use the thin fundamental groupoid $\Pi_1^{\text{thin}}(M)$, since such invariants are constant under thin homotopy but need not factor through ordinary homotopy classes.

A geometric model for the twin helix. For the purposes of geometric computation, a finite twin helix module is best modeled as a regular neighborhood of an embedded arc. Topologically, this gives

$$THM \cong D^2 \times I \cong B^3.$$

Thus, the underlying space is contractible: its ordinary homotopy type is trivial, and the undecorated fundamental groupoid carries no nontrivial loop information.

It means that the computationally relevant structure of a twin helix module is not encoded by the bare topology of the carrier space, but by the additional geometric data placed on it: phase fields, twist laws, chirality assignments, framings, and interface rules. In the symbolic and general decorated setting, one may attach labels such as chirality or twist to paths in this contractible space. In the quasi-fluidic abelian reduction developed later in this chapter, chirality is no longer treated as an actively updated path label, but instead as a choice of orientation convention for link variables.

Example 3.2 (Decorated Loops in a Finite Module). Let

$$THM \cong D^2_{(r,\varphi)} \times I_z,$$

and choose a smooth phase 1-form on the cross-sectional angular direction,

$$A = \Omega(r, \varphi, z) d\varphi.$$

For fixed (r, z) , let $\gamma_{r,z}$ denote the loop obtained by traversing once around the circle of radius r in the cross-section $D^2 \times \{z\}$. As a loop in the contractible manifold THM , $\gamma_{r,z}$ is null-homotopic in the ordinary sense. Nevertheless, it may carry nontrivial geometric data.

For example, taking

$$S = U(1) \times C_2,$$

one may define a decoration on an appropriate refined path groupoid (for example, the thin fundamental groupoid) by

$$d([\gamma_{r,z}]) := \left(\exp\left(i \int_{\gamma_{r,z}} A\right), \chi(\gamma_{r,z}) \right),$$

where the first component records the accumulated phase and the second records a chirality label determined by the chosen geometric structure [Bar91]. The resulting decorated class

$$([\gamma_{r,z}], d([\gamma_{r,z}]))$$

is computationally nontrivial even though the underlying ordinary homotopy class of $\gamma_{r,z}$ is trivial.

Thus, in a finite twin helix module, the computational content lives in the decoration rather than in the bare fundamental groupoid.

Remark 3.6. This is precisely why the general decorated picture and the later quasi-fluidic abelian reduction must be kept distinct. In the symbolic/general setting, chirality may legitimately appear as part of the decoration target S , and one works with a refined path groupoid because ordinary homotopy forgets too much. In the specialized $U(1)$ phase regime, by contrast, chirality is frozen into an orientation convention, and the active structure reduces to abelian phase transport.

Why Lie groupoids appear in the QF regime. The preceding decorated picture is intentionally broad. In the quasi-fluidic regime, however, the relevant structure simplifies dramatically. The active degree of freedom is no longer a freely manipulated chirality/twist label, but an abelian phase field. The correct object is therefore not a general decorated path groupoid, but a Lie groupoid encoding *gauge redundancy* of local $U(1)$ phase assignments on a lattice.

To provide a rigorous setting for this maximal abelian regime, we explicitly construct the corresponding Lie groupoid.

Definition 3.8 (S-decorated Lie Groupoid). If $\mathcal{G} \rightrightarrows M$ is a Lie groupoid and S is a Lie group, an *S-decorated Lie groupoid* is a smooth functor

$$d : \mathcal{G} \rightarrow \mathbf{BS},$$

where $\mathbf{BS}$ is the one-object Lie groupoid with endomorphism group S .

Definition 3.9 (Underlying Lattice/Graph Datum). Fix a finite connected graph

$$\Gamma = (V, E),$$

for example, a finite patch of a GL , where V is the set of sites and E the set of undirected adjacency links. Choose an orientation on each undirected edge, written as a subset

$$E^+ \subseteq \vec{E},$$

so that every edge $\{i, j\} \in E$ appears exactly once as either $e : i \rightarrow j$ or $\bar{e} : j \rightarrow i$ in E^+ . Write $s(e), t(e) \in V$ for the source and target of an oriented edge e .

Definition 3.10 (Core Configuration Space in the QF Regime). The quasi-fluidic core state of the lattice is a *link phase assignment*

$$M_{\text{QF}} := U(1)^{E^+} \cong (S^1)^{|E^+|}.$$

An element $U \in M_{\text{QF}}$ is a family $U = (U_e)_{e \in E^+}$ with $U_e \in U(1)$. The value on the oppositely oriented edge is determined by inversion:

$$U_{\bar{e}} := U_e^{-1}.$$

Remark 3.7 (Chirality as Orientation Convention). In this quasi-fluidic interpretation, chirality is *not* a separate state coordinate. Instead, the handedness of the underlying helix selects (or suggests) a preferred orientation for reading the link phase; reversing that convention replaces U_e by U_e^{-1} . Thus, the earlier symbolic chirality label is absorbed into the orientation convention on each link and disappears from gauge-invariant observables.

Definition 3.11 (Gauge Group and Action on Link Phases). Let the site gauge group be

$$G_{\text{QF}} := U(1)^V.$$

For $g = (g_i)_{i \in V} \in G_{\text{QF}}$ and $U = (U_e)_{e \in E^+} \in M_{\text{QF}}$, define the action

$$g \cdot U \in M_{\text{QF}}$$

by the standard link transformation law

$$(g \cdot U)_e := g_{s(e)} U_e g_{t(e)}^{-1}, \quad e \in E^+.$$

Definition 3.12 (QF Core Lie Groupoid: the Gauge Action Groupoid). The quasi-fluidic core Lie groupoid is the action groupoid

$$\mathcal{G}_{\text{QF}} := G_{\text{QF}} \ltimes M_{\text{QF}} \rightrightarrows M_{\text{QF}}.$$

An arrow is a pair (g, U) with $g \in G_{\text{QF}}$ and $U \in M_{\text{QF}}$. The source and target maps are

$$s(g, U) = U, \quad t(g, U) = g \cdot U,$$

and composition is

$$(h, g \cdot U) \circ (g, U) = (hg, U), \quad h, g \in G_{\text{QF}}.$$

For finite Γ , this is a finite-dimensional Lie groupoid.

Remark 3.8 (Gauge Invariants Live on the Quotient Stack). The physically meaningful quantities in the quasi-fluidic abelian regime are gauge-invariant functions on M_{QF} (equivalently, functions on the quotient stack

$$[M_{\text{QF}}/G_{\text{QF}}]).$$

The groupoid $\mathcal{G}_{\text{QF}}$ records *gauge redundancy*, that is, equivalences of description rather than physical time evolution. Dynamics—for example, the gradient flow of a local gauge-invariant energy built from plaquette terms—is additional structure: a G_{QF} -equivariant flow on M_{QF} that descends to the quotient. We treat this dynamical layer separately from the gauge groupoid itself; see also [CFP18] as reference.

The viewpoint adopted here is therefore two-tiered. At the general level, decorated groupoids package the local geometric and combinatorial data relevant to symbolic or hybrid computation. At the quasi-fluidic abelian level, this broad picture reduces to a concrete lattice $U(1)$ gauge groupoid whose arrows encode only local phase-frame redundancy. This distinction is intentional: transcendental or geometric data may decorate computation in general, whereas the $U(1)$ corridor isolates the simplest smooth abelian core on which the later return-map and phase-ledger constructions are built.

Conjecture 3.2 (Extended Grothendieck Homotopy Hypothesis). *Let $\mathcal{C}$ be a category of smooth or stratified manifolds equipped with additional geometric structure (for example, decorations by phase, twist, chirality, flow, or other local computational data), and let*

$$(M, \text{str}) \longmapsto \Pi_1^S(M, \text{str})$$

denote the corresponding decorated path groupoid (or, in higher form, decorated ∞ -groupoid) obtained by adjoining this structure to the ordinary homotopy data.

Then the following principle is expected to hold.

- (a) **Classical recovery.** *When the decoration is trivial, or factors entirely through the ordinary homotopy type, the decorated object reduces to the usual homotopy-theoretic one. In this sense, the classical Grothendieck Homotopy Hypothesis is recovered as the undecorated case.*
- (b) **Enriched classification.** *When the decoration records genuinely geometric or computational data not visible to the bare homotopy type, the resulting decorated groupoid refines the ordinary homotopy invariant. Thus, two spaces may be weakly equivalent as topological spaces while remaining inequivalent as decorated computational spaces.*
- (c) **Failure of undecorated classification.** *In general, the passage*

$$(M, \text{str}) \longmapsto \Pi_1(M)$$

forgets computationally relevant structure. A faithful classifier for such spaces, therefore, requires an enriched object, such as a decorated groupoid, Lie groupoid, or higher groupoid, carrying the additional geometric data explicitly.

- (d) **Computational interpretation.** *In favorable cases, decorated paths may be read simultaneously as homotopies and as computational processes: composition corresponds both to path concatenation and to execution of successive geometric updates, while invariance under decorated homotopy expresses computational equivalence.*

It would be amiss not to mention the Grothendieck Homotopy Hypothesis. However, while decorated groupoids should be viewed not as replacements for ordinary homotopy types, a natural extension of them is relevant when geometry carries computational meaning. This conjectural principle is the homotopy-theoretic background for the later in appendix A.8 when the decoration factors through ordinary homotopy, one expects to recover the classical cobordism-type picture, whereas non-factorizing decorations lead to the enriched computational case. This suggests a possible enrichment of the homotopy hypothesis, in which ∞ -groupoids may encode intrinsic computational or dynamical structure—a new bridge between geometry, logic, and computation foreshadowed by Grothendieck’s original vision [Gro21].

Definition 3.13 (Homotopy Computation). Fix a GCM lattice in the quasi-fluidic regime with configuration space M_{QF} , pulse map

$$P : M_{\text{QF}} \rightarrow M_{\text{QF}},$$

and readout

$$\text{Read} : M_{\text{QF}} \rightarrow B^V.$$

A *homotopy computation* is a finite or continuous state evolution

$$\gamma : [0, 1] \longrightarrow M_{\text{QF}}$$

such that γ is generated by admissible local dynamics of the model (for example, gradient flow segments, pulse updates, or their hybrid composition), together with its induced readout trajectory

$$\text{Read} \circ \gamma : [0, 1] \rightarrow B^V.$$

Two homotopy computations $\gamma_0, \gamma_1 : [0, 1] \rightarrow M_{\text{QF}}$ with the same endpoints are said to be *computationally equivalent* if there exists a homotopy

$$H : [0, 1] \times [0, 1] \longrightarrow M_{\text{QF}}$$

such that:

- (a) for each fixed $u \in [0, 1]$, the path $t \mapsto H(t, u)$ is again an admissible computation,
- (b) $H(t, 0) = \gamma_0(t)$ and $H(t, 1) = \gamma_1(t)$,
- (c) the relevant observable data are preserved along the deformation (for example, the same decorated invariants, or the same induced symbolic readout up to the chosen notion of equivalence).

Interpretation. This definition is intentionally regime-dependent. In the symbolic regime, a homotopy computation reduces to a path through a finite transition graph. In the quasi-fluidic regime, it is a path in the continuous phase configuration space M_{QF} , typically observed only through the stroboscopic return map and the readout Read . Thus, the same underlying idea—computation as an admissible path modulo invariant-preserving deformation—specializes to both discrete and continuous settings.

Remark 3.9. The point of the terminology is not that every computation is literally a topological homotopy, but that admissible evolutions in configuration space form path-like objects whose equivalence is controlled by the invariants one chooses to preserve. In the abelian quasi-fluidic corridor, these are typically gauge-invariant phase observables and their induced symbolic readouts.

Lie algebroid chart and Atiyah’s linearization of flow. Readers from differential geometry may reasonably ask why we do not take Lie algebras and Lie brackets as the primary algebraic language throughout. The short answer is that our computational semantics is organized by *global, composable transport data* (paths, holonomy, winding/phase, and gluing across interfaces), and these are most naturally encoded at the groupoid level rather than by infinitesimal generators. Nothing here rejects the Lie-algebraic viewpoint; rather, it appears as a *linearization chart*. Given a Lie groupoid $\mathcal{G} \rightrightarrows M$, its infinitesimal object is the Lie algebroid

$$A(\mathcal{G}) := \ker(ds)|_M \longrightarrow M, \quad \rho := dt|_{A(\mathcal{G})} : A(\mathcal{G}) \rightarrow TM,$$

whose bracket on sections encodes first-order commutator information of local bisections. This is the many-object analogue of “Lie algebra = infinitesimal Lie group” [Wei96]. Thus, whenever one restricts to a sufficiently smooth regime where a genuine local symmetry pseudogroup governs evolution, one can pass from the groupoid semantics to an algebroid bracket calculus.

Atiyah’s gauge-theoretic instance. In particular, for a principal G -bundle $P \rightarrow M$, the gauge groupoid $(P \times P)/G \rightrightarrows M$ has associated Lie algebroid

$$A \cong TP/G \rightarrow M,$$

fitting into the short exact “Atiyah sequence”

$$0 \rightarrow \text{ad}(P) \rightarrow TP/G \rightarrow TM \rightarrow 0,$$

introduced by Atiyah in his study of connections. This is precisely the setting in which the Lie bracket formalism is maximally effective: it governs infinitesimal deformations of connections and local curvature evolution [Ati57].

Why we do not build the theory around brackets. The point of the “algebra of flow” emphasized here (4.2) is that many of the invariants relevant to computation are inherently global and persist under operations for which infinitesimal linearization is either insufficient or not stable. Accordingly, we treat

Lie algebroids (and brackets) as a *derived* local chart available in the upcoming fluidic/gauge regimes, while keeping the groupoid/decorated-path level as the primitive semantic layer. Unlike Lie algebras, Lie algebroids need not integrate to global Lie groupoids without additional conditions; integrability is controlled by explicit obstructions [CF03]. This reinforces the methodological choice: our semantics is formulated on the global/compositional side, with infinitesimal calculus used when (and only when) the regime admits it.

This work presents one possible homotopy-theoretic reading of geometric computation, offered in the hope that it may be refined, challenged, or extended by the mathematical, computer science, and physics communities.

3.6 Connections and Phase Ledgers on a Trivial Helical Bundle

The finite twin-helix module used in the basic construction is locally modeled by a regular neighborhood of an embedded arc, and is therefore topologically

$$D^2 \times I \cong B^3.$$

Thus, as a D^2 -bundle over the interval I , the local helical model is topologically trivial. In particular, there is no intrinsic bundle monodromy in the D^2 fibers. This does *not* mean that the geometry is computationally trivial. Rather, it means that the information relevant to computation is not carried by the bundle class itself, but by additional geometric structure placed on this trivial scaffold: chosen phase fields, connections, deformations, and local interface rules.

This is the correct abelian analogue of the Aharonov–Bohm (AB) viewpoint [AB59]. In the AB effect, the underlying space may be locally simple while the physically meaningful information is carried by phase accumulation of a $U(1)$ connection around loops. The same lesson applies here: even on a topologically trivial helical module, one may impose a phase field on the transverse circular directions in each cross-section, and the resulting loop integral defines a nontrivial $U(1)$ response. Thus, the relevant invariant in the present setting is not topological twisting of the bundle, but phase accumulation determined by the chosen geometric control data.

We therefore use the above product structure to isolate the abelian structure that will persist in later sections: an S^1 -valued phase accumulated around transverse circles in a fixed cross-section. In modern language, this is most naturally described by a $U(1)$ connection on a trivial circle bundle over such loops (equivalently, a Hermitian line bundle with unitary connection), although in the present discussion only the elementary notions of phase, curvature, and holonomy are required. As an organizational comparison, the product structure also means that local differential data may be written globally in coordinates; in this limited sense, one avoids the extra cocycle bookkeeping that would arise on a genuinely nontrivial fibration.

Fix a longitudinal parameter value $z_0 \in I$, and work in the transverse disk

$$D^2 \times \{z_0\}.$$

For each radius r for which the circle

$$\gamma_r(\theta) := (r, \theta, z_0), \quad \theta \in [0, 2\pi],$$

lies in the interior of the disk, let

$$A := \Omega(r, \theta) d\theta$$

be a $U(1)$ connection one-form on that transverse slice. Define the associated phase ledger by

$$U(r) := \exp\left(i \int_{\gamma_r} A\right) = \exp\left(i \int_0^{2\pi} \Omega(r, \theta) d\theta\right) \in U(1).$$

Proposition 3.3 (Radial Shear Identity). *Let*

$$A := \Omega(r, \theta) d\theta \quad \text{and} \quad F := dA = \partial_r \Omega(r, \theta) dr \wedge d\theta$$

on a fixed transverse slice $D^2 \times \{z_0\}$. For $r_1 < r_2$, let γ_{r_1} and γ_{r_2} denote the circles of radii r_1 and r_2 , and let

$$\mathcal{A}_{r_1, r_2} := [r_1, r_2] \times S^1$$

denote the annulus between them. Then

$$U(r_2)U(r_1)^{-1} = \exp\left(i \int_{\mathcal{A}_{r_1, r_2}} F\right).$$

Equivalently, the relative phase between the two radii is determined by the curvature flux through the annulus. In particular, if $\partial_r \Omega \equiv 0$ (no radial shear), then $F = 0$ on the annulus and $U(r)$ is independent of r .

Proof. By definition,

$$U(r_k) = \exp\left(i \int_{\gamma_{r_k}} A\right), \quad k = 1, 2.$$

Hence

$$U(r_2)U(r_1)^{-1} = \exp\left(i \int_{\gamma_{r_2}} A - i \int_{\gamma_{r_1}} A\right) = \exp\left(i \int_{\partial \mathcal{A}_{r_1, r_2}} A\right).$$

Applying Stokes' theorem on the annulus gives

$$\int_{\partial \mathcal{A}_{r_1, r_2}} A = \int_{\mathcal{A}_{r_1, r_2}} dA = \int_{\mathcal{A}_{r_1, r_2}} F,$$

which proves the identity. If $\partial_r \Omega \equiv 0$, then $F = 0$ and therefore the phase ledger $U(r)$ is constant in r . $\square$

Example 3.3 (Gluing and Factorization of Phase Ledgers). Let A_1 and A_2 be two annular regions in transverse slices, each equipped with a $U(1)$ connection one-form

$$A^{(1)} = \Omega_1(r, \theta) d\theta, \quad A^{(2)} = \Omega_2(r, \theta) d\theta,$$

defined in compatible boundary trivializations. Suppose the outer boundary circle of A_1 is identified with the inner boundary circle of A_2 so that the connection data glues smoothly across the common interface. Then the glued annulus

$$A_{\text{gl}} = A_1 \cup A_2$$

inherits a connection A_{gl} obtained by patching $A^{(1)}$ and $A^{(2)}$.

Let γ_1 and γ_2 denote the fundamental loops in A_1 and A_2 , and let γ_{gl} denote the corresponding loop in the glued annulus. The associated holonomies satisfy the factorization law

$$\text{Hol}_{A_{\text{gl}}}(\gamma_{\text{gl}}) = \text{Hol}_{A^{(2)}}(\gamma_2) \text{Hol}_{A^{(1)}}(\gamma_1).$$

Equivalently, writing

$$U_k := \exp\left(i \int_{\gamma_k} A^{(k)}\right), \quad k = 1, 2,$$

one has

$$U_{\text{gl}} = U_2 U_1.$$

Likewise, if

$$F^{(k)} = dA^{(k)}, \quad k = 1, 2,$$

are the curvatures on A_k , then the curvature on the glued annulus is

$$F_{\text{gl}} = dA_{\text{gl}},$$

and Stokes' theorem gives additivity of flux:

$$\int_{A_{\text{gl}}} F_{\text{gl}} = \int_{A_1} F^{(1)} + \int_{A_2} F^{(2)}.$$

Thus, in the abelian $U(1)$ corridor, phase accumulation behaves exactly as a ledger: holonomies multiply under composition, while curvature contributions add under gluing.

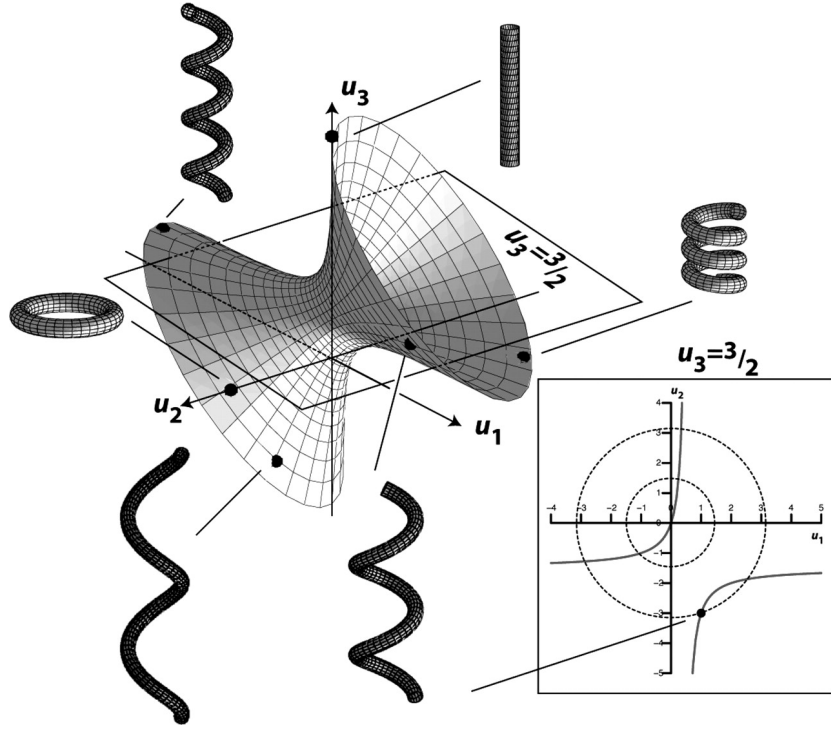


Figure 4: Homology, Helicity, and Hyperbolicity: a geometric “Rosetta Stone.” (Image Courtesy of Alain Goriely, “*Helices*” 2006)

3.7 Lancrét’s Ghost?

Helices are mathematical objects rich in dynamical variables—as explored in this section—whose properties interact synergistically but without the constraints of maximal symmetry. This flexibility has arguably obscured their structural significance. A helix is not merely a curve in space, but a framed, oriented 1-manifold carrying both axial and cyclic data. As earlier sections suggest, maximal symmetry tends to suppress geometric expressiveness, whereas physically useful computation requires enough asymmetry to support nontrivial state encoding and controlled transitions. Helices can dance with ones and zeros, but a sphere will never lie.

The helix presents a unique challenge in mathematical physics: despite its ubiquity in nature—from DNA, to protein folding, to solitons—it resists easy classification in modern foundational frameworks. The standard helix relies on *transcendental* functions (e.g., $\gamma(t) = (a \cos t, a \sin t, bt)$), and as a result, it falls outside the traditional scope of algebraic geometry, which deals natively with polynomial varieties. As a result, the helix constitutes something of a disciplinary orphan: too smooth for pure combinatorics, too transcendental for algebraic geometry, and too continuous for discrete computation. Yet, for a theory of physical computation—especially one grounded in constructible geometric objects—this specific structure must take center stage.

Emergence of helices in nature. Helices occur constantly in physics, chemistry, and biology precisely because they integrate two forms of structure at once: cyclic phase and longitudinal transport. This tension may even be computationally significant. Recent work on the internal representations of large language models [KT25] suggests that gradient-based optimization can give rise to helical or helix-like internal organization for arithmetic tasks. Though such findings do not admit a universal interpretation, they at least reinforce the plausibility that helices are not merely common physical shapes, but natural carriers of computation whenever cyclic and linear degrees of freedom must be coordinated.

And while helices offer remarkable efficiency for encoding computational structure, it is essential to note that not all helices in nature are used for computation, nor do they dominate large-scale architectures. Helical forms excel at packing information, facilitating transmission, and exploiting symmetry at small scales (e.g., DNA and chiral molecules). However, at larger scales, nature often favors other

symmetries—branching, fractal, planar, or modular structures—driven by stability, energetic, and functional constraints. Thus, the computational potential of helices arises not from their ubiquity but from their unique ability to localize and propagate information under appropriate conditions.

The helical moduli problem. Lancret’s theorem (1802) already hints that “being helical” is not a local accident: a space curve $\gamma \subset \mathbb{R}^3$ is a circular helix iff the Frenet invariants satisfy $\tau/\kappa \equiv C$. Equivalently, helices are integral curves of a geometric constraint in the 2-jet bundle of immersed curves [Bar97, Sau89]. In our setting, however, the computational objects are *thickened* and *decorated*: a helical *spine* carries a framed tubular neighborhood (cross-section data), marked boundary ports for gluing, and (in higher regimes) connection/holonomy data. The natural “moduli space” is therefore not a single finite-dimensional manifold, but a quotient of an infinite-dimensional configuration space by reparametrization and symmetry. Once one allows variation in framing, cross-section, pitch, chirality, and coupling data, the resulting parameter spaces become extremely large.

Definition 3.14 (Helical Moduli Stack, Working Model). Fix a smooth compact *template* helical body H (e.g. a capped tube with marked base disk and chosen framing class) and a regularity class C^k (or Sobolev H^s) appropriate to the regime. Let $\text{Emb}_{\text{fr}}^k(H, \mathbb{R}^m)$ be the space of framed C^k -embeddings, and let $\text{Diff}_{\text{fr}}^k(H)$ act by reparametrization (preserving the marked boundary structure), together with the ambient Euclidean group $E(m)$ acting by rigid motion. We define the moduli *stack*

$$\mathfrak{M}_H^k := \left[\text{Emb}_{\text{fr}}^k(H, \mathbb{R}^m) / (E(m) \times \text{Diff}_{\text{fr}}^k(H)) \right],$$

and its coarse moduli space M_H^k as the corresponding orbit space when it is well-behaved. Allowing variable cross-sections amounts to fibering this construction over a shape space of embedded disks (or contours), which increases the functional degrees of freedom.

This formulation is concrete enough to state what must be constructed: local models (Banach/Fréchet charts depending on k or s), a stratification by singular phenomena (self-contact, self-intersection, knotting/linking type, degeneration to the torus-like and straight-rod limits), and gluing maps induced by the marked boundary ports. In particular, the difficulty is not naming the quotient; it is extracting computable invariants from it in a way that is stable under deformation and compatible with composition [EM70].

Once one treats computation or physics as *local-to-global* with observables defined locally on helical pieces and composed by gluing, one is forced toward an integration/pushforward formalism on a stratified quotient object. In other words, the analogue of a path integral or partition function in geometric computation is a functorial assignment built from (a) passage to moduli (quotienting by symmetry), and (b) a gluing law that behaves like a Fubini theorem for composition. Developing such an “integration on moduli of helical histories” is exactly the point at which one should expect derived/stacky and constructible techniques to enter, much as they do in Kontsevich-style theories of integration over highly singular, infinite-dimensional parameter spaces [Met03].

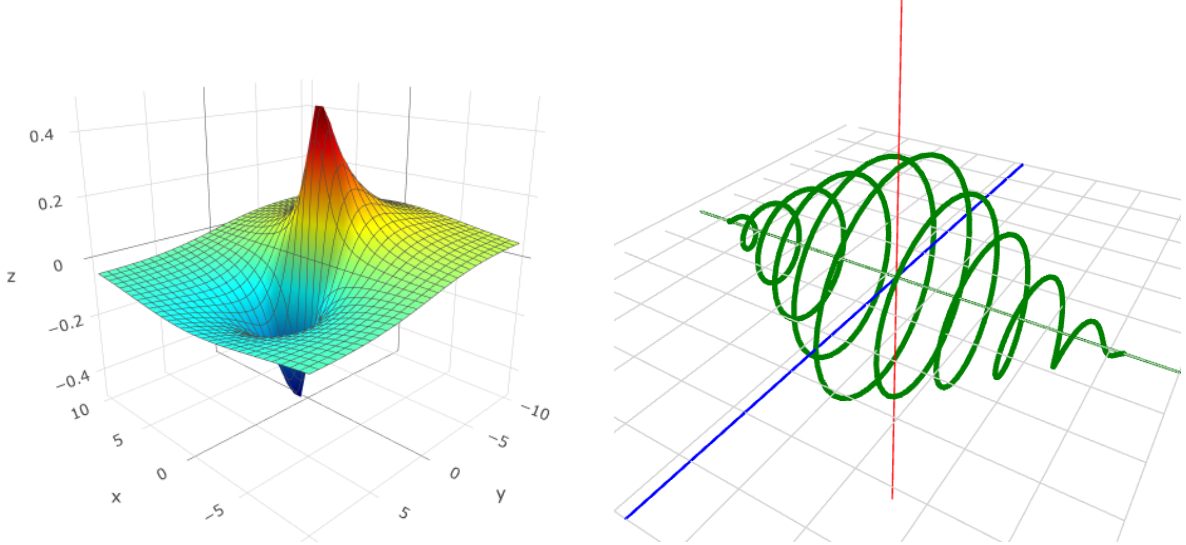
Overlooked but important reading on helicity. For a continuum-mechanics foundation on why helices are generic energy minimizers (and how curvature/torsion couple), see Gorieli [Gor17]; for a crisp, canonical derivation of when minimizers are helical with pitch/radius set by constraints, see Chouaieb et al. [CGM06]. Developable-ribbon energetics and torsion-curvature locking—which motivate “channel” formation in cross-sections—are captured by Starostin van der Heijden [SvdH07]. On the geometric side that is often missed, Adachi [Ada09] provides a leading example of how (helical) Killing fields and foliations structure flows, aligning with our base-join transport picture. Ferrández Izquierdo’s recent monograph [FI25] will also provide a focused, modern treatment of helical geometries to unify these threads for newcomers.

For all we know? Perhaps the key to modern quantum theory lurks quietly in the diagrams of a century-old German treatise [Lor02] on transcendental curves—waiting for someone to notice that helices and hyperboloids speak a universal geometric tongue.

4 Geometric Computational Field Theory

With the geometric and computational preliminaries now in place, we turn to the intersection of modern quantum physics. The aim of the present section is not to reopen the full problem of quanti-

zation, but to show that the structures developed above—holonomy, bordism, curvature, chirality, and coarse-grained state spaces—naturally dovetail with questions that have also appeared in the work of Kontsevich [Kon03], Connes [Con00], and Witten [Wit89], among others. Our claim is not that these programs are identical, but that they touch a common boundary at which geometry, topology, and computation can no longer be cleanly separated. It is from that boundary that the final section proceeds.



(a) BPST instanton: a finite-action, localized topological solution to the Yang–Mills equations, exhibiting geometric twist and winding [BPST75].

(b) Complex quantum wave: a superposition of oscillations, illustrating the intertwined, helical modes of computation.

Figure 5: Familiar examples of geometric configurations corresponding to Type 3 and Type 1 *THM*.

4.1 Curvature as Computation

After assembling *THM* in the abelian, quasi-fluidic band—where computation is well captured by finite local operations on geometric primitives—we now enter the *fully fluidic* regime. Regard the helix lattice as a bundle with a connection. Let

$$\pi : P \rightarrow B$$

be a principal G -bundle over a computably presented base B (the working lattice region), with associated state bundle $E := P \times_G F$ whose fiber F encodes the finite-dimensional internal variables of a twin helix cell. A connection $A \in \Omega^1(B; \mathfrak{g})$ prescribes parallel transport, and its curvature

$$F = dA + A \wedge A \in \Omega^2(B; \mathfrak{g})$$

measures path-dependence. In this language, a *join* is no longer purely combinatorial: it is the composition of parallel transports along adjacent interfaces. When G is abelian (e.g. $U(1) \simeq SO(2)$), joins commute; once G is non-abelian, joins become order-sensitive and composition acquires memory.

Chern–Weil readouts. Classically, Chern–Weil theory [BT82] converts the curvature of a connection into characteristic classes: invariant polynomials in F_A , such as $\text{tr}(F_A \wedge F_A)$, define closed differential forms whose cohomology classes are independent of the chosen connection. Thus, curvature is not merely local analytic data; it supplies stable global invariants of the bundle. In the present work, Chern–Weil theory gives the standard mathematical instance of curvature-as-readout: local path-dependence of geometric transport is compressed into deformation-stable global quantities. Chern–Simons theory then appears as the corresponding transgression/boundary ledger measuring how such characteristic data change across a history.

Example 4.1 (Holonomy as Computational Memory). For a loop γ based at $x \in B$, the parallel transport

$$U_\gamma = \mathcal{P} \exp \left(\int_\gamma A \right) \in G$$

is the *holonomy*. The subgroup $\text{Hol}_x(A) \subseteq G$ generated by all such U_γ is the effective symmetry seen at x . In the abelian limit $U_{\gamma_1} U_{\gamma_2} = U_{\gamma_2} U_{\gamma_1}$ and only additive flux matters; in the non-abelian case, path ordering is essential and different execution paths may yield nonconjugate elements, even for homotopic loops when $F \neq 0$. The gauge-invariant readouts are the *conjugacy classes* $[U_\gamma]$ (or class functions such as $\text{tr}_\rho(U_\gamma)$). The following now hold:

- (a) **Registers as loops.** Minimal cycles in the lattice carry holonomies $\{U_{\gamma_k}\}$ serving as state registers; the state is the family $\{[U_{\gamma_k}]\}$.
- (b) **Updates as transport.** Driving fields update A and hence $\{U_{\gamma_k}\}$; gluing cells changes the loop basis.
- (c) **Curvature as storage.** F encodes what cannot be gauged away—persistent, path-dependent context (geometric memory).

This is the geometric analogue of Wilson loops [Wil74]: the observable is the holonomy, *not* the potential itself. In the quasi-fluidic abelian band, holonomy reduces to a scalar phase (orientation logic, integer twist counters). Entering the fluidic/gauge band promotes these to matrix phases (non-abelian Berry/Wilczek-Zee transport) [Ber84, WZ84], and join symmetry breaks via $[U_{\gamma_1}, U_{\gamma_2}] \neq e$.

From abelian phases to non-abelian transport. Each twin helix cell has a finite internal state space; as parameters vary over B , these states form a vector bundle $E \rightarrow B$ (the *band bundle*). Adiabatic transport induces a connection on E . If $\text{rk } E = 1$ over a region, the induced connection is abelian (Berry phase), hence commuting joins. When degeneracy persists ($\text{rk } E \geq 2$ on an open set), the connection becomes non-abelian (Wilczek-Zee): parallel transports no longer commute and path ordering becomes a resource. Base anisotropy and helix-helix coupling are precisely the mechanisms that create protected degeneracy and thus promote $U(1)$ to $SU(2)$ -type dynamics.

Definition 4.1 (Join Commutator in the Interface Groupoid). Let $\mathcal{J}$ be the interface groupoid whose objects are boundary faces and whose morphisms are admissible joins (parallel transports along interfaces). For $J_1, J_2 \in \text{Hom}_{\mathcal{J}}(X, Y)$ with common source/target, set

$$[J_1, J_2] := J_1 J_2 J_1^{-1} J_2^{-1}.$$

Joins *commute* if $[J_1, J_2] = \text{id}$ in $\text{Hom}_{\mathcal{J}}(X, X)$.

Computation now proceeds by progressively internalizing curvature; symbolic joins become ordered transports, and the state is read off from conjugacy classes of holonomies. Expressivity exceeds purely symbolic models because time-ordering, loop creation/annihilation, and curvature redistribution generate behaviours not captured by classical finite-step semantics.⁶ Physically, the twin helix fiber is the matter degree of freedom, and A is the control field; noncommutativity (path dependence) turns the join algebra into a genuine gauge computation.

Spin textures as fluidic defects. A *THM* defines an internal orthonormal 2-frame; choosing a unit normal gives a map $n : B \rightarrow S^2$. Its defects organize memory channels:

- **Vortex/phase defects** (*abelian*). Windings in $U(1)$ along loops (π_1) ,
- **Skyrmion-type textures** (*non-abelian*). Degree of n on embedded 2-spheres $(\pi_2(S^2) \cong \mathbb{Z})$,
- **Linked textures** (*Hopf-type*). Linking of preimages $n^{-1}(p)$ and $n^{-1}(q)$ for $p \neq q$ $(\pi_3(S^2) \cong \mathbb{Z})$ in 3D thickenings).

⁶We refrain from machine-model claims. Under continuum idealizations, differential dynamics can exhibit noncomputability phenomena; our interest here is programmability via $(A, F, \text{Hol}(A))$.

These classes appear in the holonomy spectrum: abelian vortices shift scalar phases; non-abelian textures produce non-conjugate holonomies around homologous loops, yielding distinct class–function readouts. The central split in summary:

- (a) **Symbolic/Quasi–fluidic.** G is effectively $U(1)$, joins commute, memory is a scalar phase or integer flux.
- (b) **Fluidic/Gauge.** G is non-abelian (from helix symmetry and interface groupoid), joins do not commute, memory lives in holonomy conjugacy classes; curvature stores context.

The base topology controls available memory channels: on contractible B with $F=0$ holonomy is trivial up to conjugacy; nontrivial F , defects, or handles produce noncommuting generators. In the terms of lemma 2.4 and table 7, flipping across from the quasi–fluidic abelian band to the fluidic/-gauge band trades raw orientation microstates for noncommuting holonomy, where the true expressive capacity of geometric computability is realized. Fully fluidic GCMs begin to resemble endless streams of information as increasingly chaotic computational phenomena appear.

Example 4.2 (Spectral Flow as a Computable Readout). Driving the system around a control cycle S^1 promotes B to $B \times S^1$. The net change in holonomy class across a period is governed by the integral of curvature over spanned 2–cycles (a Thouless–type pump):

$$\Delta(\text{class function on } U_\gamma) = \int_{\Sigma \subset B \times S^1} \text{tr}(F) \quad (\text{abelian}), \quad \Delta(\text{spec}(U_\gamma)) \text{ by } \int \text{tr}(F \wedge F) \quad (\text{non-abelian}).$$

Thus, periodic control implements a bordism in parameter space, and evaluation of class functions on Wilson loops gives an integer (or discrete spectrum) that is stable and algorithmically retrievable in our symbolic presentation.

Symplectic vs. contact presentation. Although we have not formulated the above regime in the standard symplectic/contact dialect, much of the structure we use admits a direct translation. A fully fluidic “state space” is naturally modeled as a symplectic manifold (X, ω) (phase space), with evolution given by Hamiltonian dynamics or a controlled perturbation thereof; see [MS17]. When one restricts to an energy hypersurface $\Sigma \subset X$ of *contact type* (equivalently: there exists a primitive λ with $d\lambda = \omega$ near Σ such that $\alpha := \lambda|_\Sigma$ is a contact form), the induced characteristic dynamics on Σ is canonically described by the *Reeb vector field* R_α determined by

$$\alpha(R_\alpha) = 1, \quad (d\alpha)(R_\alpha, \cdot) = 0$$

as in [Gei08]. In this setting, Hamiltonian flow restricted to Σ is a reparametrization of Reeb flow: for suitable defining Hamiltonians, one has

$$X_H|_\Sigma = f \cdot R_\alpha \quad \text{for some positive function } f,$$

so closed Hamiltonian characteristics on Σ are the same as periodic Reeb orbits.

Periodic cycles as discrete readouts. From this perspective, the “periodic control cycles” and robust closed trajectories used as readouts in our fluidic discussion can be viewed (after choosing a contact-type boundary chart) as periodic Reeb orbits and their associated action/index data. This is precisely the regime in which symplectic/contact methods (Floer-theoretic and SFT-type constructions) extract deformation-invariant algebraic information from Reeb dynamics [EGH00]. We do *not* invoke the full machinery of symplectic field theory here; the point is only that the fluidic limit admits a recognized contact-geometric presentation, and that the periodic-orbit language provides a standard interface for readers coming from Hofer-style dynamics.

Interpretation. In our terminology, passing to a contact/Reeb chart is best viewed as a *change of presentation* (a local coordinate system on the same process space) rather than an additional ontological assumption. It supplies a conventional vocabulary for “time-like” evolution on contact-type boundaries and for the discrete invariants carried by periodic cycles.

Computational notion	Gauge-theoretic realization
Local state description	Choice of local frame or gauge representative
Interface update	Parallel transport across an interface
Sequential execution	Composition of transport operators
Closed execution path	Loop in the base or interface groupoid
Gauge-invariant readout	Class function of holonomy, such as a Wilson-loop trace
Locally represented internal data	Gauge-dependent connection or state representative
Persistent computational memory	Holonomy conjugacy class or curvature-supported sector
Equivalence of descriptions	Gauge transformation preserving invariant observables

Table 8: A computational dictionary for gauge transport. The correspondence is operational rather than merely analogical: local descriptions may depend on a chosen frame, while the observable computational output is extracted from gauge-invariant functions of the resulting transport.

4.2 The Algebra of Flow

Starting in the quasi-fluidic regime, interactions are not purely symbolic: they are geometric and dynamical—the information is encoded in the differential structures of both flow and form. In the fully fluidic regime, the cross-sectional geometry, as discussed in section 2.6, reappears. Why the focus?

Two kinds of analog computation. Computation as we know it splits as such:

- (*Algebraic analog flow*) Rank-1 bundles with abelian holonomy ($U(1)$): states are continuous but updates commute; residual symmetry imposes selection rules; behavior is predictable.
- (*Geometric analog chaos*) Rank- ≥ 2 degeneracy with non-abelian holonomy ($SU(2)$ -type): path ordering matters; joins do not commute; internal mixing encodes control flow; behavior is history-sensitive. In Felleisen’s sense [Fel91], these differ at the level of *core constructs*, not mere “data types.”

Symmetry selection rules (abelian kinematics). When a local annulus $A \simeq S^1 \times I$ around a helix cell carries a residual C_n symmetry, transport admits a harmonic decomposition $e^{im\theta}$ in which the connection couples only modes $m \equiv m' \pmod{n}$. Transport block-diagonalizes by angular-momentum class modulo n , and the induced holonomies on each block commute. In this quasi-fluidic band, joins commute and memory reduces to scalar phases or integer twist counters. Breaking C_n (or leaving a round annulus) removes these selection rules; see appendix A.9 for the calculation.

Promoting $U(1) \rightarrow SU(2)$ by degeneracy. Helix-helix coupling plus base anisotropy can maintain a rank- ≥ 2 degenerate internal subspace across a region of B . Adiabatic transport on this subbundle is then *non-abelian* (Wilczek-Zee): path-ordered products of transports do not commute, and the state carried by a loop is a conjugacy class $[U_\gamma]$ rather than a single phase. Practically, anisotropy supplies off-diagonal couplings while coupling between the twin strands stabilizes the degeneracy that upgrades $U(1)$ to $SU(2)$ -type dynamics. Let two loop transports act on a rank-two internal space by

$$U_x = \exp(i\alpha\sigma_x), \quad U_y = \exp(i\beta\sigma_y),$$

where (σ_x, σ_y) are Pauli matrices. Then

$$U_x U_y - U_y U_x - 2i \sin(\alpha) \sin(\beta) \sigma_z,$$

which is nonzero for generic (α, β) . Thus, traversing the (x) - and (y) - transport channels in opposite orders produces different internal states. By contrast, for $(U(1))$ -valued transports ($e^{i\alpha}$) and ($e^{i\beta}$), the two orders agree. The non-abelian holonomy, therefore, retains an intrinsic record of execution order.

Definition 4.2 (Macro-expressiveness). For two geometric computation models A, B , write $A \succeq_{\text{macro}} B$ if there is a finite library of local transports/joins/defect insertions in A whose *gauge-invariant readouts* (class functions on holonomy) can *macro-express* every primitive of B with bounded overhead, preserving observational equivalence at the boundary. Abelian $U(1)$ lacks path ordering and conjugation gates; the non-abelian regime supplies them, hence the jump in macro-expressiveness.

Base topology and gauge-invariant readouts. On contractible patches with $F = 0$, holonomy is trivial up to conjugacy, so no persistent state is stored. Nontrivial memory channels arise from curvature, defects, or handles. Gauge transformations act by conjugation on holonomies; the physically meaningful readouts are *class functions* of U_γ (e.g. traces in fixed representations).

Computational tradeoffs: algebraic vs. geometric analog.

- **Memory carrier.** *Algebraic analog* ($U(1)$): a scalar phase per loop. *Geometric analog* ($SU(2)$): a conjugacy class per loop (equivalently, spectra or traces $\text{tr}(U_\gamma), \text{tr}(U_\gamma^2), \dots$).
- **Registers and gates.** Registers are chosen loops $\{\gamma_k\}$. In $U(1)$ gates add phases (commuting joins); in $SU(2)$ gates are ordered transports with nontrivial commutators $[U_{\gamma_i}, U_{\gamma_j}] \neq \text{id}$.
- **Control knobs.** (a) introduce base anisotropy beyond the first harmonic; (b) maintain internal degeneracy (rank ≥ 2 band bundle); (c) add defects/handles so $\pi_1(B)$ generates noncommuting holonomies; (d) redistribute curvature to reprogram stored context.
- **Verifiability.** $U(1)$: readout is a phase/integer. $SU(2)$: readout uses class functions (e.g. traces of Wilson loops) that remain algorithmically retrievable in our symbolic presentation.

Since the base geometry controls capacity, on simply-connected regions with $F = 0$, holonomy classes are trivial; introducing curvature, handles, or protected degeneracy creates noncommuting generators and promotes the system from the quasi-fluidic abelian band to the fluidic/gauge band. This is the bridge where join algebra becomes order-sensitive and holonomy conjugacy classes serve as programmable memory.

Now the internal geometry of the cross-section becomes a new axis of calculation. Beyond circles, one can admit distinct flow channels, branch lines, and defect cores within the same cross-section; these change the *isotypic decomposition* of transport and create/maintain degeneracies. Higher symmetry (round annulus) yields fewer angular sectors and abelian blocks; broken symmetry and multi-channel shapes create protected mixing and non-commuting gates. Thus cross-section complexity controls the number and kind of “channels” available for computation: not just *which* states, but *how* transport moves *within* a state.

Remark 4.1. Only the geometric analog can outstrip algebraic field-operation models under standard readouts: chaotic/PDE dynamics can exceed polynomial/field-arithmetic semantics as shown by [PER79]. In our constructive calculus (section 4.5), such steps appear as *effects* with witnesses (no hidden oracles), preserving normalization and observational equivalence on gauge-invariant readouts.

4.3 Helicity and the Origin of Quantum Spin Networks

Although our program does not aim to adjudicate unification debates, aspects of the geometric computation model do resonate with the combinatorial, categorical, and background-independent *perspectives* emphasized in Loop Quantum Gravity [SMO04].

As developed throughout this work, helices exhibit exceptionally rich geometric behaviour. The pre-quantum computational geometry of twin helices—where information resides in flows, holonomies, and deformations—can supply a substrate from which discrete boundary data (as in spin-foam state spaces [Pen71, RS95]) may emerge after coarse-graining. Rather than postulating quantum discreteness as primitive, it is reasonable to view quantum structure as an invariant/stable phase of the underlying geometric dynamics, much as solitons or topological defects arise from continuous systems. In this picture, coarse-graining the micro-geometry of helices can yield spin-network-like graphs (cf. section 2.5); the effective pure-state manifold of a qubit is $\mathbb{CP}^1 \cong S^2$ (the Bloch sphere), while discreteness enters at the level of projective readout rather than the underlying state space.

Our viewpoint also aligns with strands of work beyond LQG: non-Markovian/correspondence ideas in quantum foundations [Bar23], relativistic quantum information [FSM05], the positive geometry

program in scattering amplitudes [AHT14], holonomy-first approaches [AG16], and deterministic explorations [tH16].⁷ In our reading, discrete quantum geometry then appears as a phase transition within a more general continuous geometric logic—without discarding the manifold substrate.

Example 4.3 (The Quantum Tetrahedron). Fix a working region B and a principal G -bundle $P \rightarrow B$ with connection A , as in the fluidic discussion. Choose a cellular decomposition dual to the helix lattice (plaquettes/handles adapted to the THM_4 pattern), and let Γ be the resulting 4-valent graph:

- **Edges** $e \in E(\Gamma)$. Homology classes of minimal loops threading twin helix channels. The nonabelian holonomy U_e around e has conjugacy class $[U_e]$; its spectrum (eigenphases) is the gauge-invariant “register value.”
- **Edge labels.** Are extracted by coarse-graining class functions of U_e . For $G = SU(2)$, the spectrum organizes into unitary irreps; we record the isotypic content by an $SU(2)$ spin $j_e \in \frac{1}{2}\mathbb{N}$. Operationally, j_e is the minimal label whose character χ_{j_e} matches the measured class functions $\{\text{tr}(U_e^k)\}$ on e .
- **Vertices** $v \in V(\Gamma)$. These arise where four channels meet (the V_4 cell). Gauge covariance imposes the Gauss law $\prod_{e \ni v} U_e \simeq \text{id}$ (up to conjugation); in representation terms this singles out an $SU(2)$ intertwiner $\iota_v \in \text{hom}(\bigotimes_{e \rightarrow v} V_{j_e}, \bigotimes_{e \leftarrow v} V_{j_e})$.

Thus each coarse cell yields a labeled pair $(\Gamma, \{j_e\}, \{\iota_v\})$, i.e. a spin network state in the standard sense. No discreteness hypothesis was used: it is the *stability* of the holonomy conjugacy classes under small deformations (and the V_4 join rule) that makes the labels robust.

Choice of group: Z_4 vs V_4 ? The two independent binary symmetries of a twin helix cell (left/right, up/down) generate $V_4 \simeq \mathbb{Z}_2 \times \mathbb{Z}_2$. Classifying surface modes or boundary excitations by $\mathbb{Z}_4$ collapses these axes; phrasing them as V_4 cleanly factors the parity data that seed the four-valent node. In practice, this matches the base 2-geobit state space and yields a sharper route from micro-geometry to the 4-valent “quantum tetrahedron” intertwiners.

Spin foams as computational bordisms. Helix updates (gluing, defect motion, curvature redistribution) generate a 2-complex $\mathcal{F}$ whose faces track the evolution of edge holonomies and whose vertices encode local join moves. Coarse-graining sends such an update to a labeled 2-complex (a spin foam [Bae98]) whose faces carry j -labels and whose edges carry intertwiners, precisely the data of LQG transition amplitudes. In our setting, amplitudes arise from products of local class functions evaluated on Wilson loops, making the construction functorial with respect to computable bordisms. The “foam” is therefore an emergent bookkeeping of gauge computation, not a fundamental discretization of spacetime.

Spin networks correctly capture the *gauge-invariant* content (conjugacy classes and intertwiners) of nonabelian holonomy on coarse loops and junctions; spin foams capture its elementary updates. In the helix picture, these are *phase data of transport* through THM_4 cells, stabilized by join rules and curvature, rather than postulated “atoms of space.” Background independence is recovered at the coarse level because all readouts are class functions (insensitive to coordinate choices), even though the micro-dynamics live on a smooth 4-manifold.

Proposition 4.1 (Coarse-graining to Spin Networks). *There is a canonical coarse-graining functor*

$$\mathcal{C} : \frac{\text{helix configs on } (B, A)}{\text{computable gauge bordism}} \longrightarrow \frac{\text{spin networks } (\Gamma, \{j_e\}, \{\iota_v\})}{\text{graph isomorphism}}$$

sending minimal loop holonomies to edge labels and join constraints to intertwiners. Moreover, $\mathcal{C}$ is compatible with the composition of updates: a computable bordism of helix configs maps to a spin foam between the corresponding spin networks.

Proof sketch. Gauge-invariant readouts are class functions of holonomy; spectral data of U_e define j_e via the $SU(2)$ character table. Local gauge constraints at V_4 junctions select the invariant subspace, i.e. an intertwiner. Elementary updates act by conjugation and fusion/splitting, which descend to foam faces/edges. Functoriality follows from the locality of joins and the multiplicativity of class functions along concatenated loops. $\square$

⁷Regarding superdeterminism: GCT rejects this. Measurement settings are treated as exogenous boundary data; at the ontic level, this induces measurement dependence via a global, time-symmetric constraint, while linear/gauge symmetries preserve operational free choice and no-signalling.

Baez’s and Barrett’s Quantum Tetrahedra inside THM_4 At a single V_4 cell, four edge labels $\{j_1, \dots, j_4\}$ and an intertwiner ι define a quantum tetrahedron. In the helix model, these labels come from the spectra of the four minimal loop holonomies, while the join law enforces ι . The usual area/angle relations are thus coarse summaries of underlying nonabelian transport around the cell.

Physical evidence of note for GCT/LQG. Recent work by [BLLN19] has demonstrated that the surface Maxwell modes at media interfaces are governed by a topological invariant valued in $\mathbb{Z}_4$. Their approach, rooted in phase winding, represents a significant advance in our understanding of topological invariants in photonic systems. However, in our view, this classification does not fully account for the independent binary symmetries present in the physical configuration—specifically, the left/right and up/down combinations encoded in a THM_4 construction. While the choice of $\mathbb{Z}_4$ as an invariant is logical within their framework, a reformulation in terms of the Klein four-group V_4 more accurately reflects the underlying symmetry structure. The viewpoint directly parallels the base 2-geobit state space and LQG tetrahedra, providing both a more granular symmetry classification and a richer foundation for both physical insight and computational applications.

Summary. Spin networks/foams are not competitors to geometric computation; they *are* its coarse invariants. The helix micro-geometry supplies the nonabelian holonomy that the LQG combinatorics were designed to encode. In this sense, LQG’s discrete data are recovered as a stable phase of a continuous, programmable gauge dynamics, and need not be elevated to fundamental kinematics.

4.4 On the Expressive Power of Physical Computation

Before stating our final claim, we collect a few canonical gauge-theoretic solutions whose geometry already exhibits the helical/twisting features emphasized in this work. The point is not to assert equivalence, but to note recurring *mechanisms*—periodicity, compactification twists, unstable saddles, and chiral transport—by which the connection A accumulates phase or matrix holonomy along preferred directions.

- **Calorons.** Periodic instantons on $\mathbb{R}^3 \times S^1$, whose Euclidean time compactification enforces a helical/phase structure along the circle [Kv98].
- **Twisted solutions.** Gauge fields on compactified or cylindrical backgrounds with nontrivial boundary conditions; the twist enforces a globally enforced phase (or matrix) winding [tH79].
- **Sphalerons.** Unstable saddle points in electroweak theory; they sit atop energy barriers and encode large, topological reconfigurations (holonomy changes) along interpolation paths [KM84].
- **Witten’s twisted constructions.** Lower-dimensional/topologically twisted reductions where the effective degrees of freedom are organized by holonomy classes and cohomological data [Wit88].
- **Non-abelian plane waves.** Chiral, order-sensitive transport in non-abelian gauge theory; parallel transport along wavefronts produces noncommuting holonomies [Col77].

Taken together, these exemplars highlight three recurring themes: (a) *periodic/compactified directions* that act as helical axes (calorons, twisted sectors), (b) *unstable saddles* that mediate large holonomy moves (sphalerons), and (c) *intrinsically chiral, non-abelian transport* (plane waves). In what follows, we leverage exactly these mechanisms—now encoded by holonomy class functions on chosen loops—to motivate the usage of geometric computation.

While this discussion does not claim to exhaust the intricate connections between geometric computation and modern field theory, it is instructive (cf. fig. 5a) that gluing two helical manifolds along a common boundary produces patching data directly analogous to the clutching that underlies BPST instantons. We now enter the final of the four computational regimes, *emergent gauge*.

Each unit helix manifold H_A, H_B is modeled by open sets $U_A, U_B \subset M$ carrying local trivializations of a principal G -bundle $\pi : P \rightarrow M$. Gluing the helices along their common base is encoded by a transition map

$$g_{AB} : U_A \cap U_B \longrightarrow G,$$

which acts on the fibers; on the overlap, the base identification is the identity. In gauge theory, g_{AB} specifies how local data patches are related; geometrically, it records how internal frames are

rotated when crossing the interface. Deformations of the gluing (and of g_{AB}) change the induced holonomies on linking loops and thus realize discrete computational updates. In this sense, a unit helix behaves solitonically, and computation in this regime naturally traffics in the same patching/moduli structures that organize the instanton sector of Yang–Mills theory [YM54, ADHM78]. This positions geometric computability as a concrete starting point for a *computational Yang–Mills* framework: a GCM presented as a bundle with connection whose updates are local in the bundle charts and functorial under gluing.

$$\begin{array}{ccc}
\text{Local helix chart } (U_A) & \xrightarrow{g_{AB}} & \text{Local helix chart } (U_B) \\
\downarrow & & \downarrow \\
\text{Trivialization } P|_{U_A} & & \text{Trivialization } P|_{U_B}
\end{array}$$

The diagram summarizes the configuration space of gluings: once a connection is chosen and equations (e.g., ASD) are imposed, one arrives at a moduli space modulo gauge—the familiar organizing object for instantons.

Example 4.4 (Minimal Bundle–Connection Data). Let $P \rightarrow M$ be a principal G –bundle with local 1–forms $A_A \in \Omega^1(U_A, \mathfrak{g})$ and $A_B \in \Omega^1(U_B, \mathfrak{g})$ related on overlaps by

$$A_B = g_{AB}^{-1} A_A g_{AB} + g_{AB}^{-1} dg_{AB}, \quad F_B = g_{AB}^{-1} F_A g_{AB},$$

with $F_A = dA_A + A_A \wedge A_A$. Viewed as a time-bordism on $M \times I$ (temporal gauge), the BPST sector corresponds to $G = SU(2)$ and anti-self-dual curvature,

$$F + *F = 0.$$

On a compact 4-manifold $\widehat{M}$ (a compactification of $M \times I$), the instanton number is

$$k = \frac{1}{8\pi^2} \int_{\widehat{M}} \text{tr}(F \wedge F) \in \mathbb{Z},$$

and the boundary Chern–Simons functional changes by $\Delta \text{CS} = \int_{\widehat{M}} \text{tr}(F \wedge F) = 8\pi^2 k$ under standard normalizations.

In the abelian/quasi–fluidic limit $G = U(1)$, a loop $\gamma \subset M$ carries the computational phase

$$d_\tau([\gamma]) = \text{Hol}_\gamma(A) = \exp\left(i \int_\gamma A\right), \quad d_\chi \in C_2 \text{ (chirality),}$$

and these labels transform covariantly under a change of patching g_{AB} . Gluing two helices along $U_A \cap U_B$ modifies the transition map ($g_{AB} \mapsto h g_{AB}$) and hence the induced holonomies on linking loops—implementing a discrete update at the interface. In the $SU(2)$ sector, a computational step can be idealized as traversing an ASD configuration on $M \times I$; the induced change in the boundary Chern–Simons functional is governed by k , providing a topological gate in the emergent-gauge regime.

At the level of *computational role*, effectively abelian regimes correspond to the stable carriage of information: state is propagated in a periodic, order-insensitive, and comparatively reversible manner, so that the dominant behavior is storage, transmission, or recurrence rather than structural transformation. By contrast, non-abelian regimes are characterized by order-sensitive evolution, where the flow itself performs the computation: composition is no longer reducible to simple commutative propagation, and state is actively reshaped through mixing, routing, and nontrivial transition. In the highest regime—*emergent gauge*—computation becomes genuinely history-dependent and topological. There, the relevant logical content is carried not only by the states themselves but by the global manner in which one state is connected to another, so that transition processes—for example, sphaleron-type [Man19] barrier crossings—become part of the computation itself. The current hypothesis in this paper, that the geometry of a Yang–Mills field is, in all likelihood, some form of analog computation comprised of helices, is outlined in table 9.⁸

⁸As to why the Universe chooses to *express itself* this way? The author kindly asks that questions be directed elsewhere (cf. [Pen89]).

Table 9: Computational regimes in GCT and the geometric structures that realize them.

Regime	Computational signature	Primary state carrier	Topological role	Representative physical exemplars
Symbolic	Discrete, stable, addressable states; local read/write behavior; robust storage.	Local occupation values, boundary conditions, or effectively commuting observables.	Topology is present but inactive; it serves as a fixed background structure rather than as the active computational variable.	Classical bits, digital memory, effectively abelian electromagnetic storage modes.
Quasi-fluidic	Cyclic or clocked evolution; recirculating transport; finite memory carried through repeated traversal of a loop.	Holonomy, phase accumulation, or other history-bearing data associated with motion around a compact direction.	Topological content is accumulated and reassembled over a cycle rather than expressed as a single isolated transition.	Calorons, periodic instantons, finite-temperature Yang–Mills configurations with nontrivial holonomy.
Fluidic	Distributed continuous transformation; order-sensitive mixing; non-commutative transport across an extended medium.	Flow profile, relative ordering of non-commuting updates, chiral or braided propagation patterns.	Topology is secondary to transport dynamics; the main effect is transformation by extended flow rather than sector change.	Non-abelian plane waves, helical or twisted Yang–Mills solutions, propagating chiral field configurations.
Gauge	Global, history-dependent update between sectors; computation by topological event rather than by local rewriting.	Bundle class, winding data, Chern-type invariants, or other global sector labels.	Topology is primary and active: the computation is identified with tunneling, obstruction crossing, or sector change.	Instantons, sphalerons, vacuum tunneling configurations, topological transition events.

Interpretation of the regime labels. The classifications in table 9 are *computational* rather than group-theoretic. They do not assert that all examples in a given column are abelian or non-abelian as gauge fields. Instead, they classify the *role* played by the configuration in the computational picture.

In particular, calorons are genuinely non-abelian Yang–Mills configurations, but they are placed in the quasi-fluidic regime because their operative structure is cyclic and holonomy-driven: state is transported around a compact S^1 direction, and memory is encoded by recurrence and phase accumulation. By contrast, instantons appear in the gauge regime when the emphasis is on isolated topological sector change.

Example 4.5 (The Computational Consequences of Non-Abelian Plane Waves). As a motivating example of one solution cited above, consider the appearance of non-abelian plane waves:

$$L = -\frac{1}{4}\text{tr}(F_{\mu\nu}F^{\mu\nu}) = \frac{1}{2}\sum_{a=1}^n [(E^a)^2 - (B^a)^2] =: \frac{1}{2}(E^2 - B^2).$$

This is discussed by [Kas83], where he notes (with some exasperation) that plane-wave solutions with equal electric and magnetic energy densities satisfy $E^2 = B^2$ and hence $L = 0$. In the non-abelian case, such waves carry internal structure: along the propagation direction, the gauge potential undergoes nontrivial parallel transport, yielding non-vanishing holonomy in the gauge group. Equivalently, the field configuration traces a nontrivial path in the group manifold even though the local scalar invariant $E^2 - B^2$ vanishes.

Within a geometric–computational viewpoint, this slice $L = 0$ may be regarded as a *switching hypersurface* in configuration space: crossing it changes the group-theoretic state realized by the parallel transport while preserving the light-like character of the wave. The family of such plane waves thus furnishes a controlled alphabet of *signal trajectories*: each solution implements a definite group action (holonomy) along its world-volume, so compositions of waves correspond to compositions of group elements, with path ordering in the non-abelian case. In this sense, classical Yang–Mills

solutions encode a concrete logic of state transitions, with the group providing the algebra and the spacetime evolution providing the flow.

4.5 Constructive Computational Gauge Theory

We feel that it is a useful exercise to frame gauge theory (and, by extension, quantum gravity) as a problem of *expressiveness* versus *verifiability*. If we allow “all histories” (arbitrary geometries, topology change, gauge redundancy, unbounded recursion in the construction of spacetimes), amplitudes become ill-defined and intractable. If we clamp down too hard, we lose physically relevant states and dynamics. **Functional programming offers a blueprint for balancing these extremes.** Our proposal is for a constructive computational gauge theory that strikes a principled middle ground: a typed, linear, total, effect-controlled calculus of geometries. Concretely, boundary data (3-geometries with gauge labels) are the types; spacetime regions (4-dimensional histories/cobordisms) are the terms; and gluing is composition. This gives a compositional semantics familiar from TQFTs but designed to scale beyond purely verifiable/topological settings (e.g., Chern-Simons [CS74]). Readers are asked to consider the correspondences in table 10. The primitives are as follows:

Table 10: Programming concepts and quantum gravity analogues

Programming Concept	Quantum Gravity Analogue
Types	Boundary states (3-geometries with gauge data)
Terms / Programs	4-geometries (cobordisms, histories)
Composition	Gluing of spacetime regions
Linear types	Conservation laws, unitarity (no-cloning of boundary data)
Totality	Termination of the “geometry evaluator” (finite amplitudes)
Effects & handlers	Coarse-graining and renormalization
Dependent types	Gauge and diffeomorphism constraints

Types. The proposed helical primitives provide the concrete generators of histories: smooth, orientable flows that carry discrete topological labels (orientation/chirality) alongside continuous geometry. This marries the “continuous versus discrete” tension: spectra and curvature are continuous objects; quantum numbers arise as stable, counted winding data. In this sense, a type is not merely a boundary condition in the ordinary functional-analytic sense, but a geometric interface condition: a declaration of what sort of flow, symmetry, and conserved structure may consistently enter or leave a region. Types therefore package both kinematic admissibility and computational resource discipline.

Terms. *Terms* are the actual geometric histories realizing computation. Concretely, a term is a decorated bordism—a glueable spacetime region assembled from helical primitives, equipped with whatever bundle, connection, flow, and readout data are required by the regime under study. In the symbolic and quasi-fluidic bands, such a term may be little more than a controlled phase-carrying cylinder or junction; in the fluidic and gauge regimes, it may carry nontrivial holonomy, curvature, or defect data. The point is that terms are not symbolic expressions standing in for geometry: they *are* the geometry, regarded operationally as a computation from one typed boundary to another. See appendix A.5 in this regard.

Composition. *Composition* is gluing. Given two terms whose outgoing and incoming interfaces match at the level of type, one composes them by identifying the relevant boundary components and extending the declared geometric structure across the seam. This is the physical and categorical content of “running one computation after another”: the output flow of the first region becomes the admissible input flow of the second. At the semantic level, composition must preserve the chosen observational equivalence on boundaries—for example, class functions of holonomy, flux, or winding data—so that gluing does not merely concatenate regions topologically, but yields a well-defined composite computation. Parallel composition is given by disjoint union, while sequential composition is

given by boundary matching; together they form the minimal compositional skeleton needed for a constructive gauge calculus.

Three design choices enforce computability and physics: linearity, totality, and effects. Linearity tracks boundary degrees of freedom as conserved resources (no cloning/erasure). At the same time, totality means the “geometry evaluator” (our state-sum/variational executor) always normalizes. The phenomena that usually force uncontrolled manipulations, such as coarse-graining, stochastic mixing, and renormalization, are modeled explicitly as algebraic effects with handlers. Dependent types encode gauge and diffeomorphism constraints at the level of well-typedness, so invariances propagate mechanically through compositions. We elaborate each in turn:

Linearity. *Linearity* is the discipline that boundary degrees of freedom are treated as conserved resources under gluing: no implicit copying, no implicit erasure [Gir87, Wad90]. This is the categorical analogue of “no cloning,” so in this way, conservation laws (charge, flux) are enforced by typing rather than imposed post hoc. In particular, it ensures that “assembling” a region from primitives is compatible with conservation constraints at interfaces: what enters a glued boundary must exit somewhere, unless an explicit effect declares otherwise.

Totality. By *totality* we mean: the *core* evaluator always normalizes. Concretely, the “geometry evaluator” (state-sum/variational executor) is required to terminate on the core fragment and return a well-defined, finite semantic value (e.g. a boundary object, a morphism, or a computable readout). This is not the claim that nature is finite; it is the claim that the *core language* is designed so that evaluation is mechanically checkable and compositional. The point is to impose guardrails: the core fragment is the part of the semantics on which we can prove locality and gluing theorems without importing the full analytic burden of continuum QFT.

Effects and handlers. The phenomena that typically force uncontrolled manipulations—coarse-graining, stochastic mixing, renormalization, metric dependence, and (in ambitious extensions) topology change—are introduced as *algebraic effects* with *handlers* [PP09, Mog91]. In this way, “renormalization” is not treated as an ad hoc subtraction, but as an explicit *program transformation*: a handler takes a well-typed geometric term and returns a new well-typed term, together with a stated observational equivalence on the boundary.

Example 4.6 (Coarse-Graining as a Handled Effect.). For example, let a geometric term $W : \Sigma_{\text{in}} \rightarrow \Sigma_{\text{out}}$ contain a family of microscopic helical loops $\gamma_1, \dots, \gamma_k$ in some interior region. A coarse-graining handler replaces this subterm by an effective loop system W_{eff} together with a certificate that the chosen boundary observables are preserved. In the abelian regime, this means preserving total phase/flux; in the non-abelian regime, it means preserving specified class functions of holonomy (for instance, Wilson-loop traces) on the boundary. The transformation is therefore not an arbitrary simplification of geometry, but a semantics-preserving rewrite of the program represented by W .

Soundness obligations for handlers. To prevent the effect layer from collapsing back into folklore, we impose the following contract on any declared handler (RG, coarse-graining, metric insertion, noise):

- (a) **Functoriality.** Handling commutes with gluing/composition: handling a composite computation agrees (up to stated equivalence) with composing the handled pieces.
- (b) **Monoidality.** Handling respects parallel composition (disjoint union): independent subsystems remain independent.
- (c) **Gauge/diffeomorphism compatibility.** Handling preserves invariance of boundary readouts: observable semantics are phrased in gauge-invariant class functions of holonomy (e.g. Wilson-loop traces), and handlers must preserve these up to the declared equivalence relation.
- (d) **Resource discipline.** Handling does not implicitly violate linearity; any creation/erasure of degrees of freedom must be explicit (an effect, not a side-effect).

Dependent types. Gauge and diffeomorphism constraints are carried as dependent typing data on boundaries and interfaces: a term is admissible only if it respects the declared constraint structure [ML84]. The payoff is the mechanical propagation of invariances through composition: once a boundary constraint is well-typed, gluing does not “forget” it.

Practically, the workflow is: specify typed boundary data; assemble regions from helical primitives; compose by gluing; evaluate the *core* semantics (the total fragment); and only then, where required, apply declared effect handlers with an explicit statement of what equivalence of boundary readouts they preserve. In this sense, renormalization and coarse-graining are treated as structured code transforms on geometric terms, rather than informal manipulations of integrals.

Our aim is a language that is *expressive* enough to describe nontrivial gauge dynamics and background independence, yet *restricted* enough to prove normalization, locality/compositionality, and anomaly-freeness in the core. Extensions (matter content, topology change, nonperturbative sectors) are added modularly as new effects or controlled type extensions, preserving verification theorems as we widen scope. In this way, constructive computational gauge theory provides a semantics where we can calculate, compose, and certify. For the foundational work on constructive quantum field theory, see [BSZ92]. Our approach here is in this spirit, but computational. Likewise, see [ALR25] for recent philosophical work on constructive spacetime axiomatics.

On QFT and foundational description. We do not dispute the empirical success of quantum field theory, nor its central role as a unifying descriptive framework in modern physics. Per the EVT 2.6, our concern is narrower. When QFT is taken as a single, continuum-based probabilistic formalism in four dimensions, it appears to exceed the class of systems that can be globally verified without encountering tractability barriers (exponential numbers of Feynman diagrams, for instance). Infinite-dimensional functional measures inherit substantial analytic and measure-theoretic difficulties, while renormalization already shows that the formalism is intrinsically scale-dependent: the effective description is repeatedly rewritten as one changes resolution. In this sense, QFT behaves less like a fixed axiomatic object than like a computational process whose semantics are only partially stable across scales [GJ87].

From this perspective, the issue is not that QFT is incorrect, but that it may therefore be better understood, for present purposes, as an emergent descriptive interface rather than as the primitive computational substrate. Our proposal is therefore not a rejection of QFT, but a change in starting point: rather than taking continuum probability and field measures as primitive, we begin from a constructive local-to-global calculus in which admissible operations are restricted from the outset, and in which probabilistic and field-theoretic behavior arise only after coarse-graining. In this way, QFT is retained as a working tool, but the foundational burden is shifted to a more controlled computational substrate that alleviates computational and time burdens on human physicists.

Compactness modulo gauge. The analytic precedent for this kind of controlled evaluation is Uhlenbeck compactness. In Yang–Mills theory, sequences of connections with bounded curvature do not usually converge naively as raw local fields; rather, after passing to suitable gauges, one obtains subsequential convergence away from a controlled singular set, with bubbling recording the failure of ordinary compactness [Uhl82]. This is exactly the kind of phenomenon that a constructive computational gauge theory must respect. Compactness is not free convergence of all geometric data, but convergence *modulo gauge*, together with an explicit account of the defects or lost energy that appear in the limit. In our language, this is the analytic analogue of a handled effect: the limiting process is admissible only when the boundary observables are preserved or when the failure of preservation is recorded as structured defect data. Though compactness should not be confused with computability. Under energy bounds, one can extract controlled subsequential limits. Computable analysis asks a different question: whether such limits are effectively accessible from finite data, with computable moduli of convergence or representations. We require both forms of control: compactness to prevent geometric escape, and effectivity to make the resulting semantics computational rather than merely existential.

From external lattices to an intrinsic, lattice gauge calculus. Lattice gauge theory discretizes a continuum on an external grid: links carry group elements, plaquettes carry curvature, and one hopes

to recover physics in a subtle continuum limit. But computability is not baked in; it is delegated to numerics.

By contrast, GCM lattices are of a different kind. The “lattice” is the 1–skeleton generated by minimal holonomy loops of the helix geometry itself; nodes are join interfaces (Gauss constraints), edges are Wilson–loop registers, faces carry curvature flux. No continuum limit is imposed from outside: the cell structure is induced by the manifold’s own group–theoretic and topological constraints. Spectral gaps and forbidden transitions then arise from intrinsic geometry rather than from a choice of mesh.

Operationally, this translates to:

- **State space.** A presheaf of local moduli (connections/holonomy classes on cells) over a helix–adapted cover; *sheafification* enforces gluing laws (Gauss, Bianchi) and yields global states.
- **Evaluation.** Amplitudes are functorial evaluations of class functions on Wilson loops (edge registers) subject to node intertwiners; normalization is a theorem of the typed core.
- **Scale change.** Refinement/coarse–graining is an explicit algebraic effect that changes the cover and updates moduli by certified handlers; proofs of invariance live in the type/effect system.

In this setting, the “sum over histories” is replaced by *sheafification + evaluation*: one computes global sections (or their invariants) of the sheafified moduli [MLM92] rather than integrating over an externally supplied configuration measure.⁹

Example 4.7 (Sheafification/Evaluation on an Intrinsic Helical Chain). Let $X = U_1 \cup U_2 \cup U_3$ be a helical working region whose open sets are centered on three consecutive intrinsic holonomy loops $\gamma_1, \gamma_2, \gamma_3$, with overlaps $U_{12} = U_1 \cap U_2$ and $U_{23} = U_2 \cap U_3$, and no triple overlap. Assume these opens as coming from the geometry of the GCM itself, rather than from an externally imposed cubic grid.

Let $\mathcal{M}(U)$ be the presheaf assigning to each open set $U \subseteq X$ the groupoid of $U(1)$ –connections on the trivial line bundle over U , modulo gauge, together with the induced Wilson–loop values on the intrinsic loops contained in U . A local section over U_i is represented by a connection A_i and its holonomy

$$u_i := \text{Hol}_{\gamma_i}(A_i) \in U(1).$$

The local data glue precisely when the restrictions align to gauge overlaps:

$$A_2|_{U_{12}} = A_1|_{U_{12}} + g_{12}^{-1} dg_{12}, \quad A_3|_{U_{23}} = A_2|_{U_{23}} + g_{23}^{-1} dg_{23}.$$

After sheafification, the admissible global states are therefore not arbitrary triples (u_1, u_2, u_3) , but compatible glued sections

$$A \in \Gamma(X, \mathcal{M}^\#).$$

Now choose the outgoing loop γ_{out} obtained by concatenating the three intrinsic channels. In the abelian corridor, the functoriality of holonomy gives

$$\text{Hol}_{\gamma_{\text{out}}}(A) = u_3 u_2 u_1.$$

Thus, the intrinsic lattice calculus does not “sum over histories” on an external mesh; it first constructs the global section by gluing local moduli, and only then evaluates the observable.

The constructive evaluator is the gauge–invariant map

$$\text{ev}_X : \Gamma(X, \mathcal{M}^\#) \longrightarrow U(1), \quad \text{ev}_X(A) := \chi(\text{Hol}_{\gamma_{\text{out}}}(A)),$$

where in the simplest abelian case one may take $\chi(z) = z$, or more generally a thresholded readout to a finite alphabet.

This also exhibits the effect–handler viewpoint. A coarse–graining handler may replace the pair (U_1, U_2) by a single effective region U_{12} with effective holonomy

$$u_{12} := u_2 u_1.$$

The rewritten term has local data (u_{12}, u_3) , but the boundary observable is unchanged:

$$\chi(u_3 u_{12}) = \chi(u_3 u_2 u_1).$$

⁹“It’s nice to have a framework where you can only have one Lagrangian, but it would be even nicer to have some reason that this particular algebra would inevitably appear as an answer to some question.” —E. Witten [Int14]

Hence, coarse-graining is not an informal simplification; it is a semantics-preserving rewrite certified by preservation of the boundary class function.

In the non-abelian lift, the same example survives with ordered transport:

$$U_{\text{out}} = U_3 U_2 U_1, \quad \text{ev}_X(A) = \text{tr}_\rho(U_3 U_2 U_1),$$

so the only change is that composition becomes path-ordered, and the evaluator passes from scalar phase to a class function on holonomy.

Remark 4.2 (Why Not Classical $\mathbb{R}$?). In the fluidic/gauge band, one is tempted to model directly over the classical continuum. However, unrestricted real computation (cf. A.6) admits hyper-Turing power and brittle pathologies (exact tests, branch-on-sign over nonrobust inputs, oracle-like primitives). This undermines the two pillars we require: *executability* (every step is effectful but total) and *verifiability* (small perturbations do not change the decision class). Our constructive stance preserves smooth geometry while preventing hidden oracles.

Geometric reals? We separate *semantics* (smooth reasoning) from *execution* (effective approximation):

- (a) **Computable reals** $\mathbb{R}_{\text{comp}}$. externally, numerical data (couplings, boundary values, readouts) live as computable reals given by fast Cauchy names (or, when possible, algebraic numbers with exact arithmetic). Equality is replaced by interval refinement; separation proofs guard branching. This is the target of all executable algorithms [Wei00].
- (b) **Synthetic/smooth reals.** $\mathbb{R}_{\text{syn}}$ internally, differential reasoning uses the smooth reals of a well-adapted SDG model (Kock–Lawvere axiom [Koc06], nilpotent $d\varepsilon = 0$), so vector calculus and curvature remain first-class without appeal to pointwise limits.¹⁰
- (c) **A bridge.** A semantic-to-execution functor sends internal smooth expressions (curvatures, class functions on holonomy) to external names in $\mathbb{R}_{\text{comp}}$, with *proof obligations* discharged by our effect handlers (for limits/coarse-graining).

The aim is not to deny the continuum but to choose a base field that respects computability. Parameters are measured to finite precision; algorithms require effective representations. Thus we use rationals/algebraics/ $\mathbb{R}_{\text{comp}}$ for execution, while $\mathbb{R}_{\text{syn}}$ supplies calculus internally. This pragmatic split aligns with the observation (cf. [Szu12]) that physical models do not require arbitrary reals to be *expressed* or *simulated*, even if smooth structure is invaluable for *reasoning*.

Design axioms: equality, limits, and effects. We enforce three constraints in the core calculus:

- **No exact branching on undecidable predicates.** Tests are certified by separation bounds (or deferred as obligations).
- **Limits are effects.** Completion, closure, and renormalization are algebraic effects with handlers that carry witnesses of convergence/stability.
- **Gauge invariants only.** Readouts are class functions of holonomy (e.g. traces), ensuring stability under small perturbations and gauge changes.

Remark 4.3 (Disclaimers on Classical QFT). Our setting concerns analog geometrodynamics on a noncompact configuration space. “Gaps” (mixing rates, spectral edges of transfer generators) are typically defined by nested sup/inf over unbounded domains. In general, these constants need not be uniformly computable from the defining data. We therefore (a) state results in terms of *semi-decidable* or *effectively approximable* quantities, or (b) confine genuinely noncomputable constants to effectful interfaces outside the core (they cannot influence totality/normalization proofs). As a result, this is a computability warning label for our own continuum models; we cannot use our model to solve the Clay Yang–Mills problem [JW00], which requires a fully rigorous quantum theory.

¹⁰Any well-adapted model suffices; no commitment to a specific topos is needed, but an implementation is provided in appendix A.8 for convenience.

On finite information and intuitionist physics. We concur with the operational point that experimental data carry finite information, and that uncountable precision should not be a primitive of algorithms, as brought up by Prof. Gisin [Gis19]. Where we differ is methodological: we retain a smooth internal semantics ($\mathbb{R}_{\text{syn}}$) to keep geometry local and functorial, while ensuring that *every* externally visible quantity is realized in $\mathbb{R}_{\text{comp}}$ with certified error control. Thus, the continuum remains a modeling tool; computation proceeds on effectively presentable numbers.

Choosing a constructive base allows us to retain the familiar analytic language where needed while ensuring that each step of the formalism remains executable. The framework developed here treats physical dynamics as smooth, often chiral/transcendental flow, but it also treats topology and quantization as mechanisms of coarse-graining that project this flow onto the discrete structures accessible to observation. In this sense, the continuum substrate and the discrete emergent layer are not competitors but complementary levels of description. What we call “physical computation” is the geometric process that occurs at their interface, with its behavior constrained by the EVT Trilemma. Nothing in the present approach depends on sub-Planckian assumptions, nor do we take uncountable structure as primitive in the dynamics. Our hope is that the fully constructive classification of topological field theories outlined (4.7) provides a more precise setting in which longstanding questions, including those surrounding Yang–Mills, can be reformulated and studied. A high-level road map toward the Yang–Mills mass gap in computational gauge theory is outlined below:

- (a) **Define computational semantics for gauge theory.** Define a typed, linear, total calculus whose types encode boundary data (3-geometries with gauge labels) and whose terms encode histories (4-dimensional “programs” composed by gluing).
- (b) **Build computable geometric bundles.** Give the calculus a denotational semantics as a computable fiber-bundle (connection, parallel transport) formulated in a constructive/synthetic differential setting over a computable base field. This preserves the differential structure while ensuring that all primitives are evaluated effectively.
- (c) **Assess the scaling limit to continuum Yang–Mills.** Prove that families of programs approximate continuum Yang–Mills in a controlled scaling limit. Concretely, a “differential computational jet calculus” provides error bounds showing convergence of correlation functions/Wilson loops to those of continuum Yang–Mills as the program lattice is refined (see appendix A.3.)
- (d) **Prove the gap and its stability.** Establish a spectral gap (uniform lower bound on excitation energies) for the constructive model using the calculus’ linear/compactness properties; then show this lower bound survives the scaling limit, yielding a continuum mass gap. Here, the constructive setting replaces global non-compact searches with local-to-global, type-theoretic invariants.

All physically relevant predictions are recoverable with parameters and observables taken from an effective number system. This keeps the continuum as a limit, not a prerequisite, and allows “proof by construction” to coexist with the analytic tools physicists use.

This suggests a computational reading of the Yang–Mills mass-gap problem. Instead of treating the gap only as a spectral property of a Hamiltonian or Euclidean correlation operator, one may ask how a compact simple gauge theory constrains the distinguishable state transitions available to a geometric computational substrate. In this language, a mass gap would correspond not merely to the absence of arbitrarily low-energy excitations, but to a structural separation between admissible computational sectors: local gauge fluctuations may vary continuously, while gauge-invariant readouts remain separated by a nonzero scale. The point is not to replace the analytic formulation of the mass-gap problem, but to recast part of its content in operational terms. **This approach allows us not just to compute solutions of the Yang–Mills equations, but also to compute *with* the solutions themselves.**

4.6 Euclidean Gauge Computation and Emergent Spacetime

The constructive calculus developed above is formulated at the level of typed geometric histories, gluing operations, and gauge-invariant readouts. To connect this formal architecture with familiar physical spacetime, however, one must explain how its Euclidean computational presentation relates to causal evolution, metric structure, and the effective Lorentzian theories used in physics. Our constructions are not introduced as an additional fundamental ontology, but as controlled presentations of the same

underlying gauge computation to show how Lorentzian spacetime, gravitational response, and particle-like excitations may emerge as effective semantics of a deeper Euclidean geometric process.

Euclideanization and clocking. The physical Yang–Mills mass gap is posed on Minkowski spacetime $\mathbb{R}^{1,3}$, but our constructive model is formulated on a *Euclidean* 4-manifold. Concretely, we work after a Wick rotation $t \mapsto -i\tau$. Hence, the background metric is Riemannian, and the fourth coordinate can be used to encode an extended, chirality-resolving structure (e.g., Type 2 *THM*) without committing to a Lorentzian time foliation.

Operationally, temporal evolution is replaced by global pulses/flow propagation that provide an internal clocking of evaluations. Lorentzian observables are recovered by analytic continuation: correlation functions are computed in the Euclidean model and (where reflection positivity holds) reconstructed to Minkowski amplitudes via the usual Osterwalder–Schrader logic [OS73]. In our calculus, “Wick rotation” is a controlled effect: Euclidean evaluation (for existence/compactness) and Lorentzian reconstruction (for scattering/causality) are separate, typed phases of the same program. This retains the computational advantages of a 4D Euclidean setting while maintaining a principled path back to $\mathbb{R}^{1,3}$ for physical predictions.

The landscape of twistor theory. While our constructions emphasize categorical and geometric composition, it would be amiss not to note that *twistor theory*, pioneered by Penrose and collaborators [PM73], gives a striking *holomorphic* encoding of four-dimensional conformal geometry and massless field equations. Concretely, the Penrose transform and the Ward correspondence identify cohomological data on projective twistor space with solutions of linear massless equations and, in the non-abelian case, (anti-)self-dual Yang–Mills on spacetime.

Reader’s compass. For our purposes, “twistor phase” means a regime where inherently noncommuting spacetime/gauge data admit a holomorphic reparametrization that turns analysis into cohomology. The symmetry group at play is the full conformal group $SU(2, 2) \cong Spin(2, 4)$; it is *not* an abelian corner. Thus, the twistor limit is best read as a complex-analytic chart in which verification improves because equations become integrable/holomorphic—even though the underlying gauge content remains non-abelian. Nothing in this viewpoint licenses literal real-valued “infinite computation”: finite-information bounds [Bou02] still apply. The twistor phase should be read as a *cohomological* linearization window, not as physical hypercomputation.

Finally, our framework is naturally compatible with holomorphic tools that also appear in parts of the Geometric Langlands story (cf. [KW07]) and with twistor approaches to scattering amplitudes; see, e.g., the mathematical discussion noted in [Woi22]. The correspondence is evidence that the gauge regime admits nontrivial equivalences between presentations of the same boundary observable algebra; we treat such dualities as chart changes. Our work does not require the full program, but it motivates the principle that physical/gauge computation should be invariant under these categorical re-factorizations. Hence, we do not claim a direct realization of Langlands duality; rather, we point to a common structural mechanism—holomorphic/complex-analytic repackaging of non-abelian field data—that GCT will interface with in future work.

On Calabi–Yau manifolds and stable loci. The present viewpoint is not a String-theoretic construction, but it shares with string theorists a geometry-first emphasis on extended objects, moduli, and the extraction of physical structure from constrained geometric deformation. On a compact Kähler manifold with $c_1 = 0$, Yau’s theorem [Yau78] guarantees a unique Ricci-flat Kähler metric in each Kähler class. This furnishes a canonical, *integrable* background with powerful linear tools (Hodge theory, period maps, variation of Hodge structure) and controlled deformations.

Reader’s compass. In our setting, Calabi–Yau (CY) [Pic19] structures are *stable bookkeeping loci*: places in the global moduli where curvature decouples enough that the effective local algebra is as close to commutative as the dimension allows. The harmonic representative of each cohomology class provides a linear “ledger” for fluxes and charges, and the Hodge decomposition/intersection form supplies rigid accounting for admissible operations. For threefolds, the $(1, 1)$ and $(2, 1)$ sectors (Kähler vs. complex moduli) control, respectively, capacities and morphisms available to the substrate; at such loci, symbolic projections are maximally checkable because key questions reduce to cohomological linear algebra.

Technically, this is the state space where:

- (a) the effective coefficient algebra is (up to mild twists) E_∞ in abelian and maximally-abelian quasi-fluidic regimes, so excision holds, and parallel composition is transparent;¹¹
- (b) twisted and equivariant K -theory encodes stable resource classes (channels, phases) and their integral constraints via periods, while variations of complex structure are recorded by period integrals and Picard–Fuchs equations;
- (c) moving away from a CY locus (e.g. conifold degenerations or large-complex-structure limits) introduces monodromy/Picard–Lefschetz transformations, which reintroduce noncommuting transport: checks become globally harder even as the space of behaviors enlarges.

Hence, CY manifolds are not a generic case; they are *integrable gate points*—symmetry-enhanced regions that abelianize the accounting and make verification accessible. They play an essential role in the lower (abelian) and quasi-fluidic phases as calibration loci and phase boundaries: near a CY stratum, the dynamics are governed by linear/cohomological data; moving away activates non-abelian holonomy and the twistor calculus native to the more non-commutative regimes.¹²

Table 11: Classification of computational regimes by data type and algebraic structure

Symmetry Regime	Algebraic Structures	Data Types
Discrete / symbolic (finite cyclic C_n)	Finite rings/Boolean algebras	Bits, n -its
Analog abelian ($U(1)$)	Real/complex fields, Lie algebra $\mathfrak{u}(1)$	Circle phases, amplitudes
Analog non-abelian ($(SU(2), SU(3))$; gauge group actions)	Principle bundles, moduli spaces, ∞ -groupoids; (co)homology operations	Global classes (characteristic classes/Chern, Floer/Donaldson-type)

Interactions of regimes within emergent gauge behavior. The presentations of the discrete and abelian regimes in section 2 and 3 are intentionally given at the level of their *readouts*—coarse projections that isolate the symbolic (C_n) and commutative ($U(1)$) components of the substrate. In the full interaction picture, where non-abelian transport and defect/gluing data are active, the resulting dynamics are *not* adequately described by a standalone $C_n/U(1)$ -valued cellular automaton. For expository purposes, we can collapse the final four-regime classification of section 4.7 into the three-band summary of table 11, and then indicate how the emergent gauge regime reorganizes the lower readouts.

Regimes as readouts of a single substrate. As emphasized in proposition 3.1, the notion of a geobit is readout-dependent, so table 11 should not be read as a list of unrelated theories, but as a hierarchy of *coarse-grained readouts* of one underlying geometric computing substrate. Concretely, let $\mathcal{S}$ denote the space of full substrate states (helical geometry, joins/defects, and transport data). The three rows of table 11 correspond to progressively richer “decoding maps”

$$\pi_{C_n} : \mathcal{S} \rightarrow \mathcal{S}_{C_n}, \quad \pi_{U(1)} : \mathcal{S} \rightarrow \mathcal{S}_{U(1)}, \quad \pi_{\text{gauge}} : \mathcal{S} \rightarrow \mathcal{S}_{\text{gauge}},$$

where π_{C_n} forgets geometric transport and retains only local symbolic degrees of freedom (a cellular automaton shadow), $\pi_{U(1)}$ retains a commutative phase/flux bookkeeping (an “abelianized” quasi-fluidic shadow), and π_{gauge} retains full non-abelian parallel transport and defect/gluing data (the fluidic/gauge regime). Thus, entering into π_{gauge} does not replace the lower regimes; it means we stop applying the early projections. The lower regimes remain present as *shadows* of the full dynamics, i.e. as the observables that survive after discarding non-commuting transport information.

¹¹ E_∞ here just means “homotopy-commutative”; readers may safely read it as “fully abelian up to coherent homotopy.” See appendix A.6.

¹²For readers fluent in string theory: one may regard a geobit as a deterministic analogue of a worldsheet defect. This viewpoint suggests a geometric mechanism reminiscent of the weak gravity conjecture [AHMNv07]; we do not attempt a derivation here.

Gauge back-reacts on lower readouts. Once non-abelian transport is active, the lower readouts inherit three systematic features: (a) *sectorization* by global topological labels (bundle/defect classes); (b) *twisting* of coefficients (fields become sections of associated bundles/local systems); (c) *center/trace observables* (the surviving abelian data is typically class-function data of holonomy). Concerning cohomology, near integrable/CY strata, cohomology acts mainly as a linear *ledger* within a sector; in the full gauge/twistor regime, it acts as *governance* (sectors, obstructions, monodromy/wall-crossing).

Example 4.8 (A Three-band Toy Model on T^3 : Abelian Ledger Twisted Gauge Readout). This example makes explicit how a non-abelian gauge regime induces (a) sector labels, (b) twisted coefficients, and (c) trace/center observables in the lower abelian/quasi-fluidic readouts.

Base. Let $X = T^3 = \mathbb{R}^3/\mathbb{Z}^3$ with coordinates $x_1, x_2, x_3 \in \mathbb{R}/\mathbb{Z}$. Then

$$H^1(T^3; \mathbb{Z}) \cong \mathbb{Z}^3, \quad H^2(T^3; \mathbb{Z}) \cong \mathbb{Z}^3, \quad H^2(T^3; \mathbb{Z}_2) \cong (\mathbb{Z}_2)^3.$$

Band A: $U(1)$ ledger (commutative transport). A $U(1)$ -connection A on a line bundle $L \rightarrow T^3$ splits into topological flux and a flat holonomy part:

- (a) *Flux ledger* (H^2). The Chern class $c_1(L) \in H^2(T^3; \mathbb{Z}) \cong \mathbb{Z}^3$ may be represented by an integral curvature form

$$\frac{F}{2\pi} = n_1 dx_2 \wedge dx_3 + n_2 dx_3 \wedge dx_1 + n_3 dx_1 \wedge dx_2, \quad n_i \in \mathbb{Z},$$

so (n_1, n_2, n_3) is a literal cohomological “ledger” of $U(1)$ fluxes.

- (b) *Holonomy ledger* (H^1). A flat connection may be written

$$A_0 = 2\pi(\alpha_1 dx_1 + \alpha_2 dx_2 + \alpha_3 dx_3), \quad \alpha_i \in \mathbb{R}/\mathbb{Z},$$

with holonomies around the coordinate 1-cycles γ_i given by

$$\text{Hol}_{A_0}(\gamma_i) = e^{2\pi i \alpha_i} \in U(1).$$

Thus, the abelian/quasi-fluidic readout is linear: integers in H^2 plus phases in $H^1(-; \mathbb{R}/\mathbb{Z})$.

Band B: Flat $SU(2)$ holonomy (non-abelian gauge readout). A flat $SU(2)$ -connection on T^3 is equivalent to a representation

$$\rho : \pi_1(T^3) = \mathbb{Z}^3 \longrightarrow SU(2),$$

i.e. a commuting triple

$$(g_1, g_2, g_3) \in SU(2)^3$$

modulo simultaneous conjugation. Since commuting elements in $SU(2)$ lie in a common maximal torus $T \cong S^1$, any such triple may be simultaneously conjugated to

$$g_i = \begin{pmatrix} e^{i\theta_i} & 0 \\ 0 & e^{-i\theta_i} \end{pmatrix}, \quad \theta_i \in \mathbb{R}/2\pi\mathbb{Z}.$$

The residual conjugation preserving the maximal torus is the Weyl group

$$W = N(T)/T \cong \mathbb{Z}_2,$$

which acts diagonally by simultaneous inversion,

$$(\theta_1, \theta_2, \theta_3) \mapsto (-\theta_1, -\theta_2, -\theta_3).$$

Hence the coarse flat moduli space is

$$\mathcal{M}_{\text{flat}}(T^3, SU(2)) \cong \text{Hom}(\mathbb{Z}^3, SU(2))/SU(2) \cong T^3/W,$$

where W acts by simultaneous inversion. Equivalently,

$$\mathcal{M}_{\text{flat}}(T^3, SU(2)) \cong (S^1)^3 / ((z_1, z_2, z_3) \sim (z_1^{-1}, z_2^{-1}, z_3^{-1})).$$

The important point is that the Weyl action is simultaneous: one does not independently identify each θ_i with $-\theta_i$, since gauge equivalence acts by a single simultaneous conjugation of the whole representation ρ .

This immediately exhibits two of the three claimed effects:

- (c) **Trace/character observables.** Gauge-invariant observables factor through simultaneous conjugacy classes, and hence through class functions of the holonomies. For the coordinate generators one obtains

$$\mathrm{tr}(g_i) = e^{i\theta_i} + e^{-i\theta_i} = 2\cos(\theta_i).$$

More generally, traces of words in the commuting holonomies, such as

$$\mathrm{tr}(g_i g_j) = 2\cos(\theta_i + \theta_j),$$

define gauge-invariant class functions on the simultaneous conjugacy class. Thus the “abelian shadow” of non-abelian transport is not the raw holonomy matrix, but character/class-function data extracted from holonomy.

- (b) **Twisted coefficients.** Fixing ρ , infinitesimal deformations of the flat connection are governed by cohomology with *twisted* coefficients. At a regular point of the flat moduli space, the reduced tangent directions are captured by

$$T_{[\rho]}\mathcal{M}_{\mathrm{flat}} \cong H^1(T^3; \mathrm{ad}_\rho),$$

where ad_ρ is the local system with fiber $\mathfrak{su}(2)$ and monodromy

$$\mathrm{Ad} \circ \rho : \pi_1(T^3) \longrightarrow \mathrm{Aut}(\mathfrak{su}(2)).$$

For a generic diagonal triple, the adjoint local system splits as

$$\mathrm{ad}_\rho \cong \mathbb{R} \cdot \sigma_3 \oplus \mathcal{L}_{2\theta},$$

where $\mathbb{R} \cdot \sigma_3$ is the trivial rank-1 Cartan subsystem and $\mathcal{L}_{2\theta}$ is the real rank-2 local system on which the coordinate loops act by rotation through angles $2\theta_i$. For generic holonomy, the rotating summand contributes no degree-1 cohomology, so one obtains

$$H^1(T^3; \mathrm{ad}_\rho) \cong H^1(T^3; \mathbb{R}) \cong \mathbb{R}^3.$$

Thus the surviving linear behavior is not ordinary untwisted cohomology from the outset; it is the Cartan component that remains after accounting for twisting by the background holonomy.

Band C: sector labels via $SO(3)$ and w_2 . On T^3 , every $SU(2)$ -bundle is topologically trivial, so bundle-class sectorization is muted at the simply-connected level. Passing to the adjoint form $SO(3) \cong SU(2)/\{\pm I\}$ restores genuine topological sectors: principal $SO(3)$ -bundles are classified by

$$w_2(P) \in H^2(T^3; \mathbb{Z}_2) \cong (\mathbb{Z}_2)^3.$$

Thus, the same H^2 -type bookkeeping that served as an abelian “flux ledger” becomes, in the emergent gauge regime, a discrete *sector label* (a lifting obstruction for holonomy data), i.e., cohomology functions as governance rather than ledger [vB96].

Interpretation. This toy model exhibits the general mechanism used throughout the paper: the abelian/quasi-fluidic regimes record linear ledger data (classes/periods), while turning on non-abelian transport forces (a) decomposition into global sectors, (b) twisted coefficient systems for linearization, and (c) passage to trace/character class functions as the stable abelian shadows of gauge-invariant content.

Diffeomorphism invariance and the forced move to moduli invariants. In the highest (gauge) regime, the effective closure is not merely a compact structure group G acting on fibers, but the full redundancy of the geometric substrate: diffeomorphisms of the base (and their induced action on bundles and connections) as gestured towards in table 7. Concretely, the relevant symmetry is the gauge–diffeomorphism groupoid generated by $\mathrm{Diff}(M)$ together with bundle gauge transformations. Once computation must be invariant under this action, coordinate-level quantities cease to be meaningful: only invariants of the resulting quotient spaces (moduli of connections/flows up to gauge and diffeomorphism) remain as stable readouts. This is precisely the point at which Floer/Donaldson-type constructions become natural [Flo88, Don83], since they are designed to extract computable cohomological data from such moduli spaces.

On Seiberg-Witten and Atiyah-Singer. In the continuum, anomaly cancellation and spectral flow are governed by index theory: the analytic index of Dirac-type operators equals a topological index [AS63]). In our framework, the corresponding Ward/typing identities can be seen as constructive index constraints: zero-mode counts and anomaly coefficients are enforced at the level of types and persist under gluing. Thus, index-theoretic stability is reflected as compositional invariants of our evaluator rather than as a posteriori analytic facts.

Likewise, the Seiberg–Witten equations [SW94] exemplify how gauge-theoretic PDEs on 4-manifolds can yield compact moduli spaces with energy bounds and computable invariants. Our Euclidean computational formulation seeks the same structural features: finite-energy solution classes generated by helical programs, compactness under refinement, and invariants derived from holonomy/typing. We emphasize that we do not claim new SW invariants; instead, we adopt the SW template with compactness and abelianized control as the target shape for our constructive certificates (e.g., gap lower bounds and well-behaved moduli).

Effective geometric layers. The framework developed here should not be read as assuming Lorentzian geometry at the substrate level. Geometric computation begins from a pre-metric configuration space whose first effective spacetime data are propagation and readout structures. In particular, causal and optical behavior may be organized by an effective constitutive tensor χ ; in near-equilibrium regimes, χ may become metric-induced, $\chi = \chi[g]$, thereby recovering Lorentzian geometry in the sense of pre-metric and constructive approaches to electromagnetism and gravity [Sch20]. Once this Lorentzian regime is available, quantum fields and observers should be interpreted with the usual curved-spacetime discipline: local observable algebras, admissible states, and horizon effects are part of the effective semantics rather than external additions [Wal95].

At still larger scales, gravitational dynamics are expected to appear as thermodynamic relations among the coarse-grained degrees of freedom of the substrate. This is consonant with Jacobson’s derivation of the Einstein equation as an equation of state [Jac95], and with Padmanabhan’s broader emergent-gravity program, in which the horizon or spacetime degrees of freedom satisfy thermodynamic/equipartition relations [PAD10]. Thus Schuller, Wald, Jacobson, and Padmanabhan occupy distinct effective layers: propagation selects causal structure, local quantum observables live on the recovered Lorentzian background, and coarse-grained horizon thermodynamics supply the macroscopic gravitational equation of state.

But we do not take Markovian effective field theory to be foundational. Rather, it is treated as an emergent approximation that becomes valid only after a suitable coarse-graining of the underlying geometric bulk. For this reason, clean IR/UV decoupling is not assumed a priori, and as a result, it is likely that questions of strongly coupled gauge dynamics will be addressed before one can reliably speak about thermodynamic or black hole observables as outlined above, and such work is reserved for the future.

Status of gravitons. We do not quantize gravity by postulating a continuum path integral over Lorentzian metrics, nor do we treat the metric as a fundamental Yang–Mills field, so in this sense, gravity is not “quantized” at the fundamental level. What appears in perturbative regimes as a graviton should instead be understood as an effective spin-2 excitation of the emergent Lorentzian metric sector, valid in the same restricted sense that phonons are valid excitations of a condensed-matter medium. Such excitations are therefore not denied, but classified as an approximation. The foundational role is played not by a quantum field of gravity, but by the constraint architecture and of the underlying computation in the context of sound gluing rules; causal, metric, and field-theoretic structures arise only as effective descriptions of their large-scale organization.

What would count as success or failure? The framework proposed here should be judged neither as a metaphor nor as a purely speculative reinterpretation, but by whether its structural claims survive contact with mathematics and physics. It gains support if its phase distinctions yield stable and non-trivial differences in observables, if its abelian and quasi-fluidic regimes recover familiar gauge-theoretic behavior in a more unified way, and if its computational predictions continue to align with known constraints on expressiveness, verifiability, and reversibility. It would be weakened, or potentially falsified, if the proposed geometric strata fail to produce mathematically meaningful invariants, if the claimed

transitions between regimes do not correspond to genuine differences in computational behavior, or if the framework proves unable to recover the standard structures it is intended to subsume.

4.7 On the Classification of Geometric Computational Field Theories

We can now classify *geometric computational field theories* by specifying three layers of structure and composing the canonical maps between them:

$$\begin{array}{c}
\text{decorated state/transition groupoid} \\
\downarrow \text{higher rep.} \\
\text{fully dualisable object in a symmetric monoidal } (\infty, n)\text{-category} \\
\downarrow \text{Cobordism Hyp.} \\
\text{fully extended } (\infty, n)\text{-TFT}
\end{array}$$

Existence–uniqueness [BD95, Lur09] identifies the last arrow once a fully dualisable object is supplied; the first arrow is where GCFT lives, turning helix–decorated paths and bordisms into higher representations with explicit operational semantics.

Operational layer (constructive evaluator). As in section 4.5, *types* are boundary data (3–geometries with gauge labels), *terms* are 4–dimensional histories (cobordisms), and *composition* is gluing. Linearity tracks conserved resources; totality guarantees normalization of evaluations in the core; coarse–graining/renormalization are treated as algebraic *effects with handlers*. For every regime below, the interface is the same: registers are Wilson loops/surfaces, readouts are gauge–invariant class functions of holonomy (e.g. traces), and evaluation is symmetric monoidal and local.

Regime Ladder Overview.

(a) Symbolic. *Input symmetry/labels:* decorated fundamental groupoid with finite abelian labels $(V_4, (C_2)^3)$. *Higher rep target:* finite-group 2-representations (objects are twist sectors, 1-morphisms are path homotopies). *Dualisability:* finite semisimple $\Rightarrow$ left/right duals exist. *Predicted TFT:* 2D Dijkgraaf–Witten [DW90]. *Ops:* commuting joins; characters χ_g as readouts.

(b) Quasi-fluidic. *Input:* $(C_2)^4$ with 1- and 2-form labels (higher-form flux). *Higher rep:* 3-representations/crossed-module representations. *Dualisability:* finite higher group $\Rightarrow$ rigid. *Predicted TFT:* 3D higher-form DW/Yetter quantum double [Yet92]. *Ops:* loops and surfaces as registers; higher Wilsons; coarse–graining as 2-sheafification.

(c) Fully fluidic. *Input:* symmetry of a THM cell $D_4 \ltimes (C_2)^4$ acting on a decorated Π_1 . *Higher rep:* 4-representations/2-connections on principal 2-bundles (toy non-abelian sector). *Dualisability:* restrict to compact structure groups; finite-depth truncations for constructivity. *Predicted TFT:* 4D BF (topological) as rigorous core; YM-term added as an *effect* in the evaluator. *Ops:* noncommuting joins; class functions of holonomy as readouts; renormalization as a certified handler.

(d) Emergent gauge. *Input:* bundle of simple Lie data (e.g. $SU(2)$) on helix-adapted charts. *Higher rep:* bundles of fully dualisable objects in Alg_∞ (e.g. E_2 -algebras of Wilson lines). *Dualisability:* compactness/rigidity (fusion) or factorization criteria, depending on dimension. *Predicted TFT:* 3D Chern–Simons (fully extended); 4D topological sector (BF/CS-like); YM handled as an *effect*. *Ops:* same interface; node constraints become intertwiners in the representation category.

Lemma 4.2 (Evaluator Factors Through the Classified TFT). *Let $\mathcal{D}$ be a symmetric monoidal (∞, n) -category and $X \in \mathcal{D}$ a fully dualisable object obtained as a higher representation of the decorated groupoid/bordism data. The constructive evaluator of section 4.5 (which is symmetric monoidal, local, and sends start/stop to coevaluation/evaluation) factors uniquely through the fully extended TFT $\mathbb{Z}_X$ classified by X .*

Proof sketch. By locality and monoidality, the evaluator is a symmetric monoidal functor on the bordism (∞, n) -category. Its values on units and caps/cups agree with the duality data of X by construction, hence it is equivalent to $\mathbb{Z}_X$ by the Cobordism Hypothesis. $\square$

Table 12: Constructive classification: regimes, dualisable inputs, and predicted TFTs.

Group Type	Fully dualisable object supplied	Predicted (n) -TFT	Comments	Dualisability guardrails
Abelian	Finite-group 2 -representation of the decorated groupoid (V_4)	2D Dijkgraaf-Witten	Factors through finite sets $\rightarrow \mathbf{Vect}$; commuting joins	Finite semisimple category; duals automatic
Maximally abelian	Discrete 3 -representation of $(C_2)^4$ with higher gauge 2-forms	3D DW / Yetter double	$\mathbb{Z}_2$ higher-form lattice models; surfaces as registers	Finite higher group; rigid monoidal structure
Non-abelian	Smooth 4 -representation of $D_4 \ltimes (C_2)^4$ inside decorated Π_1	4D BF; Yang-Mills via effect	Noncommuting holonomies; renorm as certified handler	Compact structure groups; depth truncations
Simple Lie	Bundle of dualisable objects in $\mathbf{Alg}_\infty$ (e.g. E_2 of $SU(2)$ lines)	Chern-Simons / (∞, n) -TFT; YM via effect	Matches the Cobordism Hypothesis with decorated invariants	Compactness or factorization-algebra criteria

Guardrails in 4D: Topological Core vs. Metric Effects. In four dimensions, the Cobordism Hypothesis classifies *fully extended* topological field theories by *fully dualisable* objects in an appropriate symmetric monoidal (∞, n) -category. This classification is inherently *topological*: it is driven by local-to-global gluing, duals/evaluations, and functoriality under bordism composition, rather than by metric-dependent dynamics [Fre12].

Pure Yang–Mills is therefore not an acceptable *core* object for dualisability statements: the Yang–Mills action depends on a choice of metric through the Hodge operator $*$, and it carries propagating local degrees of freedom. To keep the 4D semantics fully compositional, we isolate a metric-free *BF topological core* [BT91] and treat the metric-sensitive Yang–Mills term as an *effect* handled by the constructive evaluator. Concretely, one may regard Yang–Mills in first-order form as a deformation of BF theory:

$$S_{\text{BFYM}}(A, B) = \int_M \text{tr}(B \wedge F_A) + \frac{g^2}{2} \int_M \text{tr}(B \wedge *B),$$

where A is a connection with curvature F_A and B is an adjoint-valued 2-form. The BF term is topological (metric-free), while the second term is precisely where metric/scale enters via $*$. In the formal “topological limit” $g \rightarrow 0$, one recovers BF theory; integrating out B recovers the usual second-order Yang–Mills action.

This mirrors the Plebanski [Ple77] paradigm: four-dimensional gravity can be presented as *constrained BF theory*, [FS12] where a BF-type action is supplemented by *simplicity constraints* (or, equivalently, constraint potentials) to recover metric geometry and propagating modes. In our setting, the analogy is operational:

- **Core (dualisable).** BF-type topological semantics, used for the Cobordism Hypothesis classification layer.
- **Effects (non-topological).** Metric-dependent dynamics (YM terms, constraint potentials, coarse-graining/RG) implemented as controlled effect-handlers in the evaluator.
- **Soundness criterion.** Effect-handlers must preserve symmetric monoidality (gluing/parallel composition) and must respect boundary gauge-invariant observables (e.g., class functions of holonomy such as Wilson loop traces).

On the BF/Yang–Mills presentation. All dualisability statements in this paper pertain to the BF core; the physically relevant Yang–Mills sector is recovered by applying the declared metric effects in a way that is compositional on boundaries. The BF/Yang–Mills decomposition used here should not be read as a new field-theoretic formulation of Yang–Mills theory. First-order and BF-type presentations of Yang–Mills, as well as constrained-BF approaches in related gauge and gravity settings, are well established [dWPB96, MZ96]. Our use is semantic rather than historical; the novelty claimed here is the placement of this known decomposition based on helical types, holonomy readouts, and regime-dependent effect handling.

Effects, semantic data, and QFT obligations. In the tradition of lattice gauge theory, we can interpret the constructive tooling as a *regulator with admissibility built in*. Rather than beginning with an unrestricted formal path integral over all field histories and then imposing gauge, locality, or regularity conditions afterward, we attempt to restrict the space of admissible histories at the level of geometric formation rules. Boundary data, gluing constraints, transport data, and regime-dependent readouts are not external decorations; they are part of the syntax of the theory. In this sense, GCFT aims to make the space of possible histories smaller before analytic or quantum-field-theoretic reconstruction begins.

This should not be misunderstood as avoiding the usual obligations of quantum field theory. A constrained geometric semantics must still account for the structures that make a quantum field theory physical: a measure or weighting on admissible histories, positivity conditions sufficient for Hilbert-space reconstruction, a nontrivial continuum limit, a spectrum of excitations, anomaly and global-symmetry data, and a well-defined algebra of observables. These are not optional add-ons. They are the tests by which a proposed geometric semantics becomes a physical theory rather than merely a kinematic model.

It is therefore important to distinguish several roles that are sometimes conflated:

Structure	Role in GCFT
Measure	semantic weighting/counting of admissible histories
Positivity	consistency condition for quantum reconstruction
Continuum limit	scaling/refinement theorem
Spectrum	output of reconstructed dynamics
Anomalies	obstruction/global consistency data
Observables	readout algebra or semantic target
Effects	controlled changes of regime or presentation

In particular, the term *effect* is not meant as a catch-all for every analytic or quantum-field-theoretic obligation. Effects are controlled operations internal to the semantic framework: examples include Wick rotation, coarse-graining, measurement-induced boundary updates, quotient/readout projections, renormalization-like scale changes, and the introduction of metric-sensitive Yang–Mills dynamics over a more topological compositional core. These operations may transform the semantic data of the theory, but they do not by themselves replace that data.

Wick rotation. For example, Wick rotation may be regarded as an effect because it changes the presentation of a geometric-computational history from a Lorentzian dynamical regime to a Euclidean or moduli-sensitive regime in which instanton-like sectors, compactness phenomena, and tunneling configurations become more visible. But the legitimacy of this operation is not automatic. A Wick-rotated presentation must still satisfy the relevant reconstruction conditions, such as reflection positivity, if it is to define a physical Lorentzian quantum theory. Thus, Wick rotation may be an effect, while reflection positivity is a constraint on that effect.

Measurement, continuum limit and anomalies. Similarly, measurement may be treated as an effect in the sense that it changes boundary conditions, admissible continuations, or observer-accessible readout data. But the probabilities associated with measurement outcomes require a measure or weighting on the relevant space of admissible histories or moduli. The measure is not merely an effect; it is part of the semantic data of the theory. Likewise, the observable algebra is not an effect, but the target through which the theory exposes gauge-invariant or regime-invariant information:

holonomy traces, Wilson-loop-type quantities, defect charges, boundary operators, spectral data, or moduli invariants.

The continuum limit also has a different status. Coarse-graining or renormalization may be represented by effect-like operations, but the existence of a nontrivial continuum limit is a theorem to be proved, not an operation to be postulated. A finite or regulated helical system may exhibit gaps, stable sectors, or finite-dimensional readouts, but the physical question is whether these structures survive refinement. In the Yang–Mills setting, for instance, one must eventually show that the relevant observables converge and that any proposed spectral gap remains uniformly bounded below in the continuum limit.

Anomalies [AW84, Fre14] occupy yet another role. In the present language, an anomaly should be viewed as an obstruction to making local symmetry, gauge, or gluing data globally consistent without additional boundary or higher-categorical structure. Thus, anomaly data are naturally related to failed descent, defect terms, or nontrivial cocycles in the global organization of the theory. They are not errors to be ignored; they are part of the global consistency structure that any serious gauge semantics must reproduce.

The intended architecture is therefore as follows:

- (a) **Core geometry.** Helical local normal forms, admissible gluing, transport, and defects.
- (b) **Semantic data.** Measures, observable algebras, boundary categories, and holonomy readouts.
- (c) **Effects.** Wick rotation, measurement, coarse-graining, quotient/readout projection.
- (d) **Consistency conditions.** Gauge invariance, positivity, anomaly cancellation, or inflow, functoriality.
- (e) **Outputs.** Continuum limits, reconstructed Hilbert spaces, spectra, gaps, invariants.

This separation is essential. The effect structure organizes changes of presentation and regime, but it does not excuse the theory from reconstructing the physical content normally supplied by quantum field theory. Rather, the goal is to reorganize where these obligations enter. GCFT seeks to build admissibility into the geometric syntax of histories, then recover measures, observables, spectra, and continuum behavior from the resulting constrained semantic structure. If successful, the framework would not eliminate the analytic burdens of quantum field theory, but would relocate them to a setting in which the allowed histories, boundary data, and gauge-compatible readouts are already geometrically disciplined.

On topological Lagrangians. A useful historical precedent for our viewpoint is the Atiyah–Jeffrey [AJ90] reinterpretation of Donaldson–Witten theory via the Mathai–Quillen formalism. Their perspective shows that, at least in the cohomological/topological setting, a gauge-theoretic action need not be understood merely as a dynamical law or a formal infinite-dimensional integral; it can instead be read as a geometric localization device that extracts invariant content from the moduli space of solutions. In modern terms, the action behaves like an infinite-dimensional Thom form or Euler-class representative, concentrating the theory onto gauge-invariant data rather than raw field configurations. This is closely aligned with the present setup: we likewise distinguish a topological/core semantics from the surrounding metric-dependent machinery, and we treat evaluation as the extraction of stable geometric information rather than as brute-force functional integration. Atiyah–Jeffrey does not provide the full constructive apparatus sought here, nor does it resolve the genuinely dynamical problems of four-dimensional Yang–Mills. Still, it offers a precise and respected prototype for the claim that gauge actions can be reinterpreted geometrically as evaluators of invariant content.

Why Lagrangians appear: local scoring, gluing, and invariance. Given the above comparison, a recurring question from the physics side is why then our dynamics are presented (in the fluidic/gauge regimes) via an “action principle” in the first place. In the present framework, this is not an aesthetic choice but a structural one. Once one insists that (a) evolution is *local*, (b) composite processes are formed by *gluing*, and (c) the description is *coordinate/gauge invariant*, the natural notion of “cost” for a geometric process is forced into the Lagrangian form.

Local scoring and additivity under gluing. Let M be a spacetime region (or computational bordism) and let ϕ denote the relevant geometric field data (e.g. a connection, framing field, or helix-state field) subject to prescribed boundary conditions on ∂M . We seek a functional

$$S_M[\phi] \in \mathbb{R}$$

interpreted as the score/cost of realizing the process ϕ on M . The minimal structural requirements are:

- (a) **Locality.** $S_M[\phi]$ depends on ϕ through its local jets (a finite number of derivatives), so that modifications supported in a small region only affect the score locally.
- (b) **Gluing additivity.** If $M = M_1 \cup_\Sigma M_2$ is obtained by gluing along an interface Σ and ϕ restricts compatibly, then

$$S_M[\phi] = S_{M_1}[\phi|_{M_1}] + S_{M_2}[\phi|_{M_2}],$$

up to boundary/interface terms that depend only on the glued data on Σ [Ati88].

- (c) **Invariant symmetry.** S is invariant under the relevant redundancy group: diffeomorphisms (when applicable) and gauge transformations for connection-valued data.

Under these hypotheses, the functional must be representable (up to boundary terms) as an integral of a *local density*,

$$S_M[\phi] = \int_M \mathcal{L}(\phi, \nabla\phi, \dots),$$

where $\mathcal{L}$ is built from coordinate- and gauge-invariant contractions of the available geometric tensors (metric, volume form, curvature, covariant derivatives, etc.). This is the precise sense in which a Lagrangian is the canonical “local score” compatible with gluing and invariance.

Stationarity, not optimization. The variational condition

$$\delta S_M[\phi] = 0$$

does not assert that Nature “minimizes” S in a global sense; rather, it singles out histories that are *stable under first-order perturbations* with fixed boundary data, yielding Euler–Lagrange equations [Arn89]. This stationarity principle is the natural compatibility condition between local scoring and compositional evolution, and it is the mechanism by which continuous field descriptions produce constraint equations and conservation laws via Noether-type arguments [Noe18].

Topological core versus evaluator. In the GCT viewpoint, an action decomposes conceptually into a *compositional core* that depends only on topological/geometric structure needed for gluing and typing of interfaces, and an *evaluator term* that penalizes “difficulty” (curvature/strain/instability) and selects physically realizable histories. This mirrors our BF/Yang–Mills split. BF-type terms are first-order, local, and naturally compatible with gluing; they encode kinematic constraints (flatness, defect charges, coupling to sources). Yang–Mills-type terms introduce a metric-sensitive cost for curvature and thereby define an effective dynamics/regularization. Concretely, for a connection A with curvature F_A , one has the paradigmatic pair

$$S_{\text{BF}}(A, B) = \int_M \text{tr}(B \wedge F_A), \quad S_{\text{YM}}(A) = \int_M \|F_A\|^2 \text{vol}_g,$$

where S_{BF} supports gluing-amenable constraint semantics and S_{YM} acts as an evaluator (a resource/energy functional) selecting stable configurations. In our setting, the “fully fluidic” regime corresponds to the existence of such a local invariant evaluator; lower regimes (quasi-fluidic, symbolic) arise when only coarser readouts of the geometry are verifiable or when dissipation/coarse-graining dominates.

Dissipation and arrows. Finally, many effective evolutions used in modeling (including ours in certain regimes) are not strictly Hamiltonian. This is compatible with the above: one may pass to Euclidean signature or introduce a dissipation functional so that evolution follows a gradient-like flow for the evaluator (a Lyapunov monotone). In that case, the Lagrangian/action continues to serve as the correct local score compatible with gluing. At the same time, the *effective* time evolution becomes a semigroup on the verifiable state space.

Summary. Lagrangians arise because they are the unique local, gluing-additive, symmetry-respecting scoring rules on geometric processes, and the variational principle expresses stability of admissible histories under perturbation. In our language, they are the evaluator layer that sits atop a compositional (often topological) core.

Asymmetry of regimes. Recall again table 7. The four regimes split into “inner” (Symbolic, Fluidic) and “outer” (Quasi-fluidic, Gauge) types as detected by the join commutator (4.1):

- (a) **Quasi-fluidic (outer).** $[J_1, J_2] = e$ and the local symmetry on phases is abelian. Here *independence is a coordinate choice* via decoupling if need be.
- (b) **Symbolic (inner).** $[J_1, J_2] = e$ by construction; helices are *intentionally glued* through a shared discrete update table.
- (c) **Fluidic (inner).** generically $[J_1, J_2] \neq e$ couplings are path-dependent via non-abelian holonomy and helices are *dynamically glued*.
- (d) **Gauge (outer).** typically $[J_1, J_2] \neq e$, but there is gauge redundancy; after gauge-fixing, field lines behave as independent modules at the representation level, so *independence is a gauge choice* as the dynamics remains non-commutative.

Gluing and the sectors of ω -logic. There is a curious “outer vs. inner” asymmetry in the chart of n -logics. In the symbolic and fluidic bands, helices are defined as glued; in the quasi-fluidic and gauge bands, this need not hold. Under a C_2^4 embedding inside QF, joins commute yet are not always necessary (decoupling by normal form), whereas in the gauge band, joins play a different role altogether (gauge transformations). Thus, the observed breakdown of compulsory gluing toward the extreme ends of ω -logic should not be read as a loss of global compositional structure. Rather, it indicates that *local* gluing is no longer primitive there: in the quasi-fluidic band it becomes coordinate-dependent through decoupling and holonomy, while in the gauge band it becomes representation-dependent through gauge redundancy. What remains invariant is the global closure law governing admissible transport, transition, and regime change.

Relation to geometric field theory and structured bordism. The use of the phrase “geometric computational field theory” is meant to distinguish the present construction from existing frameworks for *geometric field theories*. In particular, Ludewig–Stoffel developed a general framework for geometric functorial field theories, in which bordisms are equipped with geometric structures and field theories are required to vary smoothly in families [LS21]. Their one-dimensional classification, where geometric field theories over a manifold are related to vector bundles with connection, is especially close in spirit to the transport-theoretic intuition used here. In the present work, however, the emphasis is not on classifying geometric field theories in general, but on isolating a particular computational enrichment of such structures.

Similarly, Grady–Pavlov’s Geometric Cobordism Hypothesis provides a broad classification framework for fully extended field theories on bordisms carrying geometric structure, including metrics, connections, complex and symplectic structures, principal bundles with connection, and geometric string structures [GP21]. This work should be viewed as complementary to that program rather than as a replacement for it. The role of GCFT is to specify a particular kind of geometric structure whose data are computational: helical state, local update laws, admissible interface gluing, holonomy readouts, regularity constraints, and resource-sensitive observational equivalence.

The classification discussed here is intentionally narrower in scope than the classification problems studied by Ludewig–Stoffel or Grady–Pavlov. We do not claim to classify all geometric field theories, nor to supply a new proof of the Geometric Cobordism Hypothesis. Rather, we identify a family

of computationally decorated bordism-like systems and describe how their boundary data, gluing operations, and gauge-invariant readouts can be organized functorially. For a complete overview of the Computational Cobordism Hypothesis, see appendix A.8.

4.8 Psi-Geometric

In light of our results, how can the existence of GCMs *help inform us* about potential interpretations of quantum mechanics?

It is plausible that the quantum state is an *ontic geometric structure*, a twin helix bundle with connection and density, while the usual wavefunction ψ is a *coordinate description* of that structure; probabilities arise from equivariant transport of density, and measurement is bordism-induced branching within the same geometry. This interprets ψ -ontic in the sense of [PBR12] (up to gauge), with an epistemic role for ψ as a representation.

Kinematic ontology. Fix a spacetime region M (nonrelativistic: $M \subset \mathbb{R}^3$ with time parameter; relativistic: a globally hyperbolic spacetime). The *beables* are:

- (O1) A twin helix fiber bundle $\pi : \mathcal{H} \rightarrow M$ with fiber carrying two oriented strands (chirality data), equipped with a $U(1)$ connection A (phase transport) and a smooth positive density ρ on M (helix-flux density).
- (O2) Global topological data: linking/holonomy classes of sections of $\mathcal{H}$ (“join” structure), which encode entanglement and AB-type effects.

Choosing a local trivialization of $\mathcal{H}$ yields fields (R, S) with

$$\psi(x) = R(x)e^{\frac{i}{\hbar}S(x)}, \quad R = \sqrt{\rho}, \quad \mathbf{v} = \frac{1}{m}(\nabla S - q\mathbf{A}),$$

and (relativistically) a conserved current J^μ with $\nabla_\mu J^\mu = 0$ and $J^\mu \sim \rho v^\mu$.

Dynamical Ontology.

- (D1) *Madelung/Schrödinger form.* Writing $\psi = Re^{iS/\hbar}$ gives the continuity and Hamilton–Jacobi pair

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad \partial_t S + \frac{(\nabla S - q\mathbf{A})^2}{2m} + V + Q = 0, \quad Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 R}{R}.$$

- (D2) *Helix-geometric form.* Q is a curvature/torsion scalar of the amplitude sheet induced by π and A ; $\mathbf{v}$ is the horizontal lift direction set by the connection. In relativistic settings, we use Tomonaga–Schwinger evolution $\Psi[\Sigma]$ on Cauchy surfaces Σ and define J^μ covariantly from the helix connection (spinorial or scalar sector as appropriate) [TV99].

A measurement is a *bordism surgery* in the coupled system–apparatus helix bundle: new, boundary/join conditions modify the allowed horizontal lifts. Dynamically, this selects an attracting sector (branch) of the global helix geometry. The post-measurement ψ is the restriction/re-expression of the same ontic object to the surviving branch; no stochastic primitive is added.

Probabilities and the Born rule. The density $\rho = |\psi|^2$ is *equivariant* under the guidance flow:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0.$$

Given typical initial ensembles (Liouville/mixing with respect to the helix dynamics), frequencies in repeated experiments converge to $\int_\Omega \rho dx$ on any detector region Ω . Thus, probabilities are *epistemic-dynamical* (ignorance over initial micro-geometry) while the underlying evolution remains deterministic [DGZ92].

Multi-particle states are single glued geometric objects: linked/braided helix fibers with shared holonomy. Bell nonlocality arises from this *shared geometry* (one global guidance field) rather than superluminal signals; operational no-signalling is preserved. AB shifts are the holonomy of A on a multiply connected base; interference channels move with $\Delta\varphi = (q/\hbar)\Phi$ while local forces vanish.

We use a covariant current J^μ and surface-based Tomonaga–Schwinger dynamics to avoid a preferred frame. Worldlines are integral curves of $v^\mu = J^\mu / \sqrt{-J^\nu J_\nu}$; statistics on any spacelike Σ agree by current conservation.

PBR and the Status of ψ . Following [HS10], we distinguish ψ -ontic models, in which distinct quantum states correspond to distinct ontic states, from ψ -epistemic models, in which ψ represents incomplete information about an underlying state. The epistemic reading has considerable explanatory power, as illustrated by [Spe07], but its limitations around Bell/Kochen–Specker [KS67] phenomena motivate the stronger ψ -geometric position adopted here. Let Λ denote the ontic state space of GCT beables:

$$\Lambda = \{(\mathcal{H} \rightarrow M, A, \rho; \text{holonomy/linking})\} / \text{gauge}.$$

- (a) **ψ -ontic (structural).** If ψ_1 and ψ_2 are not related by a gauge transformation and global holonomy equivalence, then the induced pairs $(\rho_1, \nabla S_1 - q\mathbf{A})$ and $(\rho_2, \nabla S_2 - q\mathbf{A})$ define *disjoint* subsets of Λ . Hence, distinct quantum states correspond to distinct ontic geometries—compatible with PBR.
- (b) **ψ -epistemic (representational).** A particular ψ is a *coordinate* on the gauge-equivalence class $[\lambda] \in \Lambda$ (choice of section/trivialization and coarse-graining). Different observers may use different ψ -charts for the same $[\lambda]$ without altering beables.

A Summary of Postulates.

- P1. Ontology.** Physical systems are twin helix bundles $(\mathcal{H}, A, \rho)$ over spacetime with specified holonomy/linking data. These structures are real (ontic).
- P2. Dynamics.** Evolution is deterministic guidance of defects/flux along the horizontal distribution set by A ; in coordinates, it reproduces Schrödinger/Dirac ([Mad27]) with Q a geometric scalar.
- P3. Equivariance.** $\rho = |\psi|^2$ is transported by the guidance field, yielding Born statistics for typical ensembles.
- P4. Measurement.** Coupling to an apparatus induces bordism surgery that selects an attracting branch; “collapse” is the re-expression of the same ontic geometry on that branch.
- P5. Nonlocality.** Entanglement is shared holonomy/bordism in one global object; no-signalling holds.
- P6. Relativity.** The law is tensorial (via J^μ) and surface-local (Tomonaga–Schwinger), avoiding a preferred frame.

This gives a ψ -ontic, computable, and geometric ontology (no configuration-space fluid, a concrete origin for the “quantum potential” Q , the shape operator of the helix amplitude sheet), and a unified account of AB phases, interference, and entanglement via holonomy and join structure [Boh52a, Boh52b]. What remains open and falsifiable is (a) a complete relativistic foliation-independence proof for world-lines in interacting QFT; (b) a first-principles determination of abelian coupling (fine-structure constant) from helix micro-geometry, and (c) loop-level matching of abelian β -function from helix excitations.

Summary. In GCT, ψ is *ontic as geometry and epistemic as description*. The “real” thing is the helix bundle with its connection, density, and holonomy; ψ is a chart on that object. Probabilities arise from deterministic equivariant flow, and measurement is geometric branching, giving a PBR-compatible, computable interpretation of quantum mechanics.

Closing Remarks on THM_4 and the role of V_4 . We do not identify THM_4 with particular Standard Model particle families, nor do we claim that it directly realizes continuous gauge groups. Its significance is more primitive. At the kinematic level, the twin helix substrate carries two independent involutive operations:

$$\chi : (\text{chirality flip}), \quad \omega : (\text{orientation flip}),$$

with

$$\chi^2 = \omega^2 = \text{id}, \quad \chi\omega = \omega\chi.$$

Hence, the substrate supports the minimal nontrivial abelian group generated by two commuting involutions,

$$\langle \chi, \omega \rangle \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \cong V_4.$$

Equivalently, if one labels the four helical sectors by

$$\mathcal{H}_4 = \{THM_4^{(\epsilon,\delta)} : \epsilon, \delta \in \{\pm 1\}\},$$

where ϵ records chirality and δ records orientation, then $\mathcal{H}_4$ is naturally a V_4 -torsor: the group acts freely and transitively by toggling these two binary invariants. In this sense, V_4 should be understood not as an incidental symmetry, but as the minimal algebraic skeleton of the twin helix substrate.

This point is strengthened by the fact that the same algebra arises in several canonical settings. First, the one-qubit Pauli group modulo its scalar center satisfies

$$\mathcal{P}_1/\{\pm 1, \pm i\} \cong \{I, X, Y, Z\}/\{\pm 1, \pm i\} \cong V_4,$$

so that V_4 is precisely the phase-stripped algebra of two independent binary flips. Second, the four Bell states form a natural V_4 -torsor under local Pauli operations: starting from any one Bell state, the action of two commuting $\mathbb{Z}_2$ generators traverses the remaining three. Third, on a torus, one has

$$H^1(T^2; \mathbb{Z}_2) \cong \mathbb{Z}_2 \times \mathbb{Z}_2,$$

corresponding to the four spin structures determined by periodic/antiperiodic boundary conditions along the two independent cycles. In each case, the same abstract pattern appears: two commuting binary distinctions generate a four-sector structure. In our setting, the relevance of V_4 is not that it encodes the exact P or T symmetries of the Standard Model, but that it supplies the smallest nontrivial algebra in which independent binary geometric distinctions can be composed, inverted, and transported [LM89].

This perspective may also help clarify the relation to quantum nonlocality. Historically, Einstein, Podolsky, and Rosen [EPR35] identified a genuine incompleteness in any theory satisfying locality and separability; Bell [Bel64] later showed that this package is incompatible with the observed quantum correlations. A deterministic but globally constrained theory preserves no-signalling while relaxing measurement independence, so the EPR impulse toward a deeper completion survives, even though the specific locality/separability package would, in the spirit of Bohm, yet again, be abandoned.

Our universe may be built from the smallest groups, not just the richest.

A Appendix

A.1 “Hidden” Variables?

It should be noted that two canonical join patterns for twin helices lead to distinct gauge–geometric behaviors.

Definition A.1 (Apex Twin Helix). An *apex-joined* configuration is obtained by collapsing a boundary circle to a point, producing a conical join $C(S^1)$ (a stratified space with a point singularity). Equivalently, remove the apex and view the result as $M \setminus \{p\}$ equipped with a connection whose curvature may carry a localized charge at p (e.g. a Dirac monopole in the abelian case). Such joins model sources/sinks or defect cores, such as optical vortices, but most prominently **Dirac cones**.

By contrast, our *THM* as in definition 1.8 are the objects used for deterministic logic flow, as in the **Schrödinger equation**. The operational distinction is *holonomic*:

Proposition A.1 (Base vs. Apex Holonomy). *Let A be a connection on M .*

- (a) (Base-joined) *If F is smooth near the join annulus and B is simply connected there, then for sufficiently small loops γ linking the join, the holonomy class $[\text{Hol}_\gamma(A)]$ is trivial; memory arises only from the distributed curvature, not from a singular core.*
- (b) (Apex-joined) *If the join collapses to p and A has curvature with total flux through a small linking 2-sphere S_p^2 equal to a quantized charge, then small loops γ encircling p acquire nontrivial holonomy classes. In $U(1)$,*

$$\text{Hol}_\gamma(A) = \exp\left(i \int_{S_p^2} F\right) = e^{2\pi i k} \quad (k \in \mathbb{Z}),$$

and in the nonabelian case the conjugacy class of $\text{Hol}_\gamma(A)$ is determined by the localized topological data (e.g. degree of a normalized frame map, or the G -bundle class on S_p^2).

Resolution by blow-up. An apex singularity can be *resolved* by the real blow-up of M at p , replacing p with the unit normal S_p^2 . The connection extends over the exceptional sphere with a (possibly nontrivial) bundle; its Chern/characteristic class on S_p^2 is exactly the stored charge. In this sense, apex-joined and base-joined pictures are two presentations of the same data: defects as boundary classes vs. smooth transport with an extra S^2 boundary component carrying flux.

Feature	Twin Helical Manifolds	The Apex Twin Helix
Join point	Smooth base join	Singular apex join
Topology	Globally smooth manifold	Manifold with point singularity
Flow	Bidirectional transport along an annulus	Radial (source/sink) around a defect core
Chirality	Controlled by local geometry	Emerges from defect topology
Singularities	Avoided (non-singular)	Essential (e.g. Dirac monopoles)
Field role	Symbolic/geometric logic (ordered joins)	Physical spinor dynamics
Purpose	Analog computational flow	Topological charge carriers

Summary. The “duality” is a clean dichotomy of join conditions: *smooth* joins emphasize programmable transport (deterministic logic), while *singular* joins emphasize conserved charges (defect logic). Hence, helicity plays two separate roles in physics. Holonomy conjugacy classes and curvature flux rigorously control both.

Likewise, as illustrated in fig. 5b, a complex plane wave

$$\Psi(x, t) = Ae^{i(kx - \omega t)}, \quad \Re \Psi = A \cos(kx - \omega t), \quad \Im \Psi = A \sin(kx - \omega t),$$

traces a Type 1 *THM* in space: the real/imaginary components are $\pi/2$ out of phase and form two counterwinding strands whose joints occur at equal-phase points. In Maxwell’s $U(1)$ sector (massless, no self-interaction), the physical photon inherits exactly this geometry: a pair of phase-locked, coaxial, counterwinding helices—the “helical orb.”

Question. *But why the difference? Why do photons appear as Type 1 while Yang–Mills instantons (fig. 5a) present as Type 3 dual-cone configurations?*

Definition A.2 (Helical Normal Forms). We say a localized geometric computational configuration on the twin helix substrate is:

- **Type 1 (Helical Orb Normal Form).** A coaxial pair of counterwinding helices as an abelian wave with a common propagation axis and phase-locked transverse frame; joints are equal-phase loci.
- **Type 3 (Symmetric Flow Normal Form).** A compact, oriented cobordism in spacetime whose incoming/outgoing faces carry oppositely oriented helical frames, visualized as two opposed cones joined along a finite 3-volume.

Proposition A.2 (Symmetry–Energetics–Boundary Selection Principle). *Given a gauge sector G and boundary data:*

- (a) **Abelian, propagating waves** (e.g. photons in $U(1)$) *minimize the geometric action by pure transverse frame rotation along an open worldline, yielding the Type 1 normal form.*
- (b) **Nonabelian, (anti)self-dual finite-action events** (instantons in $SU(2)/SU(3)$) *minimize by a compact, topologically charged 4D configuration interpolating between vacua with different $k \propto \int \text{tr}(F \wedge F)$. The oriented in/out faces enforce opposite frame orientations, producing the Type 3 dual-cone normal form.*

Proof sketch. (1) In the abelian case, holonomy abelianizes to a single phase; with no self-interaction or topological charge, the energetic minimizer is a traveling wave: equal-amplitude quadrature components assemble into a coaxial, phase-locked twin helix (Type 1). There are no junctions, cones, or domain walls to pin the frame. (2) For nonabelian (anti)self-dual solutions, the relevant invariant is:

$$k = \frac{1}{8\pi^2} \int \text{tr}(F \wedge F)$$

so finite action requires compact support in Euclidean spacetime and yields a vacuum-to-vacuum *cobordism*. The in/out orientations of the action density impose opposite normals on the two temporal faces of the event, and the lifted spin/transverse frame rotates accordingly, giving the dual, opposed cones (Type 3). These are *events* (tunneling) rather than asymptotic quanta. $\square$

Summary. Photons are asymptotic Type 1 helical waves in an abelian phase channel; instantons are nonabelian, topological *events*—Type 3 dual-cone cobordisms—not particles. The distinct pictures are fixed by (a) symmetry (abelian vs. nonabelian), (b) energetics (wave transport vs. self-dual soliton), and (c) boundary class (open propagation vs. vacuum-to-vacuum interpolation).

Question. *But what about the Type 2 and Type 4 twin helix configurations? What might they represent?*

Proposition A.3 (Composite Normal Forms for Type 2 and Type 4). *Unlike Type 1 and Type 3, which arise naturally as asymptotic wave and instanton events normal forms, Type 2 and Type 4 are best interpreted as composite or constrained channels of gauge-theoretic flow.*

- (a) **Type 2 (Twisted pair dual carrier form)** *is the natural normal form for an opposite-helicity paired channel: two transverse spin-1 excitations propagating along a common axis with vanishing net helicity. In QCD, this is the structural regime of scalar and pseudoscalar two-gluon composites, i.e. the 0^{++} and 0^{-+} glueball channels.*
- (b) **Type 4 (Co-propagating twins dual carrier form)** *is the natural normal form for a same-helicity co-propagating channel: two aligned vector excitations whose twist data reinforce rather than cancel. In gauge theory, this appears most directly in helical-string/flux-tube models of QCD, and structurally it matches tensorial channels built from same-helicity vector data.*

Heuristic justification. For Type 2, the geometric obstruction of lemma 1.3 already indicates that this configuration is not the generic shape of a free propagating mode in $\mathbb{R}^3$, but rather a constrained or

resolved pairing. This is precisely what one expects of helicity-cancelling two-gluon sectors: the individual vector modes remain transverse, but the composite object has scalar or pseudoscalar quantum numbers. The relevant gauge-invariant operators are schematically

$$\text{tr}(F_{\mu\nu}F^{\mu\nu}), \quad \text{tr}(F_{\mu\nu}\tilde{F}^{\mu\nu}),$$

and the two-gluon helicity formalism reproduces the observed low-lying glueball J^{PC} assignments without introducing spurious states associated with naive spin-1 constituent gluons [MBS08, BEKPM17].

For Type 4, by contrast, chirality and orientation reinforce one another. The resulting configuration is not cancellation but coherent co-propagation. This is the regime most naturally associated with a twisted flux carrier: the QCD string itself can be modeled as helix-shaped, with physically meaningful winding density and azimuthal ordering [TN14]. At the purely representation-theoretic level, the same-helicity pairing of spin-1 modes is also the canonical route to a spin-2 object: for a massless graviton, the helicity- ± 2 polarization tensor factorizes as

$$h_{\mu\nu}^{(\pm 2)} = \varepsilon_{\mu}^{\pm} \varepsilon_{\nu}^{\pm},$$

and in double-copy language, the positive-helicity graviton arises from positive-helicity gluonic data on both sides [Hol06]. We therefore regard Type 4 not as a new elementary particle, but as the correct helical normal form for coherent same-handed flux and tensorial composite channels. $\square$

Summary. The four *THM* normal forms separate naturally into two classes. Type 1 and Type 3 are *single-carrier* normal forms: abelian propagating waves and nonabelian topological events. Type 2 and Type 4 are instead *dual-carrier* normal forms. Type 2 is cancellation-dominated and therefore appears most naturally in scalar/pseudoscalar composite sectors; Type 4 is reinforcement-dominated and therefore appears in helical flux tubes, tensor channels, and same-helicity pairings. In this sense, the relative scarcity of Type 2 and Type 4 in textbook particle imagery is not an objection to the *THM* classification, but a consequence of the fact that these geometries belong primarily to bound, confined, or effective sectors rather than to free asymptotic quanta.

Sector	Object	Dominant functional	Boundary class	Form
$U(1)$	photon	wave (no self-int.)	open, asymptotic	Type 1
$SU(2), (3)$	instanton	(anti)self-dual soliton	vacuum $\rightarrow$ vacuum ($k \neq 0$)	Type 3
$SU(3)$	(pseudo)scalar glueball	opposite-helicity pairing	confined, composite	Type 2
$SU(3)$	helical flux tube	coherent same-helicity flux	confined, stringlike/effective	Type 4

Table 13: Selection of helical normal forms by symmetry, energetics, and boundary data.

Question. *But why does nature even bother with gluing helices?* A single helix’s axial $U(1)$ phase renders a gauge; with a twin, the relative phase $\Delta\theta \in S^1$ is gauge-invariant and robust. This yields an abelian “phase logic” realizing the $C_2 \times C_2$ actions used in the symbolic regime. Likewise, gluing balances torque and stabilizes the framing, thereby making the interfaces implementable. The most commonly cited continuous symmetry of the helix (i.e., rotation about the screw axis) is not amenable to computation.

Question. *Why require smooth end caps rather than allowing singular tips?* Smooth caps are not a cosmetic assumption; they are a regularity constraint. A conical or cuspidal endpoint would allow curvature, torsion, energy density, or gauge flux to concentrate at the tip, making any apparent charge partly an artifact of a singular boundary. By capping the helix smoothly, charge-like data must instead arise from holonomy, twist, gluing, or defect topology distributed through the helical body. Thus, the cap prevents endpoint singularities from masquerading as physical structure. It also turns the terminal boundary into part of the type system: the cap fixes a closure/framing condition and restricts the admissible global holonomy classes.

Disclaimer regarding the SM and GUTs. Though we do not execute such a derivation in this work, we propose that the observed Standard Model gauge group splitting

$$(U(1) \times SU(2)) \times SU(3)$$

would reflect distinct computational primitives of the substrate: the ω -logic (binary orientation plus phase rotation) underlying the electroweak sector, and the χ -logic (triality/twist structure) underlying color.¹³ We offer this as the organizing hypothesis for how different gauge behaviors would arise from distinct geometric-computational regimes. It does *not* constitute a derivation of the observed fermion content, Yukawa sector, mass hierarchies, mixing parameters, anomaly cancellation pattern, or measured coupling constants of the Standard Model.

Nor does the present theory exclude a GUT-like phase at higher energies. What it does *not* presently anticipate is that such unification *must* take the form of a single primitive Lie-group embedding. The more natural GCFT possibility is a symmetry-enhanced locus $\mathfrak{M}_{\text{unif}}$ in the moduli of geometric-computational states, equipped with a unified transport or coherence structure $\mathcal{G}_{\text{unif}} \rightrightarrows \mathfrak{M}_{\text{unif}}$. The familiar low-energy gauge sectors would then arise through distinct reductions, projections, or stabilized readouts,

$$\mathcal{G}_{\text{unif}} \longrightarrow \mathcal{G}_3, \quad \mathcal{G}_{\text{unif}} \longrightarrow \mathcal{G}_2, \quad \mathcal{G}_{\text{unif}} \longrightarrow \mathcal{G}_1,$$

with effective local descriptions modeled by $SU(3)$, $SU(2)$, and $U(1)$, respectively. A conventional simple group such as $SU(5)$ might still appear at a particularly symmetric point or chart—for example, as an isotropy group

$$G_m = \text{Aut}_{\mathcal{G}_{\text{unif}}}(m), \quad m \in \mathfrak{M}_{\text{unif}},$$

but it would represent a local algebraic compression of the phase rather than necessarily its fundamental ontology. The claim made here is therefore modest but directional: the low-energy factorization of the Standard Model reflects distinct computational readouts of a common substrate-level transport structure, while the existence and geometry of any corresponding unification phase, in addition to the full particle physics phenomenology, remain beyond the scope of the present paper.

Fermions and the Higgs. Fermionic matter enters the present field theory not as an additional class of helical bodies, but as spinorial modes supported by the framed gauge-computational substrate. In standard field-theoretic language, this means that once a spin structure and compatible gauge connection are specified, fermions are represented by sections of a spinor bundle acted on by a Dirac-type operator ($\mathcal{D}_A$). The Higgs field should then be understood as an order-parameter sector that controls which spinorial modes may be paired across left- and right-chiral components. In the Standard Model, this pairing is expressed by Yukawa couplings, whose nonzero vacuum expectation value produces Dirac mass terms after electroweak symmetry breaking. In GCFT language, this suggests that fermion masses measure the stable cost of coupling otherwise distinct spinorial defect channels through a scalar order parameter. In this way, the Higgs mechanism is not treated here as an arbitrary mass insertion, but as the effective low-energy expression of a geometric gluing rule between chiral spinorial sectors.

Question. *Proof of Yang–Mills instantons being composed of holonomic helices has so far not been confirmed. How then can we safely verify their existence?*

The issue is that the helical structure we posit resides in the *connection* A (parallel transport in the internal bundle). In contrast, direct observables are gauge-invariant functionals of the *curvature* $F = dA + A \wedge A$ or of holonomies (Wilson loops). A gauge transformation can erase any purely local feature of A ; what survives to observation are scalar densities such as $\text{tr } F_{\mu\nu} F^{\mu\nu}$ and $\text{tr } F \wedge F$, which appear as lumps, rings, or cone-like envelopes in lattice plots and experiments. The micro-handedness of a helical A thus projects to gauge-invariant *envelopes* rather than visible corkscrews.

Helical winding is a property of the *fiber* (the G internal directions) over spacetime. Detectors and standard lattice diagnostics live on the base manifold and access only fiber-invariant combinations. Projecting fiber twist onto base scalars discards orientation data but retains energy and topological charge, yielding the observed dual-cone profiles for instanton-like configurations. Real instruments and lattice operators average over space, time, and ensembles. Without a phase-locked reference to the internal connection, linear twist contributions cancel, leaving only quadratic or topological signals. Hence, the helical microstructure is washed out, while $\text{tr } F^2$ and $F \wedge F$ remain.

¹³Echoing our previous disclaimer on chirality and spinors, the word *helicity* has its standard meaning in QCD: it is the spin projection of a massless gauge excitation along its momentum, and hence a representation-theoretic quantum number. This should not be identified with the geometric handedness of twin or Y-junction-type helical manifolds as the ontic helical/framed substrate proposed in GCT. The relation, if any, is indirect: helicity is a gauge-theoretic/kinematic readout of spin transport, whereas the helical geometry in this paper is the proposed carrier of phase, framing, holonomy, and gluing data.

Gauge sector	Standard mode	GCFT interpretation
$U(1)$: abelian phase transport; scalar holonomy and phase ledger.	Photon after electroweak symmetry breaking, corresponding to the unbroken $U(1)_{\text{em}}$ direction.	Interpreted as a massless abelian connection mode: a freely propagating phase/holonomy disturbance in the surviving electromagnetic readout channel.
$SU(2)_L$: non-abelian internal frame rotation; Frenet–Serret-like transport of a two-dimensional internal frame.	Weak bosons W^1, W^2, W^3 , which after mixing with hypercharge produce $W^\pm, Z$, and the photon.	Interpreted as internal frame-rotation modes. Once the Higgs field selects a vacuum direction, the broken weak directions become massive $W^\pm/Z$ -type transport modes, while the unbroken direction stays massless.
$SU(3)_c$: non-abelian color-frame transport; path-ordered holonomy with noncommuting gluings.	Eight gluon fields G_μ^a , $a = 1, \dots, 8$, valued in $\mathfrak{su}(3)$.	Interpreted as local color-connection modes. Unlike the photon, gluons self-interact and are not freely observable in isolation; gauge-invariant readouts occur via color-singlet composites, flux tubes, Wilson loops, and confinement-scale excitations.

Table 14: Gauge sectors and their geometric-computational interpretation. The $U(1)$ sector supplies abelian phase transport; the $SU(2)_L$ sector supplies weak internal frame rotation; and the $SU(3)_c$ sector supplies non-abelian color-frame transport. In the GCFT interpretation, gauge bosons are not additional helical bodies, but connection modes of the framed substrate. The phrase “Frenet–Serret-like” is used here only as a geometric analogy: the weak ($SU(2)_L$) frame is an internal gauge frame, not the literal spatial Frenet–Serret frame of a curve. The point is that both structures describe transported frame data, but the weak frame lives in an internal representation space and is constrained by electroweak gauge symmetry.

Many diagnostic solutions (instantons, calorons) are Euclidean saddles. The helix lives in the Euclidean/complexified connection; in real time, one measures its *consequences* (selection rules, rates, envelopes), not the helical motion itself. The appropriate observables are *nonlocal* and gauge sensitive:

- **Wilson–loop tomography.** Families of loops $\mathcal{W}(\gamma) = \text{tr } \mathcal{P} \exp \int_\gamma A$ detect non-abelian holonomy; loop–shape/systematic studies can reconstruct local twist consistent with $\text{tr } F^2$ envelopes.
- **Gauge–fixed diagnostics.** In a fixed gauge one may visualize the Chern–Simons 3–form K_μ to track handedness (useful for diagnosis, though not itself gauge invariant).
- **Analog platforms.** Phase–locked probes in $SU(2)$ optical media, cold atoms with synthetic gauge fields, or superconducting circuits access non-abelian holonomy directly, providing experimental surrogates for helical transport [CZX⁺19, GJOS14].

Summary. The helical substrate is “invisible” for the same reason the vector potential is: nature reveals only gauge–invariant aggregates locally and holonomy nonlocally. Helices manifest as the *geometry of transport* (path–ordered phases), while energy/topology plots display their *envelopes*. To expose the helices, one must either reconstruct in a fixed gauge or measure holonomy via nonlocal probes.

Psi–geometric semantics and “quantum quandaries”. Our program does not attempt to explain *why* helices in nature behave as they do; that work is best left to our colleagues in adjacent departments. Rather, it takes seriously what their behavior appears to implement: something simultaneously ordinary and mildly unsettling—they enforce global consistency in a world that, operationally, only permits local access.

A useful way to read this is through the categorical perspective advocated by Baez [Bae06]: many familiar “quantum quandaries” are not additional physical mysteries, but structural consequences of working in a non-cartesian, process-theoretic setting (closer in spirit to **nCob** than to **Set**). In such a setting, composition is *gluing* rather than pairing; one should not expect a canonical diagonal (“copying”) nor a cartesian product that cleanly models independent subsystems. From this viewpoint, interference and AB-type phenomena are the generic signatures of phase transport and holonomy: the relevant compatibility data is global (encoded in connections and their path-dependence), while local descriptions are related by gauge. Entanglement, likewise, is best understood as the failure of factorization of the underlying glued state: two distant “systems” may simply be two boundary ports of a single global geometry prepared in their common past. No superluminal signalling is implied; the observed correlations reflect shared constraints on how boundary data can be capped, compared, and recombined.

A.2 Invariant Content of Geometric Computation

A recurring danger in any geometric formulation is to confuse a useful representative with the invariant object it represents. In the present framework, the twin helix is not intended to be a privileged shape in isolation. Its role is to provide a local normal form in which several computationally relevant structures become simultaneously visible: chirality, orientation, phase, winding, framing, transport, and admissible gluing. The invariant content of the theory must therefore be sought in the data that survive deformation, gauge change, coarse-graining, and change of presentation.

At the local level, a helical cell carries elementary geometric labels such as chirality,

$$\chi = \text{sgn}(\tau),$$

axial orientation ω , winding number

$$w \in \mathbb{Z},$$

compact phase $\theta \in S^1$, and framing data along the helical axis. These quantities explain why a helix can function as a local carrier of state: it combines cyclic, directed, chiral, and transport data in one geometric object. However, these are local invariants only. They do not yet define the global gauge-theoretic or computational content of the theory.

The first genuinely global invariant is the admissible gluing structure. Given local helical pieces H_A and H_B , the relevant question is whether their boundary, framing, chirality, orientation, phase, and transport data admit a coherent extension. Schematically, this produces an admissible join class

$$J(H_A, H_B),$$

together with obstruction data measuring the failure of such a join. When joins compose, their commutator

$$[J_1, J_2] = J_1 J_2 J_1^{-1} J_2^{-1}$$

detects the transition from effectively abelian computation to path-dependent, non-abelian computation. Thus the join commutator is not merely an algebraic artifact: it is a regime invariant recording when the order of geometric assembly becomes computationally meaningful.

This leads naturally to a moduli problem. One should not study isolated helical representatives alone, but rather a moduli groupoid or stack

$$\mathcal{M}_{\text{adm}}$$

of admissible helical extensions. Its objects are local configurations equipped with boundary, framing, chirality, phase, connection, and readout data; its morphisms are admissible equivalences, gauge transformations, and changes of presentation. In this sense, the helix is a generator, while the gluing law and its gauge-invariant readouts constitute the invariant structure. Two presentations should be regarded as equivalent when they determine the same admissible extension data and the same observable transport, not merely when they look geometrically similar.

In the gauge-theoretic regimes, holonomy provides the natural memory invariant. Given a connection A and a loop γ ,

$$U_\gamma = \mathcal{P} \exp \left(\int_\gamma A \right)$$

is gauge-dependent, but its conjugacy class

$$[U_\gamma]$$

and class functions such as Wilson-loop observables

$$\mathrm{tr}_\rho(U_\gamma)$$

are gauge-invariant. Thus the computational content of a gauge configuration is not stored in a local representative of the connection, but in the transport data that survive around closed execution paths. In compactified or finite-temperature sectors, the corresponding distinguished invariant is the Polyakov loop

$$P(\vec{x}) = \mathcal{P} \exp \left(\int_0^\beta A_0(\tau, \vec{x}) d\tau \right),$$

which records holonomy around the compact Euclidean-time direction and is naturally interpreted here as a compactified phase ledger [Pol87].

Curvature and characteristic classes form the next layer. The curvature

$$F = dA + A \wedge A$$

measures local failure of flatness, while characteristic forms extract global topological content. In four-dimensional Yang–Mills theory, the basic example is the instanton number

$$k = \frac{1}{8\pi^2} \int_M \mathrm{tr}(F \wedge F).$$

Accordingly, an instanton-like or Type-3 sector in a geometric-computational theory cannot be justified by visual resemblance alone. It must eventually carry, induce, or interact with the same kind of topological sector data. Boundary information may similarly be expressed through changes in the Chern–Simons functional when a four-dimensional history is interpreted as a cobordism between three-dimensional gauge states.

Differential cohomology is a natural language for organizing these invariants because it packages three layers that should not be separated:

$$\text{topological class} \quad + \quad \text{curvature} \quad + \quad \text{holonomy}.$$

A geometric-computational configuration is not merely a shape, a connection, or a topological charge. It is a package of global class data, local curvature data, and loop-level transport data. Thus a differential character or differential cohomology class

$$\widehat{c}_{\mathrm{GCT}} \in \widehat{H}^\bullet$$

is a natural candidate for the invariant object attached to a helical gauge configuration [Fre00]. This viewpoint also clarifies why defects, anomalies, and generalized symmetries should not be treated as decorative additions. Defect charges, anomaly cocycles, inflow data, and higher-form symmetry classes record failures of local gauge or gluing data to descend globally without additional structure [SN24].

At the moduli-space level, one must eventually study spaces such as

$$\mathcal{M}_{\mathrm{Type-3}}, \quad \mathcal{M}_{\mathrm{helix}}, \quad \mathcal{M}_{\mathrm{adm}},$$

not merely as sets of configurations, but as structured moduli objects equipped with deformation complexes, obstruction theories, singular strata, compactness properties, and orientations. The relevant invariants may include

$$\dim \mathcal{M}, \quad \mathrm{Ind}(D),$$

boundary strata, compactification data, counts of admissible sectors, or homology groups associated to those sectors. This is the point at which helical geometry becomes visible to gauge theory and geometric analysis: the question is whether admissible helical gluing produces invariants stable under deformation.

At the observational level, the invariant object is an algebra or category of gauge-invariant readouts. For a boundary ∂M , one may seek an observable algebra

$$\mathcal{A}_{\text{obs}}(\partial M),$$

or a functorial assignment

$$Z_{\text{GCT}}(\partial M)$$

of states, categories, or observable structures to boundary data. These readouts may include holonomy traces, Wilson-loop-type quantities, defect charges, boundary operators, spectral data, or moduli-space invariants. A theory without such invariant observables remains only a proposed ontology.

Finally, any connection to mass-gap questions requires spectral invariants. One must identify a reconstructed Hamiltonian, transfer operator, Euclidean correlation operator, or other dynamical object whose spectrum has physical meaning:

$$\text{Spec}(H_{\text{GCT}}).$$

A mass gap would then appear as a lower bound

$$\Delta = \inf(\text{Spec}(H_{\text{GCT}}) \setminus \{0\}) > 0,$$

or, in Euclidean form, as exponential decay of connected correlation functions,

$$\langle \mathcal{O}(x) \mathcal{O}(0) \rangle_c \sim e^{-m|x|}.$$

Thus the relevant invariant is not the existence of helical sectors as such, but the stability of nontrivial excitation energy, or correlation decay, under refinement and passage to the continuum limit.

The hierarchy of invariant content may therefore be summarized as follows:

Level	Invariant content
Local helical	χ, ω, w, θ , framing
Gluing	$J(H_A, H_B), [J_1, J_2], \text{Obs}(J)$
Holonomy	$[U_\gamma], \text{tr}_\rho(U_\gamma), P(\vec{x})$
Curvature/topology	$F, \text{tr}(F \wedge F), k, \Delta CS$
Differential cohomology	$\hat{c}_{\text{GCT}} \in \hat{H}^\bullet$
Moduli	$\mathcal{M}_{\text{adm}}, \dim \mathcal{M}, \text{Ind}(D)$
Defect/anomaly	defect charges, α_{GCT} , inflow data
Observables	$\mathcal{A}_{\text{obs}}(\partial M), Z_{\text{GCT}}(\partial M)$
Spectral	$\text{Spec}(H_{\text{GCT}}), \Delta$, correlation decay

Among these, the central organizing object is

$$\mathcal{M}_{\text{adm}},$$

understood as a moduli groupoid or stack of admissible helical extensions equipped with holonomy readout and differential-cohomological data. This is the natural place where local helical generators, global gluing constraints, gauge equivalences, defect data, observable algebras, and spectral questions meet. If such a structure can be isolated, then the helix is no longer merely a compelling mechanism. It becomes a local witness for a deeper invariant law governing which geometric computations may be assembled, transported, observed, and deformed without changing their physical or computational content.

A.3 Higher Dimensional Algebra and Geometric Computational Field Theory

To our knowledge, no single algebraic framework presently captures computation in GCMs across the full symbolic-to-fluidic range. By this we mean a framework that simultaneously (a) composes systems under joins—both commuting and non-commuting, (b) handles local-to-global propagation through interfaces, and (c) couples continuous transport (flows/phases) with discrete readout. Lie groupoids (and stacky variants) currently provide the most serviceable language for local symmetry, defects, and gluing. However, a general categorical semantics that bridges a GCM's geometry with its computability is still lacking.

As one passes from discrete to fluidic regimes, finite algebraic invariants rapidly lose discriminative power: phase freedoms and continuous deformations overwhelm purely combinatorial structure. This suggests a hybrid treatment in which dynamical systems methods provide the evolution laws, while higher algebraic machinery provides the compositional spine. Accordingly, we use “algebra” here in a programmatic sense: the aim is a language that unifies discrete composition, continuous flow, and topological joining within a single, compositional setting.

What is needed? A viable semantics for GCMs must be simultaneously symbolic and geometric: it must (a) compose discrete states and interfaces (information–theoretic layer), (b) transport phases and flows continuously (geometric layer), (c) express gluing along boundaries and track joins, and (d) specify how update laws transform the underlying geometry (chirality/orientation/topological data).

The key obstacle is nonlocal constraint propagation across helices: interface compatibility is not decided on a single patch, and path–dependence (holonomy) persists under coarse discretization. Global states are therefore not simple products of local states but solutions of descent/compatibility conditions along the interface groupoid (as in definition 3.1). Consequently, naive local linearization or discretization erases the very invariants that govern computation.

A workable mathematical language for GCMs must support four capabilities and tie each to a concrete compositional device.

- (a) **Continuous transformation.** Represent smooth deformations (tube radius, phase, twist) and their flows.
- (b) **Local-to-global behavior.** Track how interface–local updates assemble to global states and constraints.
- (c) **Structural invariance and laws.** Express conservation and controlled change of chirality, orientation, and topological charges.
- (d) **Rich morphism structure.** Compose fragments along boundaries; allow deformations as higher morphisms.

Definition A.3 (Minimal Compositional Spine for GCMs). Let $\mathbb{G}$ be a symmetric monoidal double category of open GCMs, defined as follows:

- **Objects** (0-cells). Interfaces/boundary components equipped with the declared readout structure.
- **Horizontal 1-cells.** Open GCM fragments with specified input/output interfaces, i.e., glueable bordisms equipped with interface rules $\mathcal{I}$.
- **Vertical 1-cells.** Structure-preserving maps of interfaces (refinements, relabelings, or symmetry-compatible identifications).
- **2-cells.** Commuting squares of GCM data, interpreted as compatible deformations/flows/homotopies between fragments.
- **Monoidal product.** Disjoint union $\otimes$ (parallel composition), with horizontal composition given by gluing along matching interfaces.

Decorate $\mathbb{G}$ by a stack (or higher stack) $\mathcal{S}$ of local state objects on the site of open GCMs, valued in the geometric ambient category $\mathcal{E}_{\text{GCM}}$. Thus, each open region U is assigned a local state object $\mathcal{S}(U)$ together with descent data along overlaps and interfaces. Dynamics consists of:

- **Pulses.** Distinguished endo-2-cells representing discrete update steps.
- **Flows.** Connections or transport laws on $\mathcal{S}$ representing continuous evolution.

The point of definition A.3 is to separate the *compositional skeleton* from the choice of local state model. The double-category structure records how GCM fragments glue and deform; the stack $\mathcal{S}$ records what state data live on those fragments. The minimal semantics above should be read as a workable core rather than a final synthesis. A fuller algebraic account of GCM behavior appears to radiate outward in several tightly related directions. We group by function:

Local-to-global semantics. The first extension concerns how local computational states glue into global geometric configurations. Here, the natural language is that of *stacks*, *descent*, and *higher gauge theory* [Lur11, BS06]. One should assign to each open set U a higher groupoid of local states and transports, then recover global configurations as descent data rather than as naive products of local coordinates. In this setting, the quotient or stack language is not merely a formal decoration. Passing to a coarse orbit space may identify gauge-equivalent representatives while forgetting residual automorphisms; these automorphisms can remain physically operative as higher, global, or non-invertible symmetries of gauge data [HK26] when local computational states are glued, the relevant global object is not only a set of equivalence classes but a structured descent object retaining the symmetries by which those equivalences are realized. In the abelian/quasi-fluidic band, this collapses to familiar local systems (our $B/T/Z$ -type readouts, torus phases, and integer charges), but in the fluidic and gauge regimes, the gluing problem becomes genuinely nonabelian: one must track connections, holonomies, and coherent homotopies of gauge data under refinement and recombination [SW13]. This is exactly the setting in which higher gauge theory becomes the minimal bookkeeping required for nonabelian transport, defect fusion, and feedback under gluing. From this perspective, global GCM states are not tuples of local values but *glued objects in a differential cohesive geometric semantics*.

Differential dynamics and update laws. The second extension concerns the internal mechanics of updates. Here *synthetic differential geometry*, *jet bundles*, and the *Cartan–Spencer formalism* [Koc06, Spe69] provide the natural language for finite-order control, prolongation, and integrability. Working internally to a differential cohesive ∞ -topos (see A.8) allows vector fields, flows, infinitesimals, and differential modalities (e.g. $\int$, $\flat$, $\sharp$) to be treated as first-class objects rather than external analytic artifacts. In parallel, a future *bordism type theory* could package the same ideas proof-theoretically: types as stacks of decorated manifolds, terms as global sections or configured geometries, identity types as homotopies/deformations, and Π - and Σ -types as homotopy right/left Kan extensions along fibrations modeling parameterized families and gluing. Followed by a sequent calculus whose cut/elimination rules match horizontal/monoidal composition in the double category of open GCMs. Our aim is not to multiply formalisms for their own sake, but to isolate a setting in which update laws, prolongations, phase slips, and defect motion can be represented as executable geometric rules rather than described informally.

Observables, defects, and stable invariants. The third extension concerns observables in the genuinely higher regimes. Once one allows defects, interfaces, and nontrivial operator insertions, ordinary bundle language quickly becomes too coarse. Here *stratified topological field theory*, *factorization homology*, *K-theory*, and *noncommutative geometry* [Waa25, Con94, AF19] enter naturally. Stratified/extended TFT organizes computations by defect type and gluing pattern; factorization homology provides the corresponding local-to-global integration of observables; (possibly twisted) K -theory classifies stable sectors and conserved charges; and noncommutative geometry becomes relevant when quotienting by holonomy/gauge data destroys any useful ordinary space of states. In the abelian subtheories, these tools reduce to comparatively tame flux and phase bookkeeping, but in the fluidic/gauge band, they are the natural repositories of the data that survive after passing to moduli and quotienting by symmetry. In other words, once the geometry is sufficiently expressive, the correct outputs cease to be “numbers at points” and become stable invariants of operator-valued or defect-decorated structures.

Equivariant and Floer-type refinements. The final extension concerns the onset of genuinely equivariant and cohomological phenomena. Our base symmetry $V_4 \simeq C_2 \times C_2$ embeds in $O(2)$, but in the regimes sensitive to spinorial or orientation-reversing data one should lift along $\text{Pin}(2) \rightarrow O(2)$; the corresponding preimage of V_4 is the quaternion group $Q_8 \subset \text{Pin}(2) \subset SU(2)$. Thus, chirality and orientation reversal are not cosmetic labels but genuine equivariant structure, aligning the fluidic/gauge band with the $\text{Pin}(2)$ -equivariant viewpoint that has become central in modern Floer theory, particularly in Manolescu’s work [Man16]. Practically, the abelian/quasi-fluidic analysis still sees only the V_4 projection, while the higher gauge regime naturally asks for the $\text{Pin}(2)$ lift. This is one reason that Floer-type and Donaldson-type refinements appear not as optional embellishments, but as the likely endpoint of any serious attempt to extract stable computational invariants from the highest regime.

Remark A.1. There are, of course, many further directions, but these four themes already indicate the scale of the task. A full algebraic description of GCM behavior likely requires a synthesis beyond the reach of any single mathematician—something closer to a coordinated language than a solitary breakthrough. Grothendieck famously spoke of a rising sea in which difficult problems might eventually be submerged and simplified. Yet when nature insists on phenomena such as quantum mechanics, sometimes no ocean is deep enough. To counter this, we may need to carve a new path, one with a river of flow.

A.4 Reversible Computing

GCT opens an avenue for *reversible computation*. This matters for physics because microscopic physical laws are, to an excellent approximation, information-preserving: classical Hamiltonian dynamics preserve phase-space volume (Liouville) and quantum dynamics evolve by unitary (or antiunitary) transformations. In both cases, the evolution can, in principle, be inverted. A computational model that respects this structure aligns directly with how physical systems propagate [Ben73].

Let Conf be the space of configurations of a computing device. Then for:

- (a) **Finite/discrete models.** A step of reversible computation is a bijection $F : \text{Conf} \rightarrow \text{Conf}$. Composition gives a group $\text{Bij}(\text{Conf})$ of updates; gates such as Toffoli/Fredkin [FT82] generate subgroups used in practice (see [Mor17] for an overview).
- (b) **Classical continuum models.** Let (M, ω) be a symplectic phase space. An evolution $\{\phi_t\}_{t \in \mathbb{R}}$ is reversible when each ϕ_t is a *symplectomorphism* and $\phi_{t+s} = \phi_t \circ \phi_s$, $\phi_0 = \text{id}$, hence $\phi_{-t} = \phi_t^{-1}$. Hamiltonian flows are the canonical example.
- (c) **Quantum models.** On a Hilbert space $\mathcal{H}$, closed-system time evolution is implemented by a one-parameter unitary group $U_t = e^{-iHt}$, so $U_{-t} = U_t^\dagger$. More generally, any unitary (or antiunitary, for time-reversal) map is logically reversible; non-unitary channels (measurement, dissipation) are not.

Reversible computation can be expressed as *isomorphisms* in a category of states and processes. If $\mathcal{C}$ is a category of configurations, a reversible step is an isomorphism $f : X \rightarrow Y$ with inverse $f^{-1} : Y \rightarrow X$ and $f \circ f^{-1} = \text{id}_Y$, $f^{-1} \circ f = \text{id}_X$. Collecting objects and isomorphisms yields a groupoid of configurations and reversible transitions. Equivalently, a reversible dynamics is an autoequivalence $F : \mathcal{C} \rightarrow \mathcal{C}$ with a quasi-inverse F^{-1} , i.e. $F \circ F^{-1} \cong \text{id}_{\mathcal{C}}$ and $F^{-1} \circ F \cong \text{id}_{\mathcal{C}}$. This captures “no information loss” at the structural level.

Example A.1 (Flows and Time-reversal). Let $\phi_t : M \rightarrow M$ denote a flow on a manifold M . If every ϕ_t is a diffeomorphism and $\phi_{-t} = \phi_t^{-1}$ for all t , the evolution is reversible. In classical mechanics, this is strengthened by symplecticity $\phi_t^* \omega = \omega$, while mechanics conservation of phase-space volume and the existence of a well-posed inverse flow. In quantum theory, the analogous statement is $U_{-t} = U_t^\dagger$ for unitary U_t .

Reversibility is not merely an aesthetic constraint on computation; it mirrors the fundamental invertibility of closed-system physics.

- (a) **Information conservation.** Reversible updates preserve *all* fine-grained information. Logical irreversibility (many-to-one maps) necessarily discards information and, when implemented physically, has thermodynamic cost.
- (b) **Microscopic reversibility vs macroscopic irreversibility.** Hamiltonian systems are still reversible (invertible maps); apparent irreversibility arises from coarse-graining, noise, and coupling to environments. Computationally, introducing erasure, measurement, or stochastic resets corresponds to non-invertible maps and semigroup dynamics.
- (c) **Design principles.** Algorithms and architectures that factor into reversible cores (bijection-s/symplectomorphisms/unitaries) with controlled, explicit points of irreversibility (enable tighter accounting of energy and error, and interface cleanly with physics-based models).

Modeling computation by invertible transformations (bijections, symplectomorphisms, unitaries) places the theory squarely inside the same structural class as classical and quantum dynamics, with logical irreversibility reserved for explicit coarse-grainings or measurements.

A.5 Topological Kleene Field Theories and Computational Geometric Bordisms

Gonzales, Peralta, and Miranda [GPMP25] have recently introduced *Topological Kleene Field Theory* (TKFT) as a smooth-topological realization of classical computability. In their framework, computation is encoded by a *dynamical bordism*: a manifold with boundary equipped with a suitably well-behaved vector field, together with embeddings of discrete input and output data into distinguished boundary components. The associated reaching map recovers a partial recursive function. In this sense, TKFT should be read as a model of computability in the Kleene–Turing sense, not merely as an analogy with it.

What is especially relevant for the present work is that, in TKFT, the topology of the bordism is not ornamental. Branching and iteration are reflected in the topology of the underlying manifold: pair-of-pants type decompositions model conditional branching, while nontrivial handles encode iterative or looping behavior. TKFT, therefore, offers a precise way to view program structure through smooth topology and dynamical flow.

Our use of bordism is related, but conceptually distinct. In TKFT, the bordism is primarily the ambient geometric stage through which a computation is routed between designated boundary components. In GCT, by contrast, the computational content is intended to be carried more intrinsically by the decorated geometry itself: by the manifold, its local interfaces, its admissible deformations, and the algebraic or gauge-theoretic structures placed upon it. This motivates the following definition.

Definition A.4 (Computational Geometric Bordism). Let $\mathcal{S}$ be a class of computational structures on smooth manifolds (for example, principal G -bundles with connection, decoration functors, or other geometric state data). An n -dimensional GCM is a pair (M, σ) , where M is an oriented smooth n -manifold, possibly with boundary, and $\sigma \in \mathcal{S}(M)$.

A *computational geometric bordism* from (M_0, σ_0) to (M_1, σ_1) is a compact oriented smooth $(n+1)$ -manifold W with boundary, equipped with a collar-preserving identification

$$\partial W \cong \overline{M_0} \sqcup M_1,$$

together with a structure $\Sigma \in \mathcal{S}(W)$ whose restrictions along the collars recover the endpoint data:

$$\Sigma|_{M_0} = \sigma_0, \quad \Sigma|_{M_1} = \sigma_1.$$

Two such bordisms are regarded as equivalent if they are related by a boundary-fixing diffeomorphism together with an isomorphism in $\mathcal{S}$.

Example A.2 (A Helical Cylinder as Bordism). Let

$$C := S^1 \times I, \quad I = [0, 1],$$

so that

$$\partial C = S^1 \times \{0\} \sqcup S^1 \times \{1\}.$$

For any fixed parameter $\lambda \in \mathbb{R}$, the map

$$F_\lambda(\theta, t) = (\cos(\theta + \lambda t), \sin(\theta + \lambda t), t), \quad (\theta, t) \in S^1 \times I,$$

is an embedding into $\mathbb{R}^3$. Its image is a helically twisted cylinder joining two copies of S^1 . Thus, the underlying bordism is still the cylinder $S^1 \times I$, but now equipped with an explicitly chiral geometric realization. In the present setting, the point is not that the bordism class changes, but that geometric structure carried by the bordism can itself become computationally meaningful.

TKFT provides a smooth-topological language in which classical computable functions may be organized by dynamical bordisms; GCT seeks a more intrinsic implementation in which computational state is carried directly by decorated geometry. The two frameworks should therefore be viewed as complementary rather than competing. Heuristically, one may prototype a computation in the TKFT setting and then seek a geometric implementation of the same computational content within a GCFT, where the relevant data are borne by the manifold itself rather than by an external symbolic encoding alone.

Decategorification and Kleene recursion. The role of classical computability in the present framework is best expressed indirectly. Rather than rebuilding Kleene recursion as an autonomous layer, we place it inside TKFT, and then regard TKFT itself as a decategorified shadow of GCT.

The point is structural. TKFT already shows that classical partial recursive computation can be realized by smooth dynamical bordisms. What GCT adds is not a different notion of classical computability, but a richer one, geometric semantics in which the computational content is carried intrinsically by decorated geometry: interfaces, local symmetries, holonomy, gauge data, and higher compositional structure.

From computational bordisms to their shadow. Let

$$\mathbf{CGBrd}$$

denote the geometric bordism category whose objects are decorated computational states

$$(M, \sigma),$$

with M a smooth manifold (possibly with boundary) and σ the geometric computational structure carried by M (for example, interface data, bundle data, connection data, chirality/orientation labels, or local state decorations). Morphisms are computational geometric bordisms

$$(W, \Sigma) : (M_0, \sigma_0) \rightarrow (M_1, \sigma_1),$$

composed by gluing.

Let

$$\mathbf{TKFT}$$

denote the bordism category underlying topological Kleene field theory, in which one remembers only the smooth bordism together with the dynamical data needed to define the associated reaching map.

We define the *TKFT shadow functor*

$$\mathbf{Sh} : \mathbf{CGBrd} \longrightarrow \mathbf{TKFT}$$

by forgetting all geometric data not visible at the level of the TKFT reaching map. Concretely, $\mathbf{Sh}$ retains the underlying manifold with boundary, the input/output boundary components, the ambient dynamical flow needed to define the induced computation and discards, or passes to π_0 , local gauge or bundle structure, holonomy and curvature data, defect and higher-form labels, nontrivial interface groupoid structure, distinctions visible only up to higher homotopy or higher representation data.

In this sense, $\mathbf{Sh}$ is a *decategorification*: it passes from decorated geometric computation to the smooth-topological computation visible at the TKFT level.

Definition A.5 (TKFT-shadow Equivalence). Two geometric computational bordisms

$$(W, \Sigma), (W', \Sigma')$$

are said to have the same TKFT shadow if

$$\mathbf{Sh}(W, \Sigma) \cong \mathbf{Sh}(W', \Sigma')$$

in $\mathbf{TKFT}$. Equivalently, they induce the same underlying dynamical bordism computation after all higher geometric decoration invisible to TKFT has been forgotten.

Proposition A.4 (TKFT as Symbolic Boundary of GCT). *Every TKFT computation is realized by a geometric computational bordism whose additional decorations are trivial, abelianized, or otherwise invisible under the shadow functor $\mathbf{Sh}$. Hence, TKFT embeds as the decategorified symbolic boundary of GCT.*

Proof sketch. A TKFT computation is given by a smooth bordism equipped with a sufficiently well-behaved vector field and designated input/output embeddings. This data can be regarded as a special case of a geometric computational bordism by equipping the same manifold with only the minimal decoration needed to recover that flow. Applying $\mathbf{Sh}$ then returns the original TKFT object. Thus, TKFT appears as the image of the minimally decorated sector of GCT. $\square$

Proposition A.5 (Strict Enrichment Beyond TKFT). *The shadow functor Sh is not faithful in general. Distinct GCT computations may determine the same TKFT computation after decategorification.*

Reason. TKFT is insensitive to geometric data that do not alter the reaching map at the level of the underlying dynamical bordism. In particular, nonabelian holonomy, gauge-equivalent but computationally distinct decorations, higher defect data, and order-sensitive interface composition may all survive in GCT while disappearing under Sh . $\square$

What GCT adds. The excess of GCT over TKFT is precisely the part of the computation carried by decorated geometry rather than by the underlying smooth dynamical bordism alone.

Corollary A.6 (Containment of Classical Computability). *Since TKFT recovers partial recursive computation, and since TKFT lies in the image of the shadow functor*

$$\text{Sh} : \text{CGBrd} \rightarrow \text{TKFT},$$

classical Kleene computability is contained in GCT as a special case.

In this way, the classical recursion-theoretic layer no longer needs to be treated as a separate foundation for the theory. It appears automatically as the symbolic decategorified boundary of geometric computation. The genuine novelty of GCT begins exactly where the shadow functor ceases to be injective: namely, where geometric decoration, gauge structure, and higher composition contribute computational content that TKFT cannot see.

A.6 Relationship to the Blum-Shub-Smale Model

The Blum–Shub–Smale (BSS) model [BSS89] extends classical computability from $\mathbb{N}$ to $\mathbb{R}$ by postulating a register machine whose instructions are field operations and branches on real values. It yields a rich theory of decision/complexity over $\mathbb{R}$, but with an *algebraic* semantics predicated on unit-cost real arithmetic.

Our framework is inspired but orthogonal. We retain smooth reasoning internally (synthetic/smooth reals) and execute externally over effective number systems (computable reals/algebraics), while moving the *computational substrate* from algebra to geometry: We therefore use BSS only as *context*: it shows that enlarging the base field can change computability classes. In contrast, GCT changes the *operational layer*: computation is controlled by the evolution of geometric state, with gauge invariants as outputs and coarse-graining/renormalization modeled as algebraic effects (4.5). This preserves smooth calculus for reasoning, avoids hidden oracles in execution, and matches the holonomy-based semantics used throughout.

Definition A.6 (Geometric E_∞ -algebra). Let $\mathcal{C}_\infty$ denote the symmetric monoidal ∞ -category of smooth (derived) manifolds and smooth maps, equipped with the Cartesian product $\times$ as its tensor product. A GCM lattice is specified by

$$X \in \mathcal{C}_\infty, \quad \{\mu_n : X^{\times n} \longrightarrow X\}_{n \geq 0},$$

where the operations μ_n encode parallel composition (disjoint gluing) of n unit helix components. The family (μ_n) is required to satisfy the coherence relations of the E_∞ -operad in $\mathcal{C}_\infty$, making $(X, \mu_\bullet)$ a commutative E_∞ -algebra object is as follows:

- (a) The unit map $\mu_0 : * \rightarrow X$ corresponds to the empty (null) computation.
- (b) The binary product $\mu_2 : X \times X \rightarrow X$ realizes the geometric join of two independent lattice components.
- (c) Higher μ_n encodes n -way concurrent composition, commutativity holds up to coherent homotopy, giving the required flexibility for fluidic deformations.

In this language, a classical BSS machine is recovered by restricting to the full sub- ∞ -category of finite-dimensional real vector spaces with polynomial maps, whose objects form discrete E_∞ -algebras (strict commutative monoids). The GCM lattice generalizes this: it is still an E_∞ -monoid, but now

geometry (i.e., smooth structure, group actions, bordisms) replaces uncountable real registers as the primitive data store.

$$\text{BSS (discrete commutative monoid)} \longrightarrow \text{GL (geometric } E_\infty\text{-algebra)}.$$

See [Lur17] for the general theory of E_∞ -algebras in symmetric monoidal ∞ -categories. This synthesis of higher category theory and geometry moves us closer to a physical theory of computation.

A.7 On Causal Fermion Systems

Causal Fermion Systems (CFS), developed by Prof. Felix Finster and collaborators [FKT25], is, from the GCT perspective, the current leading attempt, bar none, to rewrite the foundations of physics in a way that is *geometric-first* and *variational-first*, while treating standard quantum mechanics as a *limiting* (and in a sense, *effective*) description rather than an irreducible axiom. A brief treatment is now possible, as our journey through the computational landscape is complete.

A CFS starts from an operator-theoretic object (a Hilbert space and a measure on a space of operators), and declares that space-time itself is not given a priori: it is reconstructed as the *support* of the measure. Dynamics is not postulated in the usual “Hamiltonian on a background manifold” way, but instead emerges from minimizing a nonlinear action functional (the *causal action*). In the appropriate continuum limit, one recovers familiar structures: Lorentzian geometry, classical gauge fields, and (in further approximations) the quantum-mechanical and QFT toolkits as effective formalisms.

The core data of a causal fermion system are often packaged as a triple

$$(\mathcal{H}, \mathcal{F}, \rho),$$

where:

- $\mathcal{H}$ is a Hilbert space of “physical” spinorial states (think: a space of Dirac-type solutions, but *not* necessarily tied to a pre-existing manifold in the fundamental formulation);
- $\mathcal{F}$ is a set of (typically finite-rank) self-adjoint operators on $\mathcal{H}$, with an imposed signature constraint (this is part of how “spin dimension” enters);
- ρ is a positive measure on $\mathcal{F}$ (the *universal measure*).

The conclusion is then:

$$M := \text{supp}(\rho)$$

is defined to be space-time. One then extracts causal relations, metric information, and spin structure from spectral data of operator products xy for $x, y \in M$. The dynamics are: vary ρ (subject to constraints) to minimize a positive action functional constructed from the eigenvalues of the products xy . The theory is therefore “equations of motion as an optimization principle on measures,” not “equations of motion as PDEs on a background manifold.”

A Fisherman of the Dirac Sea? In our regime language, CFS is best read as living predominantly in the gauge band with an essential side channel into the fully fluidic band. The most faithful placement is that CFS is a gauge theory of computation: non-commutative dynamics with a considerable quotient by redundancy, such that apparent modular independence is a gauge choice rather than a literal factorization. This is precisely the hallmark of the gauge regime in our classification: joins do not commute in general, but one regains a notion of “independence” after gauge-fixing; the state is stored in gauge-invariant aggregates (holonomies/curvature/spectral invariants) rather than in raw potentials.

CFS is often perceived as intimidating because it is broad, and in addition, Finster is, by training, also a mathematician. The same broadness could be said for our own program, but as a result, the overlap is clean at the conceptual level:

- (a) **Spacetime is derived, not assumed.** In GCT, we treat the manifold structure as constructed from computational geometry. CFS likewise treats spacetime as emergent (support of the universal measure) rather than a fixed stage.

- (b) **Physics is “action and constraints” at the fundamental layer.** In GCT, we repeatedly re-encounter the principle that what matters are invariants that are stable under the allowed deformations (gluing, transport, holonomy, charge). CFS makes this explicit by encoding dynamics as a variational principle on the measure itself.
- (c) **Gauge invariance is not a cosmetic symmetry: it is the computational quotient.** Our gauge regime is precisely the regime where the machine’s *actual* state is not the raw configuration, but the equivalence class under gauge transformations. CFS is philosophically the same move, except made at the operator level: local unitary freedom is built in, and only gauge-invariant spectral aggregates are physically meaningful.
- (d) **Quantum formalism as a readout/effective theory.** CFS is often read as “de-quantizing” quantum mechanics: the linear Hilbert-space formalism is not the fundamental law, but a limiting description arising from coarse-graining and continuum approximations. This resonates strongly with the GCT posture that many “quantum” features are emergent bookkeeping for geometric microstructures.

CFS Feature	GCT Interpretation
Universal measure ρ on operator space	Microstate distribution/ensemble over admissible geometric configurations (a “state-selection” layer)
Spacetime $M = \text{supp}(\rho)$	Working manifold (or effective base) as an emergent support of stable computational configurations
Spectral data of xy defines causal structure	Causality/communication constraints extracted from transport/holonomy invariants
Causal action principle (minimize action)	“Physics as compilation”: admissible computations selected by global consistency/extremal principles
Local unitary freedom (gauge)	Gauge regime: independence after gauge-fixing; observables are gauge-invariant aggregates
Continuum limit yields classical gauge fields + gravity	Gauge regime semantics: computation stored in curvature/holonomy classes; classical fields as macroscopic limit of microstructure

We need not adopt the full CFS formalism to benefit from it. What we all can take away is the *structural lesson*:

If one treats quantum theory as an effective linearization of a deeper geometric object, then the correct primitive is not “wavefunctions on spacetime” but rather a *geometric substrate* whose macroscopic shadows look like Hilbert-space quantum mechanics.

In this sense, CFS is not a competitor but a strong sanity-check: it shows that a long-running research program exists which is already comfortable with (a) emergent spacetime, (b) gauge-as-quotient, and (c) quantum mechanics as a derived language rather than a first principle. All three are compatible with the GCT regime taxonomy, and CFS can be read as a sophisticated inhabitant of the gauge band.

Summary. CFS is best understood here as an operator-level realization of the gauge regime: non-commutative dynamics with ample redundancy, where the physically meaningful information is the gauge-invariant residue. This makes it a natural “bridge theory” between our regime picture and conventional QFT language.

A.8 The Computational Cobordism Hypothesis

The algebraic enrichment required by GCFT asks for a formal setting that can remember both geometry and computation. The purpose of this section is not to construct a completed higher-categorical classification theorem, but to isolate the categorical architecture suggested by the preceding constructions. The basic layers are:

geometric substrate $\longrightarrow$ decorated bordisms $\longrightarrow$ semantic evaluation $\longrightarrow$ executable readout.

We begin by specifying the ambient geometric category in which GCMs live.

Definition A.7 (Embedded Smooth Manifolds with Enrichments). Let **Man** denote the category of smooth manifolds and smooth maps. The *master category*

$$\mathcal{ESM}^+$$

of embedded smooth manifolds with enrichments is a symmetric monoidal category whose data are as follows.

(a) **Objects.** Pairs (M, str) , where M is a smooth manifold, possibly with boundary or stratification, and str is a finite tuple of auxiliary geometric or computational structures over M . These may include orientation data, principal bundles and connections, distinguished vector fields or flows, boundary/interface labels, phase data, discrete state labels, or regularity constraints.

(b) **Morphisms.** A morphism

$$(M, \text{str}) \longrightarrow (N, \text{str}')$$

is a smooth map $f : M \rightarrow N$ preserving the declared structure in the appropriate sense: pulling back bundles and covers, intertwining specified flows, preserving boundary/interface labels, and respecting the declared admissibility conditions.

(c) **Monoidal product.** The monoidal product is disjoint union:

$$(M, \text{str}) \otimes (N, \text{str}') := (M \sqcup N, \text{str} \sqcup \text{str}'), \quad \mathbb{1} := (\emptyset, \emptyset).$$

There is a faithful forgetful functor

$$U : \mathcal{ESM}^+ \longrightarrow \mathbf{Man}.$$

We suppress set-theoretic size issues throughout.

Remark A.2. The choice $\otimes = \sqcup$ models parallel composition of independent systems. Other geometric operations, such as gluing along matched boundary interfaces or fibered sums over boundary data, enter later as composition laws inside decorated bordism categories rather than as the primary monoidal product on $\mathcal{ESM}^+$.

We next separate the smooth geometric semantics from the effective readout semantics. The former remembers bundles, connections, curvature, holonomy, and higher geometric structure. The latter records the computable observables extracted from those structures.

Definition A.8 (Differentially Cohesive Ambient Topos for GCMs). Let $\mathcal{E}_{\text{GCM}}$ be a higher differentially cohesive topos, such as a smooth ∞ -topos of sheaves on Cartesian spaces, suitable for reasoning internally about smooth spaces, bundles, connections, curvature, and holonomy. We assume it carries the usual cohesive modalities

$$\Pi \dashv \flat \dashv \text{id} \dashv \sharp$$

together with a de Rham shape modality

$$\text{dR} : \mathcal{E}_{\text{GCM}} \rightarrow \mathcal{E}_{\text{GCM}},$$

so that principal bundles, connections, differential characters, and holonomy-type observables can be expressed internally [Sch13].

Definition A.9 (GCM Semantic Target). Let $\mathcal{C}_{\text{GCM}}$ denote a symmetric monoidal higher category of GCM-structured states and processes, internal to or presented in $\mathcal{E}_{\text{GCM}}$. Its objects are admissible boundary state data, and its morphisms are admissible geometric-computational processes between such boundary data. The monoidal product is induced by disjoint union of independent systems.

A geometric computational field theory will land in $\mathcal{C}_{\text{GCM}}$, while $\mathcal{E}_{\text{GCM}}$ serves as the ambient semantic universe in which the relevant smooth and differential geometric structures are interpreted.

Definition A.10 (Execution Ambient and Observable Extraction). Let $\mathcal{D}_{\text{exec}}$ be a symmetric monoidal category of effective observables and executable transformations, whose numeric objects are effectively representable, for example by rationals, algebraic numbers, or computable reals.

Let

$$\mathcal{C}_{\text{GCM}}^{\text{obs}} \subseteq \mathcal{C}_{\text{GCM}}$$

denote the subcategory of admissible gauge-invariant observables and observable-preserving transformations. An *extraction functor* is a left-exact symmetric monoidal functor

$$\text{Ex} : \mathcal{C}_{\text{GCM}}^{\text{obs}} \longrightarrow \mathcal{D}_{\text{exec}}$$

which sends gauge-invariant functionals, such as class functions of holonomy, differential-cohomological observables, or stable boundary readouts, to computable outputs.

Thus the semantic pipeline is:

$$\text{Bord}_n^{\mathcal{S}} \xrightarrow{Z} \mathcal{C}_{\text{GCM}} \xrightarrow{\text{Ex}} \mathcal{D}_{\text{exec}}.$$

Remark A.3 (Modalities as Semantic Projections). The cohesive modalities organize different levels of readout:

- Π extracts homotopy-level or topological content.
- $\flat$ extracts flat or discrete data, useful for symbolic verification.
- dR records de Rham or differential-cohomological observables.
- $\sharp$ suppresses topology when analytic or codiscrete control is required.

Operationally, the slogan “path integration is replaced by sheafification” means that global states are assembled as sheafified compatible local data in $\mathcal{E}_{\text{GCM}}$, while externally visible outputs are obtained only after applying Ex to gauge-invariant observables.

We now pass from the state/transition layer to the control-flow layer. The decorated groupoids or action groupoids encode the state/transition layer; decorated bordisms encode the control flow layer.

Definition A.11 (Decorated Bordism Category). Fix an integer $n \geq 1$, a decoration monoid or group $\mathcal{S}$, and a class of n -dimensional compact smooth manifolds with boundary carrying $\mathcal{S}$ -decoration data compatible with gluing.

The $\mathcal{S}$ -decorated n -bordism category

$$\text{Bord}_n^{\mathcal{S}}$$

has:

- (a) **Objects.** Closed $(n - 1)$ -manifolds equipped with admissible $\mathcal{S}$ -decorations on their boundary state/transition data.
- (b) **Morphisms.** Compact n -manifolds W with collared boundary

$$\partial W \cong \overline{M_0} \sqcup M_1,$$

equipped with compatible $\mathcal{S}$ -decoration data extending the decorations on M_0 and M_1 .

- (c) **Composition.** Composition is gluing along matched collars, with the corresponding decoration data composed by the monoidal law in $\mathcal{S}$.
- (d) **Symmetric monoidal product.** The monoidal product is disjoint union.

When a geometric substrate $(M, \text{str}) \in \mathcal{ESM}^+$ is fixed, we write

$$\text{Bord}_n^{\mathcal{S}}(M, \text{str})$$

for the full decorated bordism category of bordisms embedded in $M \times I$ whose decorations agree with the declared structure str on boundary interfaces.

Remark A.4 (Thin and Decorated Path Data). In genuinely geometric examples, decoration data need not factor through the ordinary fundamental groupoid Π_1 . Connection holonomy with nonzero curvature is invariant under thin homotopy, but not under arbitrary homotopy. Thus, the natural source for such decorations is often a thin path groupoid

$$\Pi_1^{\text{thin}}(M),$$

or a more general decorated path groupoid, rather than the bare $\Pi_1(M)$. The ordinary fundamental groupoid appears only in flat or topological reductions.

Conjecture A.7 (The Computational Cobordism Hypothesis). *Let $(M, \text{str}) \in \mathcal{ESM}^+$ be a fixed enriched geometric substrate, and let $\mathcal{C}_{\text{GCM}}$ be a suitable symmetric monoidal higher category of GCM-structured states and processes.*

A geometric computational field theory of dimension n , or GCFT, is a symmetric monoidal functor

$$Z : \text{Bord}_n^{\mathcal{S}}(M, \text{str}) \longrightarrow \mathcal{C}_{\text{GCM}}.$$

In a fully extended version of the theory, one expects such functors to be controlled by appropriate dualizability data, namely, by suitable $\mathcal{S}$ -decorated fully dualizable objects in a symmetric monoidal higher-categorical completion of the GCM semantic target. In this sense, GCFTs should admit a classification principle analogous to the Baez–Dolan Cobordism Hypothesis, but enriched by computational decoration data [BD95]. When the decorations are trivial, or when they are flat and factor through ordinary homotopy data, the construction should reduce to the corresponding topological or functorial field-theoretic case.

Three-line operational sketch.

- (a) **States/transitions.** Local updates are encoded by an $\mathcal{S}$ -decorated groupoid, Lie groupoid, or action groupoid $(\mathcal{G}, d)$ over boundary cuts.
- (b) **Control-flow.** A computation is modeled as a decorated bordism

$$W \in \text{Bord}_n^{\mathcal{S}}(M, \text{str}),$$

with composition given by gluing of bordisms and monoidal composition of the associated decorations.

- (c) **Evaluation.** A GCFT is the symmetric monoidal functor Z assigning boundary data to objects of $\mathcal{C}_{\text{GCM}}$ and assigning a decorated bordism W to a morphism

$$Z(W) : Z(\partial_{\text{in}} W) \longrightarrow Z(\partial_{\text{out}} W),$$

thereby providing denotational semantics for the computation.

Remark A.5 (Reduction to Classical Regimes). If the decoration data descend to Π_1 and $\mathcal{S}$ is finite or discrete, then the construction enters the same formal regime as a finite group topological field theories. If $\mathcal{S}$ carries continuous abelian data, such as $U(1)$ -phase, the resulting semantics is gauge-like but still commutative. Non-abelian decoration data lead toward genuinely gauge-theoretic models, where the appropriate observables are gauge-invariant functions of holonomy, such as Wilson-loop traces or other class functions.

Example A.3 (Helical Bordism Computation). Let

$$M \cong D_{(r, \varphi)}^2 \times I_s$$

be a finite helical working region, modeled abstractly as the tubular neighborhood of a helical core arc. Thus M is topologically a 3-ball, equipped with a distinguished longitudinal coordinate $s \in I$, transverse polar coordinates (r, φ) on each disk fiber, and a fixed chirality label

$$\chi \in C_2$$

recorded as part of the declared structure str . The chirality label is not a homotopy invariant of M ; it is geometric/computational structure placed on the helical module.

Let A be a $U(1)$ -connection on the trivial circle bundle over M . Since M is contractible, the bundle is topologically trivial, but the connection may still have nonzero curvature. Its holonomy defines a functor on the thin path groupoid

$$\text{Hol}_A : \Pi_1^{\text{thin}}(M) \longrightarrow BU(1).$$

Equivalently, holonomy is invariant under thin homotopy, but need not factor through the ordinary fundamental groupoid $\Pi_1(M)$ unless the connection is flat.

Set

$$\mathcal{S} := U(1) \times C_2, \quad (z, \epsilon) \cdot (z', \epsilon') = (zz', \epsilon \oplus \epsilon').$$

In the present example, the $U(1)$ -component is the genuine path decoration, while the C_2 -component records the chirality sector of the helical module. Thus, the local decoration of a loop γ in the module is

$$d(\gamma) = (\text{Hol}_\gamma(A), \chi).$$

The functorial part is carried by holonomy:

$$\text{Hol}_{\gamma_2 \circ \gamma_1}(A) = \text{Hol}_{\gamma_2}(A) \text{Hol}_{\gamma_1}(A), \quad \text{Hol}_{\gamma^{-1}}(A) = \text{Hol}_\gamma(A)^{-1}.$$

The chirality label is transported as fixed sector data across the bordism.

Now fix a radius $0 < r < 1$. The surface

$$W_r := S_\varphi^1 \times I_s \subset D^2 \times I$$

is a 2-dimensional bordism from the transverse circle

$$S_\varphi^1 \times \{0\}$$

to the transverse circle

$$S_\varphi^1 \times \{1\}.$$

Geometrically, W_r is the cylindrical surface swept out by the radius- r circle as it moves along the helical core.

If one wants an explicit embedded representative in $\mathbb{R}^3$, let $\Gamma(s)$ be the helical core and let $(N(s), B(s))$ be a chosen normal frame along it. Then

$$(\varphi, s) \longmapsto \Gamma(s) + r(\cos(\varphi + \lambda s)N(s) + \sin(\varphi + \lambda s)B(s))$$

gives a helical embedding of W_r , where λ records the imposed twist of the transverse frame.

Along a transverse loop

$$\gamma_{s,r}(\varphi) = (r, \varphi, s), \quad \varphi \in [0, 2\pi],$$

suppose the connection is locally represented by

$$A = \Omega(s, r, \varphi) d\varphi.$$

The corresponding one-turn phase gate is

$$U_s(r) = \exp\left(i \int_0^{2\pi} \Omega(s, r, \varphi) d\varphi\right) \in U(1).$$

Thus the local decoration assigns

$$d(\gamma_{s,r}) = (U_s(r), \chi).$$

Although all transverse loops $\gamma_{s,r}$ are null-homotopic in $D^2 \times I$, their holonomies need not agree. Indeed, if the curvature

$$F_A = dA$$

has a nonzero $dr \wedge d\varphi$ component, then Stokes' theorem gives

$$U_s(r_2)U_s(r_1)^{-1} = \exp\left(i \int_{[r_1, r_2] \times S^1} F_A\right),$$

so two ordinarily homotopic loops at different radii can yield distinct phase readouts:

$$U_s(r_1) \neq U_s(r_2).$$

Thus, the computational outcome depends on geometric control data, not on the bare homotopy class of the loop. This is precisely why the thin path groupoid, or a decorated path groupoid, is the natural source of the decoration in the geometric regime.

A 2-dimensional GCFT

$$Z : \mathbf{Bord}_2^S(M, \text{str}) \longrightarrow \mathcal{C}_{\text{GCM}}$$

sends a transverse boundary circle to an object

$$Z(S^1) \in \mathcal{C}_{\text{GCM}},$$

and sends the decorated cylinder W_r to a morphism

$$Z(W_r) : Z(S^1 \times \{0\}) \longrightarrow Z(S^1 \times \{1\}).$$

In this example, $Z(W_r)$ acts on the relevant abelian phase component by multiplication by $U_s(r)$, together with the fixed C_2 -chirality sector recorded by χ .

If A is flat, then the holonomy around every contractible transverse loop is trivial. In that case, the example collapses to the purely topological semantics of the underlying 3-ball and no nontrivial phase computation is obtained from ordinary homotopy data alone. The nontrivial computation in this example is therefore genuinely geometric: it is carried by connection, curvature, and controlled helical framing rather than by the topology of M .

Remark A.6 (Scope of the Categorical Formulation). The preceding conjecture should be read as a proposed semantic envelope for GCFT is not a complete proof of a classification theorem. The role of the categorical language is to organize the data already present in geometric computation: boundary states, admissible histories, gluing, holonomy, and gauge-invariant readout. The fully extended classification problem requires additional higher-categorical infrastructure, including a precise choice of target $\mathcal{C}_{\text{GCM}}$, a notion of decorated dualizability, and a treatment of higher morphisms between computational histories. These are left as part of the future formal development of the theory.

A.9 Fourier Mode Calculations

Here, we examine kinematic selection effects in the idealized computational flow on the L^2 field of a trivial bundle. Let the cross-section be a disk D^2 in the xy -plane. The velocity field $\vec{v}(r, \theta)$ at each cross-section can be written in polar coordinates as

$$\vec{v}(r, \theta) = v_r(r, \theta) \hat{r} + v_\theta(r, \theta) \hat{\theta},$$

where v_r and v_θ are the radial and angular components, respectively.

Suppose the flow is smooth and incompressible, and consider the angular dependence of v_r and v_θ . Each component admits a Fourier expansion:

$$v_r(r, \theta) = \sum_{k=-\infty}^{\infty} a_k(r) e^{ik\theta}, \quad v_\theta(r, \theta) = \sum_{k=-\infty}^{\infty} b_k(r) e^{ik\theta}.$$

Imposing rotational symmetry of order n means requiring invariance under rotation by $2\pi/n$:

$$v_r(r, \theta + 2\pi/n) = v_r(r, \theta) \quad \forall \theta.$$

This forces all Fourier coefficients $a_k(r)$ and $b_k(r)$ with $k \notin n\mathbb{Z}$ to vanish. Thus, only harmonics with k divisible by n remain:

$$v_r(r, \theta) = \sum_{m=-\infty}^{\infty} a_{nm}(r) e^{inm\theta}.$$

In the absence of symmetry ($n = 1$), all $2K + 1$ modes with $|k| \leq K$ are permitted. Under n -fold symmetry, only those with $k = 0, \pm n, \pm 2n, \dots, \pm Mn$, with $M = \lfloor K/n \rfloor$, survive—yielding $2M + 1$ allowed modes.

We define the *relative expressiveness* as the fraction of permitted angular degrees of freedom:

$$E(n) := \frac{\text{Number of allowed modes for symmetry } n}{\text{Number for } n = 1} = \frac{2M + 1}{2K + 1}.$$

For example, with $n = 4$ and $K = 8$, we have $M = 2$ (i.e., $k = 0, \pm 4, \pm 8$), so $E(4) = 5/17 \approx 0.29$.

As $n \rightarrow \infty$, $M \rightarrow 0$ and $E(n) \rightarrow 0$; i.e., only the constant mode survives. The only flow invariant under arbitrary rotation is uniform, and all angular structure is lost. This directly parallels the earlier result that the only smooth closed 1-manifold with infinite rotational symmetry is the circle.

Thus, increasing symmetry necessarily reduces the space of physically expressible patterns. The tradeoff is exact, quantitative, and determined by the order of symmetry imposed on the system.

Remark A.7. If a field $F(r, \theta)$ on an annulus is invariant under all rotations $\theta \mapsto \theta + \phi$ ($U(1)$ symmetry), then in its Fourier expansion $F(r, \theta) = \sum_{m \in \mathbb{Z}} F_m(r) e^{im\theta}$ one has $F_m \equiv 0$ for $m \neq 0$. Equivalently, only the θ -independent ($m = 0$) mode survives; axisymmetric azimuthal components may remain, but no angular structure ($|m| \geq 1$) does.

A.10 Disclaimers

Regarding Scope and Fundamental Assumptions

This paper does not challenge the empirical validity of quantum field theory, Quantum Chromodynamics, Yang-Mills theory, or the Standard Model. Its purpose is instead to investigate whether some of the geometric and computational structures underlying gauge theory can be reformulated in terms of prequantum geometry. Our work here is best viewed as a candidate semantics for organizing known physical structures, rather than as a competing phenomenological theory.

Glossary and Scope of Terminology

To avoid ambiguity, we distinguish three related but non-identical layers of the present program.

- (a) **Geometric Computability Theory (GCT).** The broad research program concerns computation realized through geometric, topological, and gauge-theoretic structures, as well as the logical, complexity-theoretic, and physical consequences of such realizations. In this sense, GCT is the umbrella field of study. It includes, for example, the classification of computational regimes, the role of helicoidal substrates, the EVT hypothesis, and *possible* future connections to number theory/arithmetic geometry (e.g., Berry-Keating), physics, and pedagogy.
- (b) **Geometric Computational Field Theory (GCFT).** The constructive (total) formalism (4.5) developed within GCT, in which geometric histories are interpreted as composable computational processes. In a GCFT, boundaries carry typed state data, gluing corresponds to evaluation or composition, gauge equivalence is handled at the level of observable output, and physical observables arise as gauge-invariant, coarse-grained results of computation. GCFT is thus the most structured and formal layer of the program: it is to GCT what a functorial field theory is to the broader study of field-theoretic phenomena.

Author’s Note on Naming Conventions

If the twin helix manifold (*THM*) described here ultimately proves to be of lasting foundational significance, the author would prefer that it not be given an eponymous name. This preference is not intended as a rejection of historical naming conventions. Still, it reflects the view that *geometric* objects of genuinely fundamental scope are often better served by descriptive terminology that emphasizes mathematical content rather than personal attribution. This is in contrast to *theorems*, *conjectures/hypotheses*, and *lemmas*, which would be more appropriate.

Still, the emphasis of this paper is therefore not on personal legacy, but on the development of Geometric Computability Theory and its possible mathematical, physical, and pedagogical consequences. The work is written from the perspective of the computational sciences, and readers are encouraged to interpret its terminology in that spirit. Likewise, the author must disclose that he is not a classically trained physicist, and as such, it would be inappropriate to identify him in this manner. The best term of description for the author would be “physical mathematician and/or computer scientist” [Moo14]

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Conflicts of Interest

None.

Data Availability

There is no associated data with this manuscript.

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AI tools were also used to assist in formulating exploratory experimental designs for detecting helical signatures.

References

- [AB59] Yakir Aharonov and David Bohm. Significance of electromagnetic potentials in the quantum theory. *Physical Review*, 115(3):485–491, 1959.
- [ABV⁺05] Juan A. Acebrón, Luis L. Bonilla, Conrad J. Pérez Vicente, Félix Ritort, and Renato Spigler. The kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of Modern Physics*, 77(1):137–185, 2005.
- [AC04] Samson Abramsky and Bob Coecke. A categorical semantics of quantum protocols. In *Proceedings of the 19th Annual IEEE Symposium on Logic in Computer Science (LICS 2004)*, pages 415–425. IEEE Computer Society, 2004.
- [Ada09] Toshiaki Adachi. Foliation on the moduli space of helices on a real space form. *International Mathematical Forum*, 4(35):1699–1707, 2009.
- [ADHM78] Michael F. Atiyah, Vladimir G. Drinfeld, Nigel J. Hitchin, and Yuri I. Manin. Construction of instantons. *Physics Letters A*, 65(3):185–187, 1978.
- [AF19] David Ayala and John Francis. A factorization homology primer. *arXiv preprint*, 2019. arXiv:1903.10961.
- [AG16] Johannes Aastrup and Jesper Möller Grimstrup. Quantum holonomy theory. *Fortschritte der Physik*, 64(10):783–818, sep 2016.
- [AHMNV07] Nima Arkani-Hamed, Luboš Motl, Alberto Nicolis, and Cumrun Vafa. The string landscape, black holes and gravity as the weakest force. *Journal of High Energy Physics*, 2007(06):060, jun 2007.
- [AHT14] Nima Arkani-Hamed and Jaroslav Trnka. The amplituhedron. *Journal of High Energy Physics*, 2014(10):30, 2014.
- [AJ90] M.F. Atiyah and L. Jeffrey. Topological lagrangians and cohomology. *Journal of Geometry and Physics*, 7(1):119–136, 1990.
- [ALR25] Emily Adlam, Niels Linnemann, and James Read. *Constructive Axiomatics for Spacetime Physics*. Oxford University Press, Oxford, United Kingdom, 2025.
- [Arn89] V. I. Arnold. *Mathematical Methods of Classical Mechanics*. Springer, 2 edition, 1989.
- [AS63] Michael F. Atiyah and Isadore M. Singer. The index of elliptic operators on compact manifolds. *Bulletin of the American Mathematical Society*, 69(3):422–433, 1963.
- [Ati57] M. F. Atiyah. Complex analytic connections in fibre bundles. *Transactions of the American Mathematical Society*, 85(1):181–207, 1957.
- [Ati88] Michael Atiyah. Topological quantum field theories. *Publications Mathématiques de l’IHÉS*, 68:175–186, 1988.
- [Ati02] M. Atiyah. Mathematics in the 20th century. *Bulletin of the London Mathematical Society*, 34:1–15, 2002.
- [AW84] Luis Alvarez-Gaumé and Edward Witten. Gravitational anomalies. *Nuclear Physics B*, 234(2):269–330, 1984.
- [Bae98] John C. Baez. Spin foam models. *Classical and Quantum Gravity*, 15(7):1827–1858, 1998.
- [Bae06] John C. Baez. Quantum quandaries: A category-theoretic perspective. In Steven French, Dean Rickles, and Juha Saatsi, editors, *The Structural Foundations of Quantum Gravity*, pages 240–265. Oxford University Press, 2006.
- [Bae22] John C. Baez. Getting to the bottom of noether’s theorem, 2022.

- [Bar91] John W. Barrett. Holonomy and path structures in general relativity and yang–mills theory. *International Journal of Theoretical Physics*, 30(9):1171–1215, 1991.
- [Bar97] Manuel Barros. General helices and a theorem of lancet. *Proceedings of the American Mathematical Society*, 125(5):1503–1509, 1997.
- [Bar23] Jacob A. Barandes. The stochastic-quantum correspondence, 2023.
- [BB99] John C. Baez and John W. Barrett. The quantum tetrahedron in 3 and 4 dimensions, 1999.
- [BD95] John C. Baez and James Dolan. Higher-dimensional algebra and topological quantum field theory. *Journal of Mathematical Physics*, 36(11):6073–6105, nov 1995.
- [BEKPM17] Zvi Bern, Alex Edison, David Kosower, and Julio Parra-Martinez. Curvature-squared multiplets, evanescent effects, and the $u(1)$ anomaly in supergravity. *Physical Review D*, 96(6), sep 2017.
- [Bel64] John S. Bell. On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3):195–200, 1964.
- [Ben73] Charles H. Bennett. Logical reversibility of computation. *IBM Journal of Research and Development*, 17(6):525–532, 1973.
- [Ber84] Michael V. Berry. Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 392(1802):45–57, 1984.
- [Bis75] Richard L. Bishop. There is more than one way to frame a curve. *The American Mathematical Monthly*, 82(3):246–251, 1975.
- [BLLN19] Konstantin Y. Bliokh, Daniel Leykam, Max Lein, and Franco Nori. Topological non-Hermitian origin of surface Maxwell waves. *Nature Communications*, 10, 2019.
- [Boh52a] David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. i. *Physical Review*, 85(2):166–179, 1952.
- [Boh52b] David Bohm. A suggested interpretation of the quantum theory in terms of “hidden” variables. ii. *Physical Review*, 85(2):180–193, 1952.
- [Bou02] Raphael Bousso. The holographic principle. *Rev. Mod. Phys.*, 74(3):825–874, Aug 2002.
- [BPST75] A. A. Belavin, A. M. Polyakov, A. S. Schwartz, and Yu. S. Tyupkin. Pseudoparticle solutions of the yang-mills equations. *Physics Letters B*, 59(1):85–87, 1975.
- [bru13] brussup. Amazing water and sound experiment #2. YouTube, 2013. Accessed: 2025-07-01.
- [BS06] John C. Baez and Urs Schreiber. Higher gauge theory, 2006.
- [BS10] J. Baez and M. Stay. *Physics, Topology, Logic and Computation: A Rosetta Stone*, pages 95–172. Springer Berlin Heidelberg, 2010.
- [BSS89] Lenore Blum, Mike Shub, and Steve Smale. On a theory of computation and complexity over the real numbers: NP-COMPLETENESS, recursive functions and universal machines. *Bulletin (New Series) of the American Mathematical Society*, 21(1), jul 1989.
- [BSZ92] John C. Baez, Irving E. Segal, and Zhengfang Zhou. *Introduction to Algebraic and Constructive Quantum Field Theory*. Princeton University Press, Princeton, NJ, 1992.
- [BT82] Raoul Bott and Loring W. Tu. *Differential Forms in Algebraic Topology*, volume 82 of *Graduate Texts in Mathematics*. Springer, 1982.

- [BT91] Matthias Blau and George Thompson. Topological gauge theories of antisymmetric tensor fields. *Annals of Physics*, 205(1):130–172, 1991.
- [CF03] Marius Crainic and Rui Loja Fernandes. Integrability of lie brackets. *Annals of Mathematics*, 157(2):575–620, 2003.
- [CFL28] Richard Courant, Kurt Friedrichs, and Hans Lewy. ”Über die partiellen differenzengleichungen der mathematischen physik. *Mathematische Annalen*, 100:32–74, 1928.
- [CFP18] Bruno T. Costa, Michael Forger, and Luiz Henrique P. Pêgas. Lie groupoids in classical field theory i: Noether’s theorem, 2018.
- [CGM06] Nadia Chouaieb, Alain Goriely, and John H. Maddocks. Helices. *Proceedings of the National Academy of Sciences*, 103(25):9398–9403, 2006.
- [CH88] Thierry Coquand and Gérard Huet. The calculus of constructions. *Journal of Symbolic Computation*, 6(2):113–129, 1988.
- [CMPSP21] Robert Cardona, Eva Miranda, Daniel Peralta-Salas, and Francisco Presas. Constructing turing complete euler flows in dimension 3. *Proceedings of the National Academy of Sciences*, 118(19), may 2021.
- [Col77] S. Coleman. Nonabelian plane waves. *Physics Letters B*, 70(1):59–60, 1977.
- [Con94] Alain Connes. *Noncommutative Geometry*. Academic Press, 1994.
- [Con00] Alain Connes. Non-commutative geometry and the Standard Model of elementary particle physics. *Journal of High Energy Physics*, 2000.
- [Coo04] Matthew Cook. Universality in elementary cellular automata. In Andrew Northern and Paola Carrasco, editors, *Proceedings of the First Annual Wolfram Science Conference 2003*, Champaign, IL, 2004. Wolfram Media.
- [CPGW15] Toby S. Cubitt, David Perez-Garcia, and Michael M. Wolf. Undecidability of the spectral gap. *Nature*, 528(7581):207–211, 2015.
- [CS74] Shiing-Shen Chern and James Simons. Characteristic forms and geometric invariants. *Annals of Mathematics*, 99(1):48–69, 1974.
- [CZX⁺19] Yuntian Chen, Ruo-Yang Zhang, Zhongfei Xiong, Zhi Hong Hang, Jensen Li, Jian Qi Shen, and C. T. Chan. Non-abelian gauge field optics. *Nature Communications*, 10(1), jul 2019.
- [dB78] Carl de Boor. *A Practical Guide to Splines*, volume 27 of *Applied Mathematical Sciences*. Springer-Verlag, New York, NY, 1 edition, 1978.
- [DGZ92] Detlef Dürr, Sheldon Goldstein, and Nino Zanghì. Quantum equilibrium and the origin of absolute uncertainty. *Journal of Statistical Physics*, 67(5–6):843–907, 1992.
- [Don83] S. K. Donaldson. An application of gauge theory to the topology of 4-manifolds. *Journal of Differential Geometry*, 18:279–315, 1983.
- [DW90] Robbert Dijkgraaf and Edward Witten. Topological gauge theories and group cohomology. *Communications in Mathematical Physics*, 129(2):393–429, 1990.
- [dWPB96] Mark de Wild Propitius and F. Alexander Bais. Discrete gauge theories, 1996.
- [EGH00] Yakov Eliashberg, Alexander Givental, and Helmut Hofer. Introduction to symplectic field theory. In Noga Alon, Jean Bourgain, Alain Connes, Mikhael Gromov, and Vitali Milman, editors, *Visions in Mathematics*, pages 560–673. Birkhäuser / Springer, 2000.
- [Ehr59] Charles Ehresmann. Catégories topologiques et catégories différentiables. In *Colloque de Géométrie Différentielle Globale*, page 137, Bruxelles, 1959. Centre Belge de Recherches Mathématiques.

- [EM70] David G. Ebin and Jerrold Marsden. Groups of diffeomorphisms and the motion of an incompressible fluid. *Annals of Mathematics*, 92(1):102–163, 1970.
- [EPR35] A. Einstein, B. Podolsky, and N. Rosen. Can quantum-mechanical description of physical reality be considered complete? *Physical Review*, 47(10):777–780, 1935.
- [Fel91] Matthias Felleisen. On the expressive power of programming languages. *Science of Computer Programming*, 17(1-3):35–75, 1991.
- [FI25] Angel Ferrández Izquierdo. *Helices*. RSME Springer Series. Springer Cham, 2025. Edition 1.
- [FKT25] Felix Finster, Sebastian Kindermann, and Jan-Hendrik Treude. *Causal Fermion Systems: An Introduction to Fundamental Structures, Methods and Applications*. Cambridge University Press, oct 2025.
- [Flo88] Andreas Floer. Morse theory for lagrangian intersections. *Journal of Differential Geometry*, 28(3):513–547, 1988.
- [Fon15] Brendan Fong. Decorated cospans, 2015.
- [Fre00] Daniel S. Freed. Dirac Charge Quantization and Generalized Differential Cohomology. In *Surveys in Differential Geometry, Volume VII*, volume 7 of *Surveys in Differential Geometry*, pages 129–194. International Press, Somerville, MA, 2000.
- [Fre12] Daniel S. Freed. The cobordism hypothesis. *arXiv preprint*, 2012. Expository; arXiv:1210.5100.
- [Fre14] Daniel S. Freed. Anomalies and invertible field theories. In *String-Math 2013*, volume 88 of *Proceedings of Symposia in Pure Mathematics*, pages 25–45. American Mathematical Society, 2014.
- [FS12] Laurent Freidel and Simone Speziale. On the relations between gravity and bf theories. *SIGMA*, 8:032, 2012.
- [FSM05] I. Fuentes-Schuller and R. B. Mann. Alice falls into a black hole: Entanglement in noninertial frames. *Physical Review Letters*, 95(12), sep 2005.
- [FT82] Edward Fredkin and Tommaso Toffoli. Conservative logic. *International Journal of Theoretical Physics*, 21(3–4):219–253, 1982.
- [GAP] The GAP Group. *GAP – Groups, Algorithms, and Programming*. Version 4.
- [Gei08] Hansjörg Geiges. *An Introduction to Contact Topology*, volume 109 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2008.
- [Gir87] Jean-Yves Girard. Linear logic. *Theoretical Computer Science*, 50(1):1–101, 1987.
- [Gis19] Nicolas Gisin. Real numbers are the hidden variables of classical mechanics. *Quantum Studies: Mathematics and Foundations*, 7(2):197–201, oct 2019.
- [GJ87] James Glimm and Arthur Jaffe. *Quantum Physics: A Functional Integral Point of View*. Springer, 2 edition, 1987.
- [GJOS14] N Goldman, G Juzeliunas, P Ohberg, and I B Spielman. Light-induced gauge fields for ultracold atoms. *Reports on Progress in Physics*, 77(12):126401, nov 2014.
- [Gor17] Alain Goriely. *The Mathematics and Mechanics of Biological Growth*, volume 45 of *Interdisciplinary Applied Mathematics*. Springer-Verlag, New York, 2017.
- [GP21] Daniel Grady and Dmitri Pavlov. The geometric cobordism hypothesis. *arXiv preprint*, 2021. arXiv:2111.01095.

- [GPMP25] Angel Gonzalez-Prieto, Eva Miranda, and Daniel Peralta-Salas. Topological kleene field theories as a model of computation, 2025.
- [Gra04] Alfred Gray. *Tubes*, volume 221 of *Progress in Mathematics*. Birkhäuser, Basel, 2 edition, 2004.
- [Gro21] Alexander Grothendieck. Pursuing stacks, 2021.
- [HK26] Po-Shen Hsin and Ryohei Kobayashi. Automorphism in gauge theories: Higher symmetries and transversal non-clifford logical gates. *SciPost Physics*, 20(6):159, 2026.
- [Hol06] Barry R. Holstein. Graviton physics. *American Journal of Physics*, 74(11):1002–1011, nov 2006.
- [HS10] Nicholas Harrigan and Robert W. Spekkens. Einstein, incompleteness, and the epistemic view of quantum states. *Foundations of Physics*, 40(2):125–157, 2010.
- [Int14] Int’l Centre for Theoretical Physics. Conversation: Salam, sciama, witten and budinich, Oct 2014. Accessed: 2025-06-18.
- [Jac95] Ted Jacobson. Thermodynamics of spacetime: The einstein equation of state. *Physical Review Letters*, 75(7):1260–1263, aug 1995.
- [JW00] Arthur Jaffe and Edward Witten. Quantum Yang–Mills Theory. In James Carlson, Arthur Jaffe, and Andrew Wiles, editors, *The Millennium Prize Problems*, pages 129–152. Clay Mathematics Institute, 2000.
- [Kam26] M. Benjamin Kaminsky. Expressiveness, verifiability, tractability: Toward a computational thermodynamics, March 2026. Provided mostly for reviewers/readers needing context; access at <https://doi.org/10.5281/zenodo.19217011>.
- [Kas83] H. A. Kastrup. Canonical theories of lagrangian dynamical systems in physics. *Physics Reports*, 101(1-2):151, 1983.
- [Kle52] Stephen Cole Kleene. *Introduction to Metamathematics*. North-Holland, Amsterdam, 1952.
- [KM84] F. R. Klinkhamer and N. S. Manton. A saddle-point solution in the Weinberg-Salam theory. *Physical Review D*, 30(10):2212–2220, 1984.
- [Koc06] Anders Kock. *Synthetic Differential Geometry*. Cambridge University Press, 2 edition, 2006.
- [Kog79] John B. Kogut. An introduction to lattice gauge theory and spin systems. *Reviews of Modern Physics*, 51(4):659–713, 1979.
- [Kon03] Maxim Kontsevich. Deformation quantization of poisson manifolds, I. *Letters in Mathematical Physics*, 66(3):157–216, 2003.
- [KS67] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.
- [KT25] Subhash Kantamneni and Max Tegmark. Language models use trigonometry to do addition, 2025.
- [Kur84] Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*, volume 19 of *Springer Series in Synergetics*. Springer, Berlin, Heidelberg, 1984.
- [Kv98] T. C. Kraan and P. van Baal. Periodic instantons with non-trivial Holonomy. *Nuclear Physics B*, 533(1-3):627–659, 1998.
- [KW07] Anton Kapustin and Edward Witten. Electric-magnetic duality and the geometric langlands program, 2007.

- [LM89] H. Blaine Lawson and Marie-Louise Michelsohn. *Spin Geometry*, volume 38 of *Princeton Mathematical Series*. Princeton University Press, 1989.
- [Lor02] Gino Loria. *Spezielle algebraische und transscendente ebene Kurven. Theorie und Geschichte*. B. G. Teubner, Leipzig, 1902.
- [LS21] Matthias Ludewig and Augusto Stoffel. A framework for geometric field theories and their classification in dimension one. *SIGMA*, 17:072, 2021.
- [Lur09] Jacob Lurie. On the classification of topological field theories. *Current Developments in Mathematics*, 2008:129–280, 2009.
- [Lur11] Jacob Lurie. Derived algebraic geometry xi: Descent theorems, sep 2011. Preprint, September 28, 2011.
- [Lur17] Jacob Lurie. Higher algebra, 2017. Available at <https://www.math.ias.edu/~lurie/papers/HA.pdf>.
- [Mac05] Kirill C. H. Mackenzie. *General Theory of Lie Groupoids and Lie Algebroids*, volume 213 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 2005.
- [Mad27] Erwin Madelung. Quantentheorie in hydrodynamischer form. *Zeitschrift für Physik*, 40(3–4):322–326, 1927.
- [Man16] Ciprian Manolescu. $\text{Pin}(2)$ -equivariant seiberg–witten floer homology and the triangulation conjecture. *Journal of the American Mathematical Society*, 29(1):147–176, 2016.
- [Man19] N. S. Manton. The inevitability of sphalerons in field theory. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 377(2161):20180327, nov 2019.
- [MBS08] Vincent Mathieu, Fabien Buisseret, and Claude Semay. Gluons in glueballs: Spin or helicity? *Physical Review D*, 77(11), jun 2008.
- [Met03] David Metzler. Topological and smooth stacks, 2003.
- [ML84] Per Martin-Löf. *Intuitionistic Type Theory*. Bibliopolis, 1984.
- [MLM92] Saunders Mac Lane and Ieke Moerdijk. *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Universitext. Springer, New York, 1992.
- [Mog91] Eugenio Moggi. Notions of computation and monads. *Information and Computation*, 93(1):55–92, 1991.
- [Moo14] Gregory Moore. Physical mathematics and the future. Written version of a talk given at Strings 2014, Princeton University. Available at <https://www.physics.rutgers.edu/~gmoore/PhysicalMathematicsAndFuture.pdf>, 2014.
- [Mor17] Kenichi Morita. *Theory of reversible computing*. Springer, 2017.
- [MS17] Dusa McDuff and Dietmar Salamon. *Introduction to Symplectic Topology*. Oxford University Press, 3 edition, 2017.
- [MVK⁺22] Chiara Marletto, Vlatko Vedral, Laura T. Knoll, Fabrizio Piacentini, Ettore Bernardi, Enrico Rebufello, Alessio Avella, Marco Gramegna, Ivo Pietro Degiovanni, and Marco Genovese. Emergence of constructor-based irreversibility in quantum systems: Theory and experiment. *Physical Review Letters*, 128(8), feb 2022.
- [MZ96] Maurizio Martellini and Mauro Zeni. The bf formalism for yang-mills theory and the ‘t hooft algebra, 1996.
- [NC10] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th anniversary edition, 2010.

- [Noe18] Emmy Noether. Invariante variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, pages 235–257, 1918.
- [OS73] Konrad Osterwalder and Robert Schrader. Axioms for euclidean green’s functions. *Communications in Mathematical Physics*, 31(2):83–112, 1973.
- [PAD10] T. PADMANABHAN. Equipartition of energy in the horizon degrees of freedom and the emergence of gravity. *Modern Physics Letters A*, 25(14):1129–1136, May 2010.
- [PBR12] Matthew F. Pusey, Jonathan Barrett, and Terry Rudolph. On the reality of the quantum state. *Nature Physics*, 8(6):475–478, 2012.
- [Pen71] Roger Penrose. Angular momentum: An approach to combinatorial space-time. In Ted Bastin, editor, *Quantum Theory and Beyond*, pages 151–180. Cambridge University Press, Cambridge, 1971.
- [Pen87] R. C. Penner. The decorated Teichmüller space of punctured surfaces. *Communications in Mathematical Physics*, 113:299–339, June 1987.
- [Pen89] Roger Penrose. *The Emperor’s New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford University Press, Oxford, UK, 1989. Foreword by Martin Gardner.
- [PER79] Marian Boykan Pour-El and Ian Richards. A computable ordinary differential equation which possesses no computable solution. *Annals of Mathematical Logic*, 17:61–90, 1979.
- [Pic19] Sébastien Picard. Calabi–yau manifolds with torsion and geometric flows. In Daniele Angella, Leandro Arosio, and Eleonora Di Nezza, editors, *Complex Non-Kähler Geometry: Cetraro, Italy 2018*, volume 2246 of *Lecture Notes in Mathematics*, pages 57–120. Springer International Publishing, 2019.
- [Ple77] Jerzy F. Plebański. On the separation of einsteinian substructures. *Journal of Mathematical Physics*, 18(12):2511–2520, 1977.
- [PM73] R. Penrose and M. A. H. MacCallum. Twistor theory: An approach to the quantisation of fields and space-time. *Physics Reports*, 6(4):241–315, 1973.
- [Pol87] A. M. Polyakov. *Gauge Fields and Strings*, volume 3 of *Contemporary Concepts in Physics*. Harwood Academic Publishers, 1987.
- [Pos21] Emil L. Post. Introduction to a general theory of elementary propositions. *American Journal of Mathematics*, 43(3):163–185, 1921.
- [PP09] Gordon Plotkin and Matija Pretnar. Handlers of algebraic effects. In *Programming Languages and Systems*, volume 5502 of *Lecture Notes in Computer Science*, pages 80–94. Springer, 2009.
- [Pra66] Jean Pradines. Théorie de lie pour les groupoïdes différentiables: relations entre propriétés locales et globales. *C. R. Acad. Sci. Paris Sér. A-B*, 263:A907–A910, 1966.
- [Ric53] H. G. Rice. Classes of recursively enumerable sets and their decision problems. *Transactions of the American Mathematical Society*, 74(2):358–366, Mar 1953.
- [RS95] Carlo Rovelli and Lee Smolin. Spin networks and quantum gravity. *Physical Review D*, 52(10):5743–5759, 1995.
- [Sau89] David J. Saunders. *The Geometry of Jet Bundles*, volume 142 of *London Mathematical Society Lecture Note Series*. Cambridge University Press, 1989.
- [Sch13] Urs Schreiber. Differential cohomology in a cohesive infinity-topos, 2013.
- [Sch20] Frederic P. Schuller. Constructive gravity: Foundations and applications, 2020.

- [Sel11] Peter Selinger. A survey of graphical languages for monoidal categories. In Bob Coecke, editor, *New Structures for Physics*, volume 813 of *Lecture Notes in Physics*, pages 289–355. Springer, 2011.
- [Sip13] Michael Sipser. *Introduction to the Theory of Computation*. Cengage Learning, 3 edition, 2013.
- [SMO04] LEE SMOLIN. An invitation to loop quantum gravity. In *Quantum Theory and Symmetries*, pages 655–682. WORLD SCIENTIFIC, Oct 2004.
- [SN24] Sakura Schäfer-Nameki. ICTP Lectures on (Non-)Invertible Generalized Symmetries. *Physics Reports*, 1063:1–55, 2024.
- [Spe69] Donald C. Spencer. Overdetermined systems of linear partial differential equations. *Bulletin of the American Mathematical Society*, 75(2):179–239, 1969.
- [Spe07] Robert W. Spekkens. Evidence for the epistemic view of quantum states: A toy theory. *Physical Review A*, 75:032110, 2007.
- [ST06] Jeremy G. Siek and Walid Taha. Gradual typing for functional languages. In *Scheme and Functional Programming Workshop*, pages 81–92, 2006.
- [Str94] Steven H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. Addison-Wesley, Reading, MA, 1994.
- [SvdH07] E. L. Starostin and G. H. M. van der Heijden. The shape of a Möbius strip. *Nature Materials*, 6(8):563–567, Aug 2007.
- [SW94] Nathan Seiberg and Edward Witten. Monopoles, duality and chiral symmetry breaking in N=2 supersymmetric Yang–Mills theory. *Nuclear Physics B*, 431(3):484–550, 1994.
- [SW13] Urs Schreiber and Konrad Waldorf. Connections on non-abelian gerbes and their holonomy. *Theory and Applications of Categories*, 28:476–540, 2013.
- [Szu12] Matthew P. Szudzik. *The Computable Universe Hypothesis*, pages 479–523. WORLD SCIENTIFIC, oct 2012.
- [Tao18] Terence Tao. On the universality of the incompressible Euler equation on compact manifolds. *Discrete and Continuous Dynamical Systems*, 38(3):1553–1565, 2018.
- [TCK⁺20] TB Thompson, P Chaggar, E Kuhl, A Goriely, and Alzheimer’s Disease Neuroimaging Initiative. Protein-protein interactions in neurodegenerative diseases: A conspiracy theory. *PLoS Computational Biology*, 16(10):e1008267, Oct 2020.
- [tH79] G. ’t Hooft. A property of electric and magnetic flux in nonabelian gauge theories. *Nuclear Physics B*, 153:141–160, 1979.
- [tH16] Gerard ’t Hooft. *The Cellular Automaton Interpretation of Quantum Mechanics*, volume 185 of *Fundamental Theories of Physics*. Springer International Publishing, Cham, 2016.
- [THF06] Sam Tobin-Hochstadt and Matthias Felleisen. Interlanguage migration: From scripts to programs. In *Companion to the 21st ACM SIGPLAN Symposium on Object-Oriented Programming Systems, Languages, and Applications*, OOPSLA ’06, pages 964–974, New York, NY, 2006. Association for Computing Machinery.
- [TM87] Tommaso Toffoli and Norman Margolus. *Cellular Automata Machines: A New Environment for Modeling*. MIT Press Series in Scientific Computation. MIT Press, Cambridge, MA, 1987.
- [TN14] Šárka Todorova-Nová. Quantization of the qcd string with a helical structure. *Physical Review D*, 89(1), jan 2014.

- [TS97] Walid Taha and Tim Sheard. Multi-stage programming with explicit annotations. In *Proceedings of the 1997 ACM SIGPLAN Symposium on Partial Evaluation and Semantics-Based Program Manipulation*, PEPM '97, pages 203–217, New York, NY, 1997. Association for Computing Machinery.
- [TV99] Charles G Torre and Madhavan Varadarajan. Functional evolution of free quantum fields. *Classical and Quantum Gravity*, 16(8):2651–2668, jul 1999.
- [Uhl82] Karen K. Uhlenbeck. Connections with L^p bounds on curvature. *Communications in Mathematical Physics*, 83(1):31–42, 1982.
- [vB96] Pierre van Baal. Instanton moduli for $t_3 \times r$. *Nuclear Physics B - Proceedings Supplements*, 49(1–3):238–249, jun 1996.
- [Waa25] Lukas Waas. Presenting the topological stratified homotopy hypothesis, 2025.
- [Wad90] Philip Wadler. Linear types can change the world! In *Programming Concepts and Methods*, pages 347–359, 1990.
- [Wal95] Robert M. Wald. Quantum field theory in curved spacetime. *arXiv preprint*, 1995. arXiv:gr-qc/9509057.
- [Wei96] Alan Weinstein. Groupoids: Unifying internal and external symmetry. *Notes of the American Mathematical Society*, 43:744–752, 1996. Also available as arXiv:math/9602220.
- [Wei00] Klaus Weihrauch. *Computable Analysis: An Introduction*. Texts in Theoretical Computer Science. Springer, Berlin, 2000.
- [Whi44] Hassler Whitney. The self-intersections of a smooth n -manifold in $2n$ -space. *Annals of Mathematics*, 45(2):220–243, 1944.
- [Wil74] Kenneth G. Wilson. Confinement of quarks. *Phys. Rev. D*, 10(8):2445–2459, Oct 1974.
- [Wit88] E. Witten. Topological Quantum Field Theory. *Communications in Mathematical Physics*, 117(3):353–386, 1988.
- [Wit89] Edward Witten. Quantum field theory and the Jones polynomial. *Communications in Mathematical Physics*, 121(3):351–399, 1989.
- [Woi22] Peter Woit. Notes on the twistor $\mathbf{p}^1$, 2022.
- [WZ84] Frank Wilczek and A. Zee. Appearance of gauge structure in simple dynamical systems. *Physical Review Letters*, 52(24):2111–2114, 1984.
- [Yau78] Shing-Tung Yau. On the Ricci curvature of a compact Kähler manifold and the complex Monge-Ampère equation, I. *Communications on Pure and Applied Mathematics*, 31(3):339–411, 1978.
- [Yet92] David N. Yetter. Topological quantum field theories associated to finite groups and crossed g-sets. *Journal of Knot Theory and Its Ramifications*, 1(1):1–20, 1992.
- [YM54] C. N. Yang and R. L. Mills. Conservation of isotopic spin and isotopic gauge invariance. *Physical Review*, 96(1):191–195, 1954.
- [Zad65] Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.